TM Forum Standard

Information Framework (SID)

Service Domain Business Entities

Information Framework Suite

GB922 Domain Business Entities

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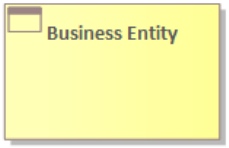
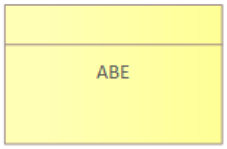
1 General Information

To find the Information framework figure, refer to “GB991 Core Frameworks Concepts and Principles” guidebook.

2 Typographic Conventions

- Relationships starting by a “/” correspond to shortcut and represent navigation relationships (aka derived relationship). A derived relationship corresponds to several relationships replaced by a unique derived relationship to give a synthetic view. It isn’t needed to implement such relationships.
- In diagrams, on Business Entities from a different ABE than the ABE being referred to in the diagram, the name after “from” represents the ABE where the Business Entity is located.
- A Business Entity is represented by a *decorator* located to the top left. 

- The Aggregated Business Entity (ABE) is represented similarly to ArchiMate. 

3 Glossary

Name	Description
ABE	Aggregate Business Entity
BE	Business Entity
eTOM	enhanced Telecom Operation Map (TMF Framework)
ODA	Open Digital Architecture
SID	Shared Information Data model (TMF Framework) is the previous name of Information Framework
TAM	Telecom Applications Map (TMF Framework) is the previous name of Application Framework

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4 Service Domain

One of the goals of the SID model is to enable a seamless transition from business concepts and definitions to other domains, such as networking. This addendum covers the definition of services.

In looking at '"Service", the first problem is one of terminology. Service means many things to many people. For example, many people use the words "Product" and "Service" interchangeably. In addition, though Resources are usually not confused with Services, the relationships between different types of Resources and a Service are confusing to many people.

Business entities in the Product domain represent what is offered to the market. This domain deals with such things as the development, promotion, pricing, and retirement of product offerings, as well as instances of offerings procured by customers and others. Business entities in the Service and Resource domain represent (among other concepts) how products offered to the market are implemented.

This Addendum is built with the express purpose of serving as a foundation for other Service (sub-) models to fit into. Conceptually, this Addendum is a framework, with other Service models (e.g., an MPLS VPN model) "plugging into" it.

4.1 Service Domain Overviews

The Service domain represents roles, information and activities carried out by parties (e.g. individuals / organizations) playing roles that are involved in the strategic planning, definition, development and operational aspects of Services that are used to realize Product offerings to the market. Activities include management of: strategies, capabilities, lifecycles, catalogs, inventories, installations, activations, problems, performance, guiding, mediation, usage statistics and support of customer-facing services that are offered to customers and resource-facing services that are presented to resources by an enterprise.

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Figure [SeD-01] Service ABEs Level 1

The diagram below lists the Service Domain's ABEs.

Service Party Roles ABE

- The Service Party Roles ABE contains all PartyRoles related to the Service Domain such as ServiceManager.

Service Specification ABE

- The Service Specification ABE contains entities that define the shared characteristics of both types of Service entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the characteristics defined in its associated specification.

Service Order ABE

- The Service Order ABE contains all entities that represent a type of Request of one or several services for any type of Service Specification. A Service Order decomposes a Product Order's products into the services associated with a ServiceOrder through which the products are realized.

Service ABE

- A Service represents an instance of ServiceSpecification. The Service ABE contains entities that are used to represent both types of Services: CSP Know-Hows instances (a.k.a. Customer Facing Services or CFS) and Technical solution (a.k.a. Resource Facing Services or RFS).
- The Service carries the place where the Service is in use if relevant, as well as configuration characteristics, such as assigned telephone numbers and internet addresses. The Service also tracks the links to Product realized and/or Resources used.

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Service Configuration ABE

- The Service Configuration ABE represents the specification of the different possible Service configurations as well as the Service Configurations instantiated for a Service.

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Service Test ABE

- The Service Test ABE defines measurement including units and values and how the values are determined. It also includes thresholds used to evaluate the measure and the consequences of violating the thresholds as well as the Service Test measures that reflect the functioning of the Service under test.

Service Usage ABE

- The Service Usage ABE represents the specification of Service Usage types and collects Service Usage records with their characteristics (a.k.a. consumption data).

Service Performance ABE

- The Service Performance ABE collects information used to correlate, consolidate, and validate various performance statistics and other operational characteristics of Know-Hows (a.k.a. CFS) and Technical Solutions (a.k.a. RFS).

Service Problem ABE

- The Service Problem ABE manages faults, alarms, and outages from a Service point-of-view. This is then correlated to trouble tickets, regardless of whether the cause is physical or logical.

Service Capacity ABE

- The Service Capacity ABE specifies ability of Service to provide capability measured in quantity and units of quantity over an extended period. It also manages the Service Capacity Demands.

Service Strategy and Plan ABE (not fully developed)

- The Service Strategy and Plan ABE contains entities that are used to address the need for enhanced or new Services, as well as the retirement of existing Services, by the enterprise. These entities have a strong dependency to both entities in the Resource and Product domains.

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Note: TIP Service Management ABE is likely to be deprecated.

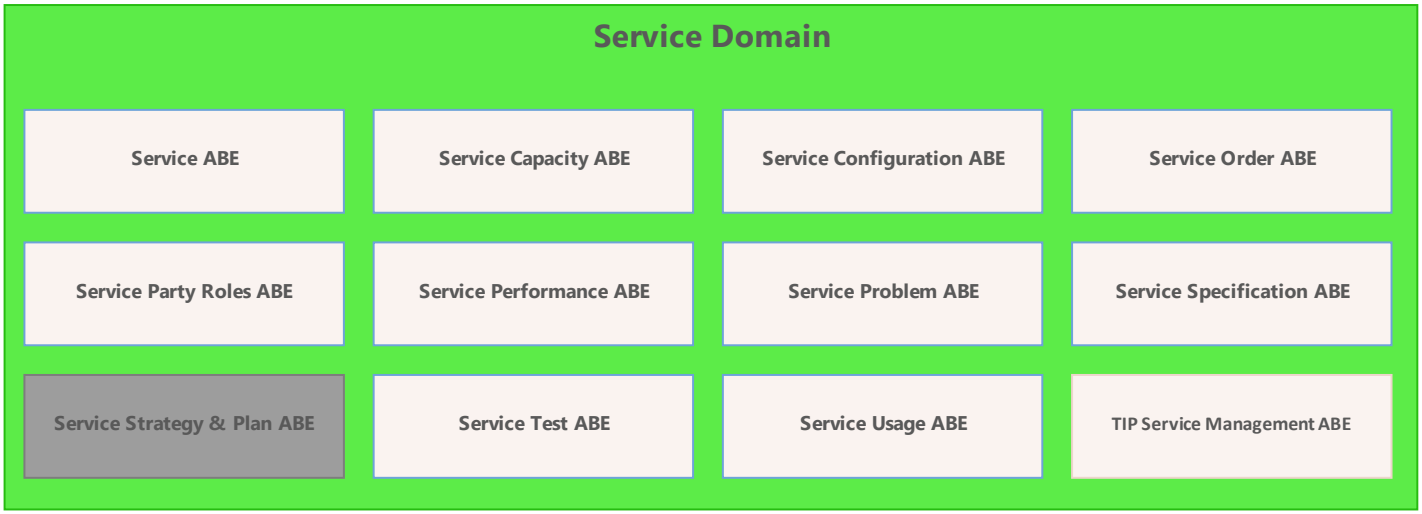

Figure [SeD-01] Service ABEs Level 1

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Figure [SeD-02] Service ABEs Level 1 & 2

The diagram below lists the Customer Domain's ABEs Level 1 and Level 2.

Customer Facing Service Spec ABE

- The Customer Facing Service Spec ABE defines the common characteristics for a set of CustomerFacingServices. This scales by enabling multiple CustomerFacingServiceSpecs to be defined to model the common characteristics of different groups of CustomerFacingServices.

Resource Facing Service Spec ABE

- The Resource Facing Service Spec ABE defines the common characteristics for a set of ResourceFacingServices. This scales by enabling multiple ResourceFacingServiceSpecs to be defined to model the common characteristics of different groups of ResourceFacingServices.

Service Catalog ABE

- A collection of entities that represent the specification of ServiceCatalogs and catalogs described by them that contain ServiceSpecifications.

Customer Facing Service ABE

- The Customer Facing Service ABE defines the characteristics of a particular CustomerFacingService that represents a realization of a Product within an organization's infrastructure.

Resource Facing Service ABE

- ResourceFacingService is an abstraction that defines the characteristics of a particular Service which support the network/infrastructure facing part of the service. For example, a VPN is an example of a CustomerFacingService, while the sub-services that perform different types of routing between network devices making up the VPN are examples of ResourceFacingServices. ResourceFacingServices are "internal" Services that are required to support a CustomerFacingService.

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Service Level Spec ABE

- This ABE includes a third type of Service Specification entity: that of a ServiceLevelSpecification. This type of specification templatizes Services that are bound to a Service Level Agreement.

Service Performance Specification ABE

- The Service Performance Specification ABE defines measure of the manner in which a Service is functioning.

Service Usage Specification ABE

- The Service Usage Specification ABE specifies types and characteristics of usage of ServiceSpecifications.

Service Usage Entity ABE

- The Service Usage Entity ABE collects service usage information, and generates ServiceUsage according to ServiceUsageSpecification for use by other business entities.

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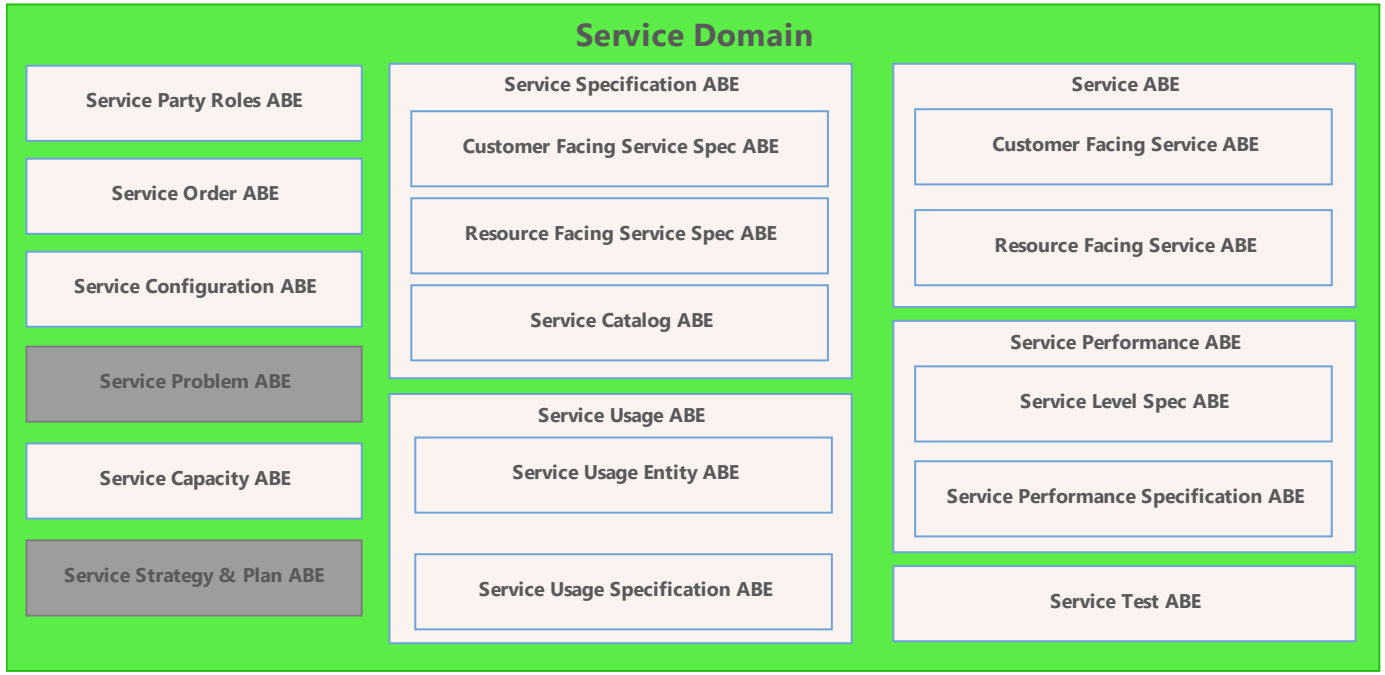

Figure [SeD-02] Service ABEs Level 1 & 2

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4.2 Service Party Roles ABE

The **Service Party Roles ABE** contains all PartyRoles related to the Service Domain such as ServiceManager.

Figure SO.00 - Service Party Roles

The Service Domain party roles presently identified are:

- **ServiceDesigner**: A ServiceDesigner is a party role which is responsible for the design of the Service Provider' know-how (CustomerFacingServiceSpecs). To be valid a know-how needs to align with one or more technical solutions (ResourceFacingServiceSpec or Supplier's ProductSpec).
- **TechnicalSolutionDesigner**: A TechnicalSolutionDesigner is a party role which is responsible for the design of technical solutions (ResourceFacingServiceSpec). To be valid a technical solution needs to align with one or more Resources (ResourceSpec or Supplier's ProductSpec).

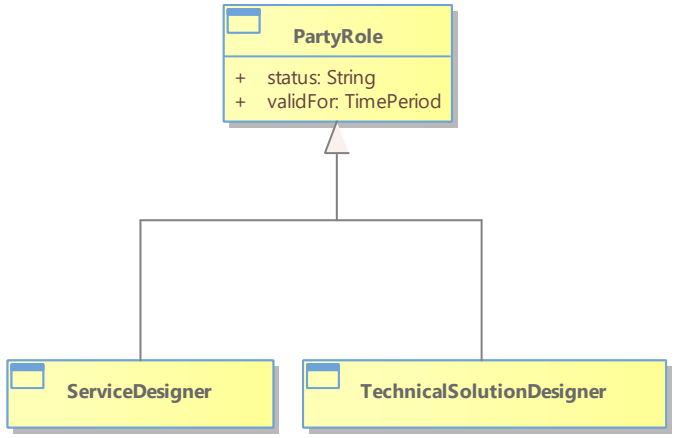

Figure SO.00 - Service Party Roles

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4.3 Service Specification ABE

The **Service Specification ABE** contains entities that define the shared characteristics of both types of Service entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the characteristics defined in its associated specification.

The Service Specification ABE is composed of two main concepts:

1. The CSP Know-Hows specification a.k.a. Customer Facing Services Specification or CFSSpec that describe only functional characteristics and capabilities that the CSP can provide to a customer however the CSP builds on its own all or part of the technical solution. Note that it corresponds only to non-tangible goods.
2. The Technical solution specification a.k.a. Resource Facing Services Specification or RFSpec that describes only technical characteristics and capabilities.

Entities in this ABE focus on adherence to standards, distinguishing features of a Service, dependencies (both physical and logical, as well as on other services), quality, and cost. In general, entities in this ABE enable Services to be bound to Products and run using Resources.

Figure SO.03 - Subclassing ServiceSpecification (OLD SO.10)

ServiceSpecification is the base entity for defining the ServiceSpecification hierarchy.

Each instance of a ServiceSpecification is made up of changeable as well as invariant attributes, relationships, and constraints. A ServiceSpecification defines the shared characteristics of a Service. It can be conceptually thought of as a template that different Service instances can be instantiated from. Each of these Service instances will have the same shared characteristics. However, the characteristics' values of the instantiated Service will be specific to each instance.

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This class can be thought of as a template, which represents a generic specification for implementing a particular type of Service. A ServiceSpecification may consist of other ServiceSpecifications supplied together as a collection.

ServiceSpecification inherits from EntitySpecification, which inherits from RootEntity.

ServiceSpecification is sub-classed as CustomerFacing and ResourceFacing ServiceSpecifications, types of ServiceSpecifications.

CustomerFacingServiceSpec

A CustomerFacingServiceSpec defines the properties (characteristics) common to a particular CustomerFacingService used to realize the associated Product(s). This entity serves as a common basis to build any set of CustomerFacingServices that the service provider needs.

The CustomerFacingServiceSpec represents all the Service Provider's know-how of intangible goods at a functional level. It would be more appropriate to name it KonwHowSpec or ProductFacingServiceSpec, but the name is not changed as it is well known.

The ProductSpecification corresponds to a sub-set of the Service Provider's know-how according to what marketing people want to sell. Therefore, a ProductSpecification is a restriction of CustomerFacingServiceSpecs.

ResourceFacingServiceSpec

The ResourceFacingServiceSpec represents the technical solutions that the Service Provider can implement for CustomerFacingServiceSpec.

Each technical solution (ResourceFacingServiceSpec) requires ResourceSpecifications to be realized and might require buying part of the solution to a supplier (for example the Service Provider might have to buy the Local Loop to provide a broadband access).

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This is the base entity for defining different types of ResourceFacingServiceSpecs. A ResourceFacingServiceSpec is an abstraction that defines the shared characteristics of a particular ResourceFacingService. The shared portion serves as a single common basis to build a set of variable ResourceFacingServices that all use this common ResourceFacingServiceSpec.

ServiceSpecification represents a generic means for implementing a particular type of Service. In essence, a ServiceSpecification defines the common portions of a set of Services, while Service defines a specific instance that is based on a particular ServiceSpecification. This relationship is represented by the SpecifiesService dependency.

It is important to remember that the Specification classes are used to define attributes shared by each associated Entity, whether it is a Service, a Resource, a Product, or something else. This is a generic pattern that is used throughout the SID. Similarly, the Entities themselves, whether they are Services, Resources, Products, or something else, represent the instances that are derived from their associated Specification classes. This is why the cardinality on the “specification” side (e.g., CustomerFacingServiceSpec) of each association (e.g., SpecifiesCustomerFacingService) is defined to be “1” – the SID spec mandate the use of Specifications.

A ServiceSpecification can classify a set of Services using the SpecifiesService association. More specifically, a CustomerFacingServiceSpec classifies a set of CustomerFacingServices created from it and a ResourceFacingServiceSpec classifies a set of ResourceFacingServices created from it.

SpecifiesCustomerFacingService and SpecifiesResourceFacingService are derived relationships coming from SpecifiesService relationship.

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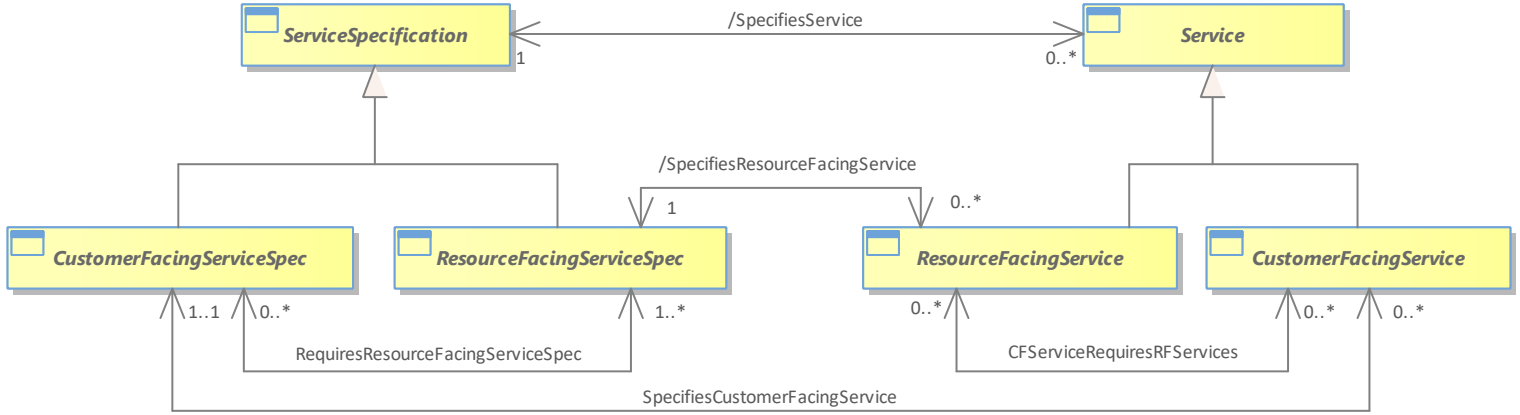

Figure SO.03 - Subclassing ServiceSpecification (OLD SO.10)

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Figure SO.04 - Simplified SID Product Model

The following Figure is a simple model of how a Product is modeled.

A ProductOffering represents what is externally presented to the market for the market's use and can be simple or bundled (atomic / composite pattern).

A ProductOffering represents how the ProductSpecification is sold (i.e. packaging rules, prices, alterations, commitments...).

The ProductSpecification represents a product specification as perceived by the business user and specifies what the marketing operator wants to sell at a functional level (i.e. what are the capacities, which usages are available...).

It can represent:

- material goods (Terminal, phone, mobile, fax, modem)
- or intangible goods (like an Anti-Spam service for email).

A ProductOffering makes available at least one ProductSpecition directly or indirectly (through contained ProductOfferings).

Vice versa, a ProductSpecification may be made available as one or more simple ProductOfferings.

A Product represents exclusively the instantiation of a ProductOffering or ProductSpecification by a Party playing a PartyRole, such as a Customer.

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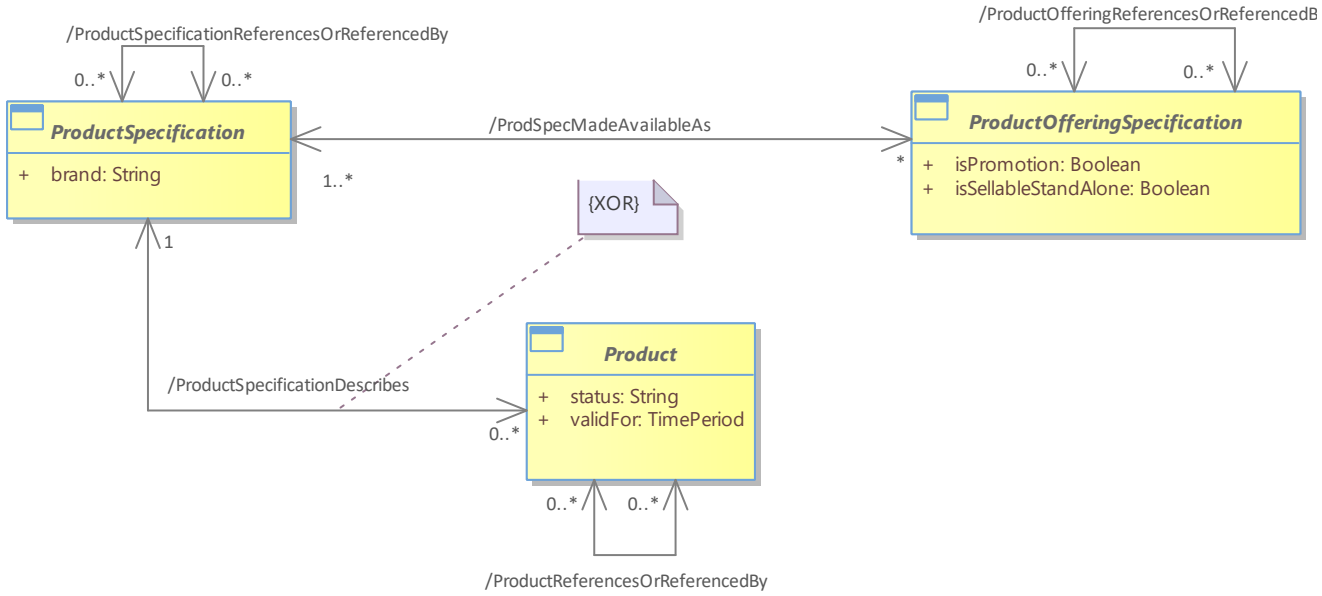

Figure SO.04 - Simplified SID Product Model

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Figure SO.08 - Service Specifications and other Domains

The Product guidebook refers to the concept of a ProductSpecification.

As a reminder, ProductOfferings provide tangible or intangible Products that organizations, such as Service Providers and Enterprises, market, sell or lease to Customers.

A ProductSpecification is a detailed description of a tangible or intangible object made available externally in the form of a ProductOffering to Customers or other Parties playing a particular PartyRole. The information described by a ProductSpecification is a set of invariant data that is common to all realizations of that ProductSpecification. A ProductSpecification may exist as a single ProductSpecification, or consist of a set of ProductSpecifications supplied together as a collection.

Note that a direct association between Product and Service does not exist. Instead, the ProductSpecRealizedAsCFSSpec association is defined between ProductSpecification and CustomerFacingServiceSpec.

Remember that the purpose of a Specification is to serve as a template. This enables specific instances of a desired entity to be more easily and consistently created.

The ProductSpecRealizedAsCFSSpec association enables a given ProductSpecification to specify CustomerFacingServiceSpecs restricted by the ProductSpecification. This relationship is only used when the ProductSpecification corresponds to intangible goods.

Note in particular that there is no association defined between a ProductSpecification and a ServiceSpecification. This is because such an association would include ResourceFacingServiceSpecs, and that violates the effort to separate CustomerFacingServices from ResourceFacingServices. Thus, the ProductSpecRealizedAsCFSSpec is defined to enforce this separation.

The ProductSpecRealizedAs association enables a ProductSpecification to define zero or more ResourceSpecifications. This is analogous to the ProductSpecRealizedAsCFSSpec association for ProductSpecification corresponding to tangible goods.

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The ProductSpecRealizedAsCFSSpec and ProductSpecRealizedAs relationships are mutualy exclusives.

A Service may depend on one or more specific PhysicalResources. For example, the Customer connects to the VPN through a PhysicalPort on a Card of a PhysicalDevice. This is represented using the RFServiceConfiguredUsing association, whose purpose is to bind Resources to Services without identifying what type of Service or Resource is being bound.

For example, this enables services to be “reserved” on a PhysicalResource for resource planning and consumption purposes, without specifically identifying the low-level system resources required.

The above discussion has been conducted in terms of “instances” of these objects. For the most part we will talk about these concepts more generically using the concept of a “specification”.

A ResourceSpecification can classify a set of Resources using the SpecifiesResource association, so its subclasses, PhysicalResourceSpec, LogicalResourceSpec and CompoundResourceSpec, inherit this association.

A Product is realized by zero or more CustomerFacingServices through the ProductRealizedAsCFService association. This enables a Product to in effect customize a CustomerFacingService already defined by a CustomerFacingServiceSpec. In other words, the ProductSpecRealizedAsCFSSpec association is used to define the set of CustomerFacingServiceSpecs for a given ProductSpecification, which means that all Products based off of that ProductSpecification will use that set of CustomerFacingServiceSpecs. Different Products have the ability to add their own features to these CustomerFacingServices, and do so through the ProductRealizedAsCFService association.

Similarly, a Product realizes zero or more Resources through the ProductRealizedAsResource association. Note that the cardinality on the Product side of both of these associations is optional, which means that a Product doesn’t have to implement these two associations. This is because not all Products are realized through CustomerFacingServices and Resources.

CustomerFacingServices cannot be implemented without ResourceFacingServices to support them. This is reflected by the CFServiceRequiresRFServices association. This relationship is templatized by the RequiresResourceFacingServiceSpec association, which defines the set of ResourceFacingServiceSpecifications that are required by a particular CustomerFacingServiceSpecification. Again, note

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that the cardinality on the ResourceFacingServiceSpec side is 1,*, meaning that a CustomerFacingServiceSpec cannot be instantiated unless it is bound to at least one ResourceFacingServiceSpec.

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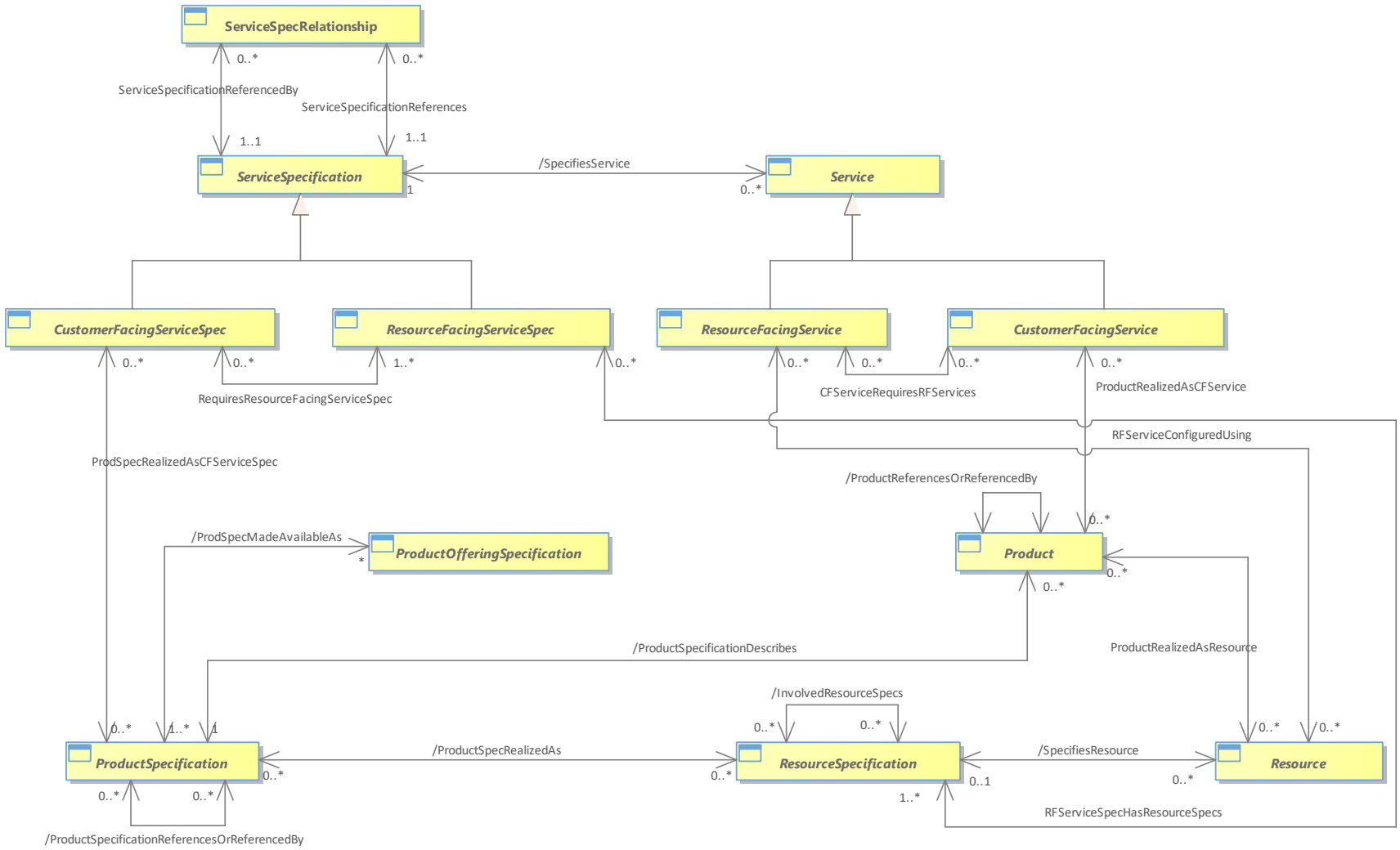

Figure SO.08 - Service Specifications and other Domains

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Figure SO.09 - VPNService – VPNServiceSpecification Example

Let's now examine a practical example. A ProductSpecification will be used to definea ProductOffering. A ServiceSpecification will define how to build a VPN to implement the defined in the ProductSpecification. Now, there are many different types of VPNs. For example, there are network-based VPNs, VPNs based on tunneling, VPNs based on encryption, hybrid VPNs such as MPLS VPNs, and others. All of these VPNs have certain things in common, which can be represented by a VPNServiceSpecification.

Consider an MPLS VPN built to the RFC2547bis specification. The MPLS VPN is an example of a CustomerFacingService. This particular type of MPLS VPN is characterized by having three different types of routers. Each of these – the CustomerPremise, ProviderEdge, and Provider routers – can run a different set of protocols, perform a different function, and be managed in a different way.

An MPLS VPN is implemented in a significantly different manner than other types of VPNs. Thus, an MPLSVPNService is a different type of object than (for example) an IPsecVPNService. However, they have three important things in common. First, they are both defined through a ProductSpecification and realized as a Product. Second, they can both be related to the same CustomerFacingServiceSpec. This is useful if, for example, the Service Provider is defining multiple types of VPNs that the Customer could use. Third, it is possible that they could both use the same CustomerFacingServices and/or ResourceFacingServices.

Note that a more complete picture would also show that the two VPNs could use the same PhysicalResources and/or LogicalResources as well. However, this complicates the above figure and is out of scope for this Addendum.

The difference between the MPLSVPNService and the IPsecVPNService is in the protocols and architecture used, which in turn cause additional system-specific differences that don't show up in a business model. Rather, the SID identifies these two CustomerFacingServices and relates them to zero or more ResourceFacingServices through the CFServiceRequiresRFServices association (as well as the RequiresResourceFacingServiceSpec association from the template point-of-view).

A ResourceFacingServiceSpec is related to a ResourceSpecification by the RFServiceSpecHasResourceSpecs association. Furthermore, a ResourceSpecification can be used to define the set of Resources that it requires using the SpecifiesResource association. This combination

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of associations means that conceivably, the same Resource could be used to support both types of VPNServices, assuming that it has the software and/or hardware to do so.

Note that in the following Figure, the VPNServiceSpecification is an abstract class that contains the basic attributes and relationships and constraints for defining a VPNService without specifying the specific type and technology used (e.g., Ipsec vs. MPLS) to implement the VPNService. For example, the VPNServiceSpecification can be used to define the set of SLSs that will be used to measure performance of the VPN. Both an MPLS VPN as well as an Ipsec VPN specification are derived from the VPNServiceSpecification class.

The MPLSVPNServiceSpecification and IPsecVPNServiceSpecification are related to the MPLSVPNService and the IPsecVPNService through the SpecifiesMPLSVPNServices and the SpecifiesIPsecVPNServices associations, respectively. This enables different MPLSVPNServices (as well as IPsecVPNServices) to be specified by the same MPLSVPNServiceSpecification (or IPsecVPNServiceSpecification, respectively). This enables the Service Provider to define a number of characteristics for each of the CustomerFacingServices and thereby templatize the definition of each. Combined with the LogicalResource and QoS Addenda, a complete roadmap will be developed for configuring this templatized definition.

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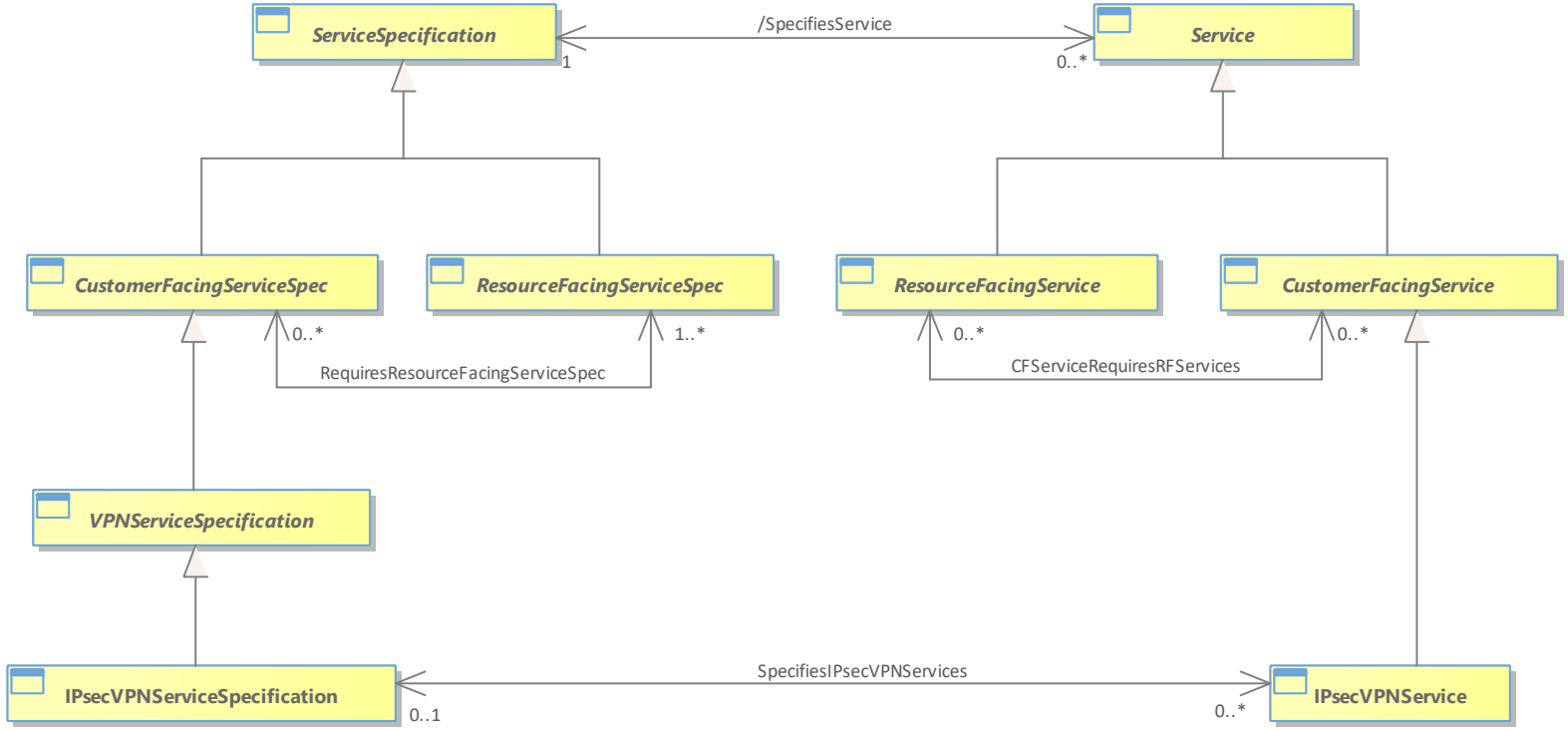

Figure SO.09 - VPNService – VPNServiceSpecification Example

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Figure SO.11 - Modeling Stand-Alone and Aggregate ServiceSpecifications

There are two basic ways to define and use a ServiceSpecification. One is to define a stand-alone ServiceSpecification, while the other is to define an aggregate ServiceSpecification. The difference is straightforward – aggregate ServiceSpecifications are used to define collections of ServiceSpecifications, whereas stand-alone ServiceSpecifications define all the functionality that is needed by themselves.

If we did this, then the ServiceSpecAtomic would be defined as an abstract base class for defining ServiceSpecifications that do not have any subordinate ServiceSpecifications. These ServiceSpecifications are stand-alone in definition and use, and don't require any supporting ServiceSpecifications to define the common characteristics of Services that it serves as a template for.

Similarly, a ServiceSpecComposite class would be defined as an abstract base class for defining ServiceSpecifications that are formed by aggregating other ServiceSpecifications. The types of ServiceSpecifications that are aggregated may be ServiceSpecAtomic and/or ServiceSpecComposite instances. A ServiceSpecComposite collectively defines all of the common characteristics of Services that it serves as a template for.

However, we already have defined two subclasses of ServiceSpecification, called CustomerFacingService and ResourceFacingService. We can just as easily apply the composite pattern to these classes instead of defining a ServiceSpecAtomic and a ServiceSpecComposite class. This is exactly what is done in the model shown in the following Figure. We can reserve the right to define generic ServiceSpecAtomic and ServiceSpecComposite classes that are not CustomerFacingServices or ResourceFacingServices in the future. Note that at this time, no examples of such ServiceSpecifications exist.

Therefore, we will proceed with the definition of CustomerFacingServiceSpecs and ResourceFacingServiceSpecs using the composite pattern to model this.

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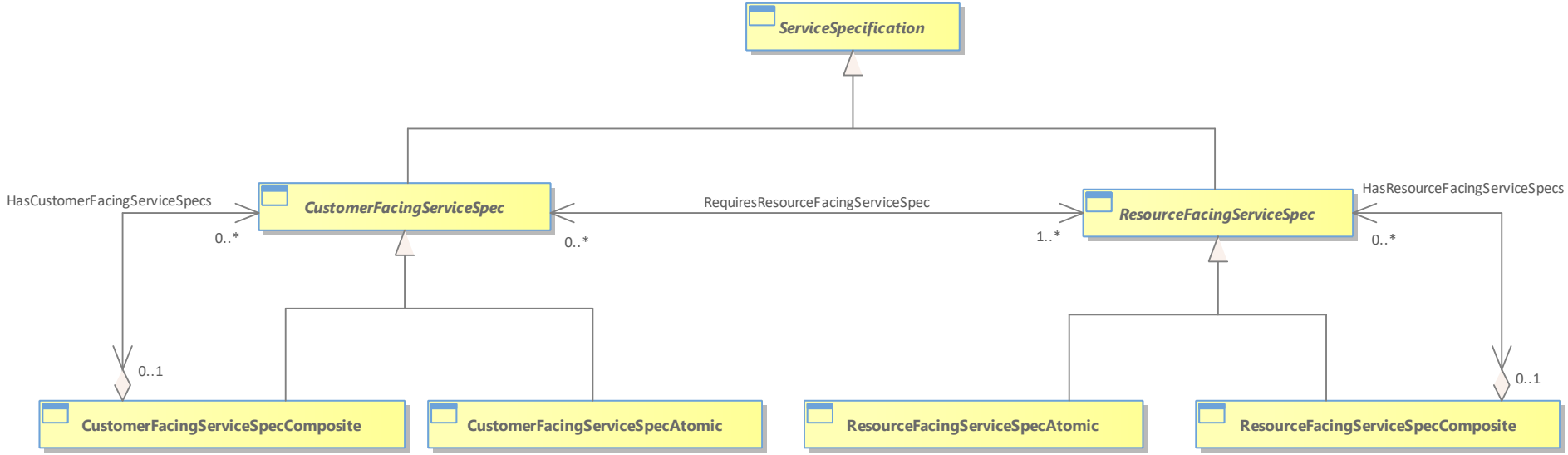

Figure SO.11 - Modeling Stand-Alone and Aggregate ServiceSpecifications

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Figure SO.13 - Modeling the Versioning of a ServiceSpecification

Applications often require the ability to distinguish between different versions of objects in general. This Addendum applies this principal to ServiceSpecifications. The ServiceSpecVersion class represents a particular form or variety of a ServiceSpecification that is different from others or from the original. The form represents differences in attributes, relationships, and/or constraints that characterize this particular ServiceSpecification, but which are not enough to warrant creating a new ServiceSpecification.

The CustomerFacingServiceSpecVersion and ResourceFacingServiceSpecVersion subclasses are defined to enable an application to track different versions of CustomerFacingServiceSpecs and ResourceFacingServiceSpecs independently.

The ServSpecModifications aggregation represents the set of versions of this ServiceSpecification. The semantics of this aggregation are implemented with the ServiceSpecVersionDetails class.

The two version details association classes are both derived from the ServSpecModifications aggregation. They add additional detail to the generic concept of defining a version of a ServiceSpecification.

The same exact principle applies to tracking different versions of the more generic ServiceSpecAtomic and ServiceSpecComposite objects. This is not shown because the focus of this Addendum is on the definition of CustomerFacingServices and ResourceFacingServices.

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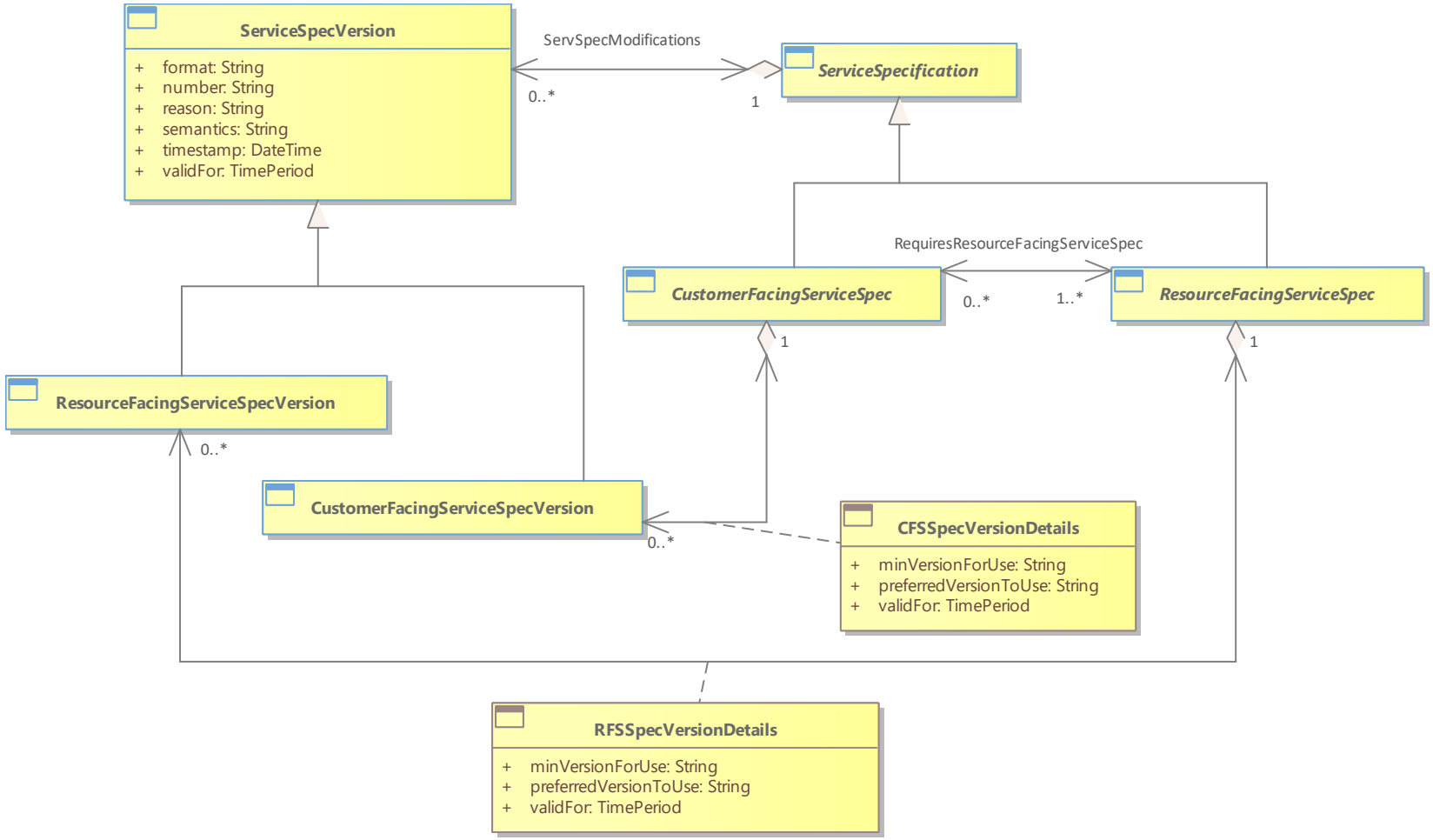

Figure SO.13 - Modeling the Versioning of a ServiceSpecification

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Figure SO.14 - The Concept of a ServiceSpecificationType

The preceding discussion has shown that multiple ServiceSpecifications can share common characteristics. This can be more efficiently represented using the concept of a ServiceSpecificationType.

The ServiceSpecificationType defines a generic category of ServiceSpecifications. Each ServiceSpecificationType serves to group a set of particular ServiceSpecifications that can be used together. The result of this is a more efficient ability to define a set of related Services that can be grouped together to form a higher-level Service. For example, a given higher-level Service might include VPN and QoS Services. If these Services are always used together, then they can be categorized using a common type.

The ServiceSpecificationTypeCategorizes association defines how ServiceSpecificationTypes categorize different ServiceSpecifications. The InvolvedServiceSpecTypes is a derived association as indicated by the "/" prefix. It is derived from an association between EntitySpecificationTypes which enables hierarchies of specification type entities to be defined.

Note that in principle, ServiceSpecificationType could be subclassed, so that the subclasses of ServiceSpecificationType could be used to categorize different subclasses of ServiceSpecification. This is not recommended for the purposes of this document, as the ServiceSpecificationType object is used solely to help categorize objects. The only reason a ServiceSpecificationType object should be subclassed is if that subclass adds additional semantics (in the form of attributes, relationships, and/or constraints). Thus, it makes perfect sense for applications that are based on the SID to subclass ServiceSpecificationType.

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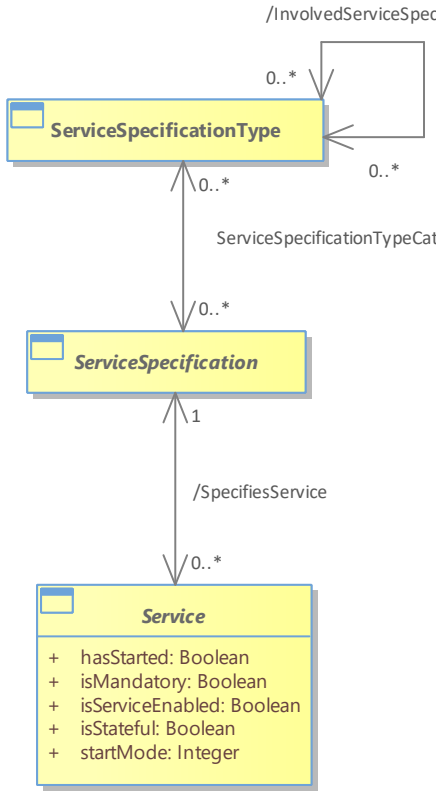

Figure SO.14 - The Concept of a ServiceSpecificationType

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Figure SO.17 - The Concept of ServiceSpecificationRoles (OLD SO.21)

ServiceSpecificationRole is an abstract base class that defines a ServiceSpecification in terms of a set of roles. The roles are then used to characterize the shared functionality of the ServiceSpecification, regardless of whether it is a resource or a customer facing service specification.

ServiceSpecificationRoles represent the shared functionality of a ServiceRole. Representing a Service in terms of ServiceSpecificationRoles enables the functionality of the Service to be defined independently of BusinessActor, PhysicalResource, LogicalResource, or ot her Services.

The ServiceSpecificationRole defines a ServiceSpecification in terms of possible roles.

The roles are then associated for both CustomerFacing and ResourceFacing services through the CustomerFacingServiceSpecificationRole and ResourceFacingServiceSpecificationRole subclasses. The utility of the ServiceSpecificationRole class is to present a single point for accumulating common relationships that its subclasses can inherit.

The RequiresResourceFacingServiceSpecRoles relationship is used to define the set of ResourceFacingServiceSpecRoles that are required by a particular CustomerFacingServiceSpecRole. Conceptually, this enables roles to be used to define the common characteristics of a set of ResourceFacingServiceSpecs that are required by a particular CustomerFacingServiceSpec.

Note that the cardinality of the RequiresResourceFacingServiceSpecRoles is * on the CustomerFacingServiceSpecificationRole side and 1..* on the ResourceFacingServiceSpecRole (component) side. This is because a ResourceFacingServiceSpecRole can exist without being bound into a CustomerFacingServiceSpecRole (e.g., in testing the network), but a CustomerFacingServiceSpecRole requires at least one ResourceFacingServiceSpecRole to function. Note also that this cardinality is symmetric with the cardinality defined in the RequiresResourceFacingServiceSpec association.

The InvolvedServiceSpecRoles association is used to define a set of ServiceSpecificationRoles that are involved with, or related to, each other in order to build a particular type of ServiceSpecification.

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This association applies equally to CustomerFacingServicesSpec and ResourceFacingServiceSpec.

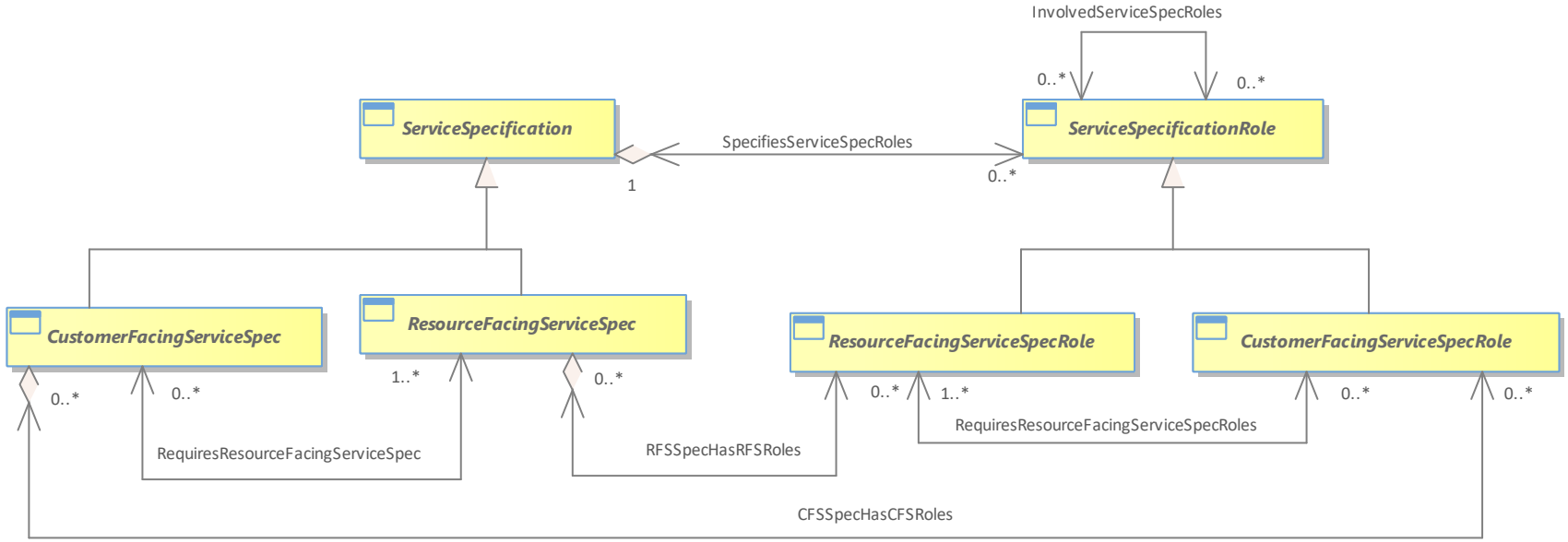

Figure SO.17 - The Concept of ServiceSpecificationRoles (OLD SO.21)

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Figure SO.18 - Interaction Between ServiceSpecRoles and ResourceSpecRoles

ServiceSpecificationRole is an abstract base class that defines a ServiceSpecification in terms of a set of roles. The roles are then used to characterize the shared functionality of the Service, regardless of whether it is a resource or a customer facing service.

ServiceSpecificationRoles represent the shared functionality of a ServiceRole. Representing a Service in terms of ServiceSpecificationRoles enables the functionality of the Service to be defined independently of BusinessActor, PhysicalResource, LogicalResource, or other Services.

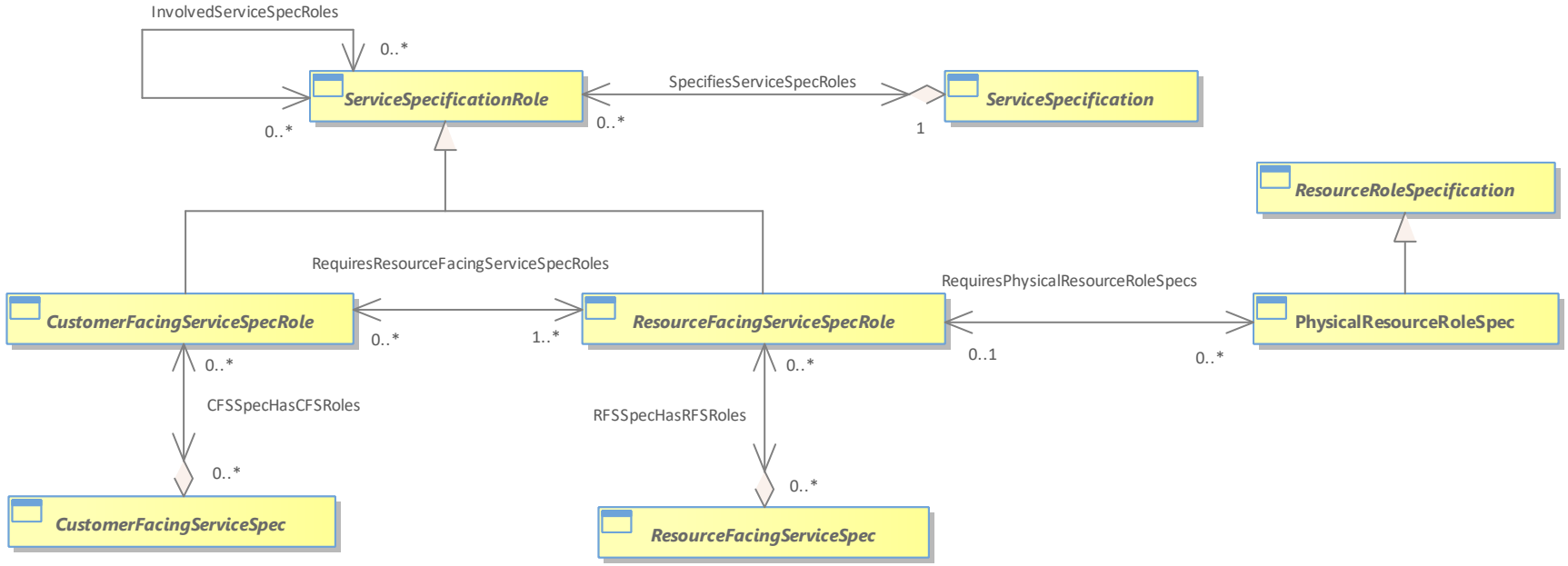

Figure SO.18 - Interaction Between ServiceSpecRoles and ResourceSpecRoles

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Figure SO.22 -The Concept of ServiceSpecCharacteristic

A major focus of marketing programs is to represent the key features of a Product. This was documented in the Product domain guide book as ProductSpecCharacteristics. Since a Service is the realization of a Product, it makes sense to reuse this concept and define the notion of a ServiceSpecCharacteristic.

The concept behind a ServiceSpecCharacteristic and a ServiceSpecCharacteristic Value is similar in nature to those of a ProductSpecCharacteristic ProductCharacteristic Value, respectively. Services have a large variety of important properties (which we call characteristics), such as bandwidth, type of QoS, and so forth. Each of these characteristics is used at the business level to characterize a Service. This Addendum models these characteristics as a set of entities, attributes, and relationships that help characterize a ServiceSpecification and Service, respectively.

Additional information about the CharacteristicSpecification/CharacteristicValue pattern can be found in the Common guide book subsection Root Business Entities ABE.

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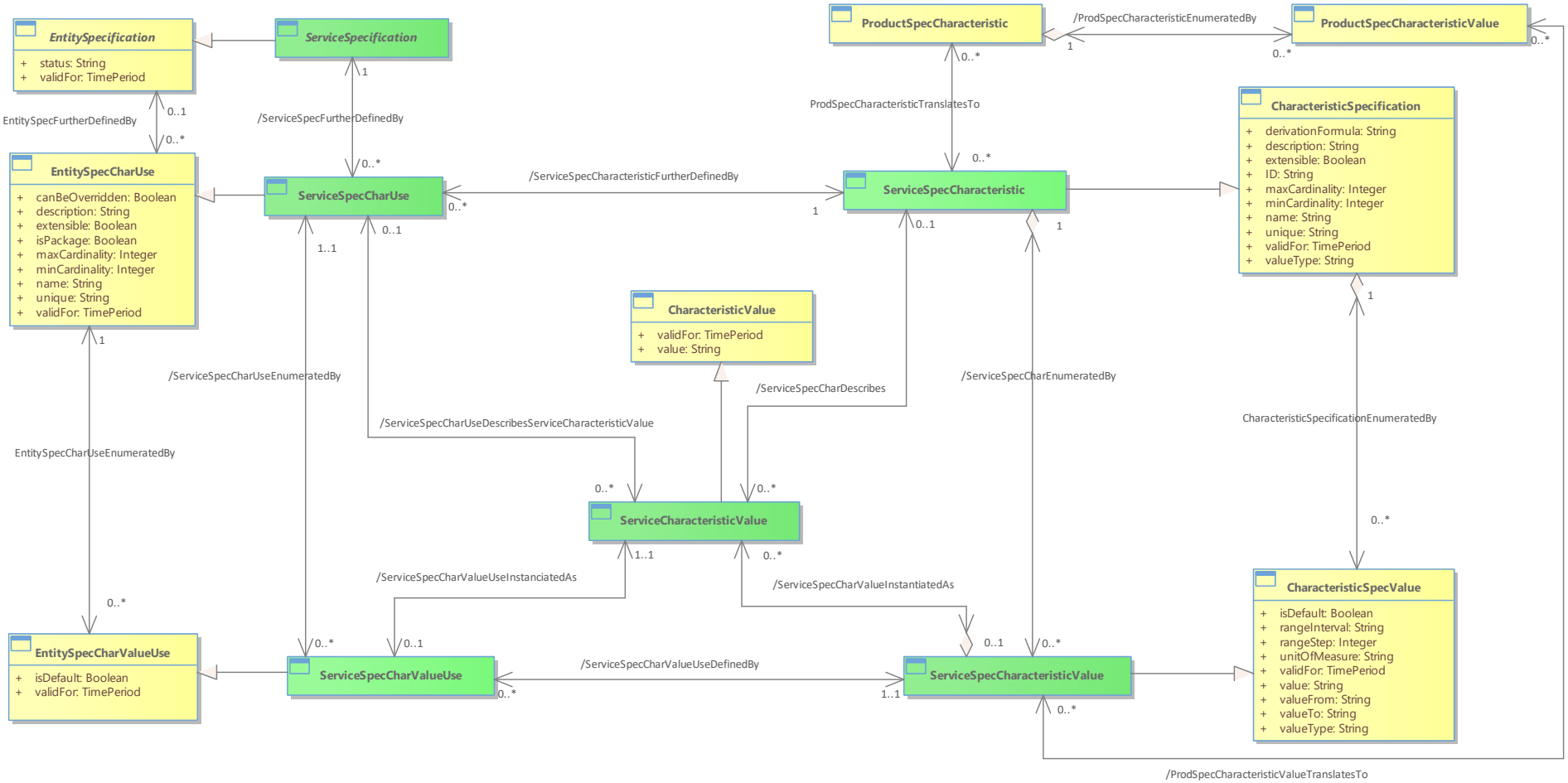

Figure SO.22 -The Concept of ServiceSpecCharacteristic

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Figure SO.22a - Service and Resource Characteristic Associations

Service Features and Service Parameters are two other synonyms for a ServiceSpecCharacteristic, and are used in models from other TM Forum member companies.

ServiceSpecCharacteristics and ServiceSpecCharacteristicValues capture the differentiating features of a particular Service. However, ServiceSpecCharacteristics are not versionable objects. This is because ServiceSpecCharacteristics do not add any information in and of themselves – rather, they are used to emphasize key features of a ServiceSpecification.

There are cases when an instance of a ServiceSpecCharacteristic and/or a ServiceSpecCharacteristicValue must be translated to a corresponding ResourceSpecCharacteristic and/or a ResourceSpecCharacteristicValue. For example, there may be a ServiceSpecCharactertistic that represents the size of an email box as defined by associated 10MB instance of SerivceSpecCharacteristicValue. This value may be translated into the actual size of 10.5MB as defined by the ResourceSpecCharacteristicValue associated to an email box size ResourceSpecCharacteristic.

The following Figure shows the associations between characteristics in the Service and Resource domains that enable this translation.

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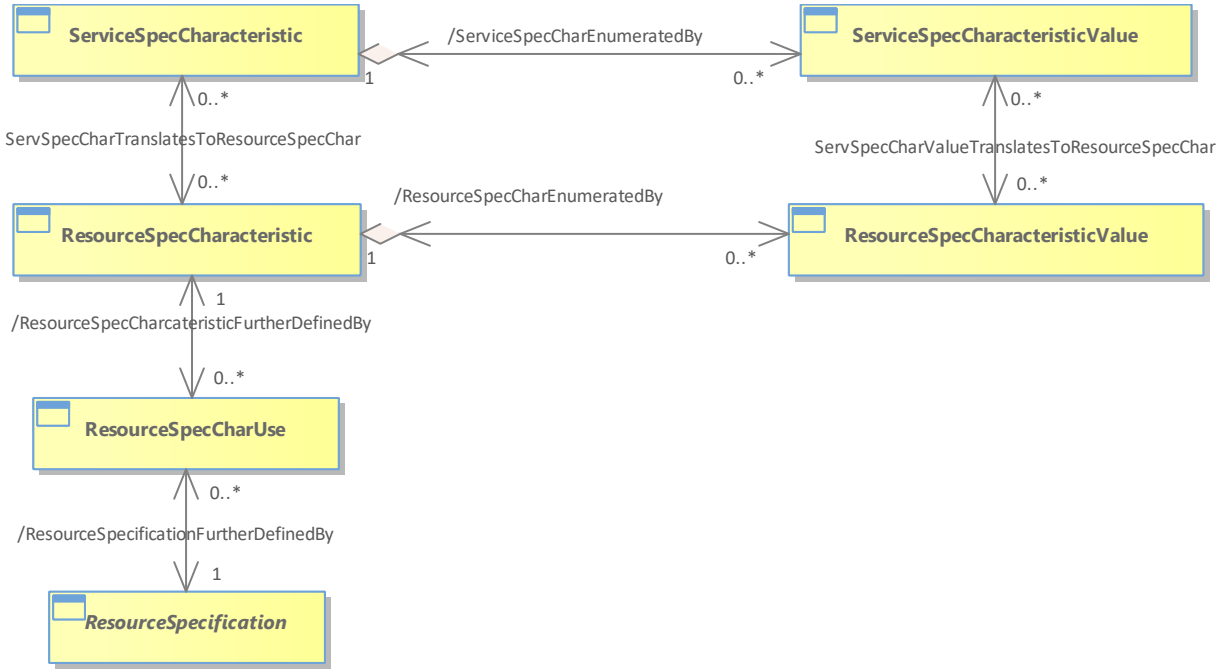

Figure SO.22a - Service and Resource Characteristic Associations

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Figure SO.23 - ServiceCharacteristicValue and ServiceSpecCharacteristicValue

The purpose of the ServiceCharacteristicValue and ServiceSpecCharacteristicValue objects is to define how instances of these objects can be populated. Again, it was influenced from the earlier work done on similar Product objects. The ServiceCharacteristicValue and the ServiceSpecCharacteristicValue classes, shown in the following Figure define this concept.

A ServiceSpecCharacteristicValue object is used to define a set of attribute values, each of which can be assigned to a corresponding attribute in a ServiceSpecCharacteristic object. This is represented by the ServiceSpecCharEnumeratedBy association. The values of the attributes in the ServiceSpecCharacteristicValue object describe the values of the attributes that a corresponding ServiceCharacteristicValue object can take on.

Note that we have elected to use classes instead of OCL to model these constraints. Therefore classes can be used to create reusable “constraints”, which are in effect what the ServiceSpecCharacteristicValue are.

The following Figure also shows how a ProductCharacteristicValue has been translated to a ServiceCharacteristicValue.

For example, consider the definition of a VPNService, as previously shown in Figure SO.09. A set of ServiceSpecCharacteristicValue could, for example, be used to define the types of VPN that are being offered. A corresponding ServiceCharacteristicValue would then represent the salient features of a particular instance of a Service VPN. For example, a Service Provider may be constrained to offer either an MPLS or an Ipsec VPN (as opposed to other types) of VPNs. These would be examples of ServiceSpecCharacteristicValues. Similarly, perhaps the MPLS VPN could only support a defined upper limit of CE devices. This upper limit would be an example of a ServiceCharacteristicValue.

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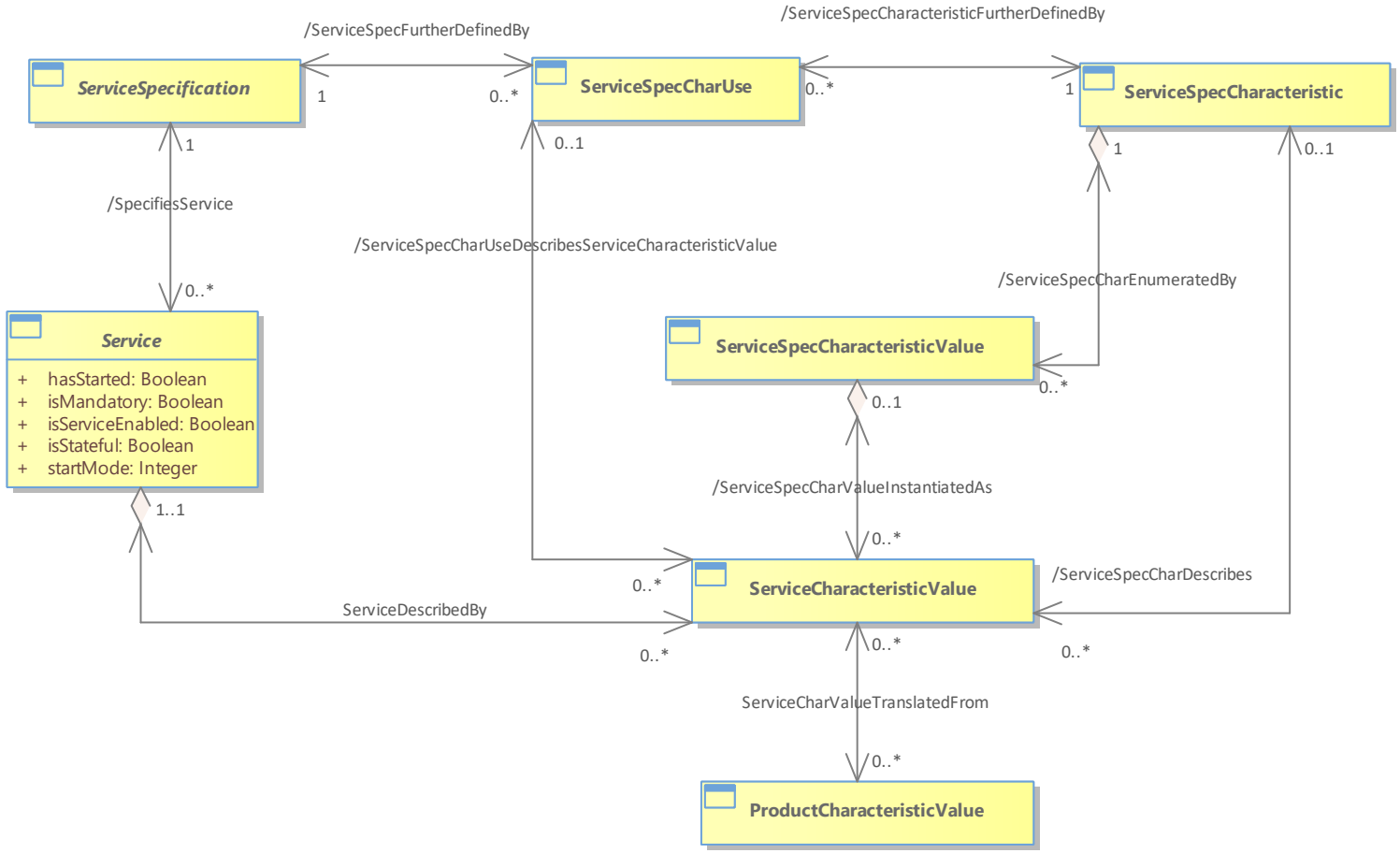

Figure SO.23 - ServiceCharacteristicValue and ServiceSpecCharacteristicValue

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Figure SO.24 - Multiple Command Syntaxes and Programming Models

Linking to QoS – ServicePackage and ServiceBundle

The Service QoS guidebook defines of the QoS model. The purpose of this model is to satisfy one of the use cases of the Service Domain in general, by describing how QoS mechanisms on a device are configured. Here, “mechanisms” refers to modeling the end-user features of the device.

Conceptually, what makes this a daunting task is that Services involve multiple devices which can have heterogeneous commands and programming models, as is shown in Figure SO.24 below.

Current network devices have different programming models using different commands. This makes it very difficult for common tasks to be performed, because in effect the network operator must know different languages. If we also take into account the fact that commands change per OS version, the analogy grows to a single person having to know multiple dialects of multiple languages. This cannot scale.

The Figure above illustrates the use of different commands and programming models to perform the same high level task. The device on the left has configuration modes, each mode defining a set of appropriate commands. The device on the right has no such modes. Furthermore, the syntax used to perform the same task is drastically different.

Note that each network device supports a Service through configuring a DeviceInterface and/or its LogicalDevice. This is worse for two reasons. First, the thought of doing this manually, for thousands of networks, is no longer viable, due to both the complexity of networks, the complexity of services that networks support, and the interaction between services supported on a network. Second, since each Device in general has a different programming model, there is no way to tell in advance which way (or ways) a Device can be programmed.

Clearly, manually configuring each Service that can run on a given device is not a viable process. This Addendum introduces two innovative classes – ServicePackageSpec and ServiceBundleSpec – that serve to link the business requirements of QoS to its implementation. Note that this is done at the specification level, to enable this to be templatized. Of course, this is yet another instance of the Entity-EntitySpecification pattern, which is used throughout the SID. (This in and of itself is important, because it shows how the existing Service model can be easily extended to bridge new concepts using the same set of standardized mechanisms. This is an indicator of the robustness and inherent extensibility of the SID.)

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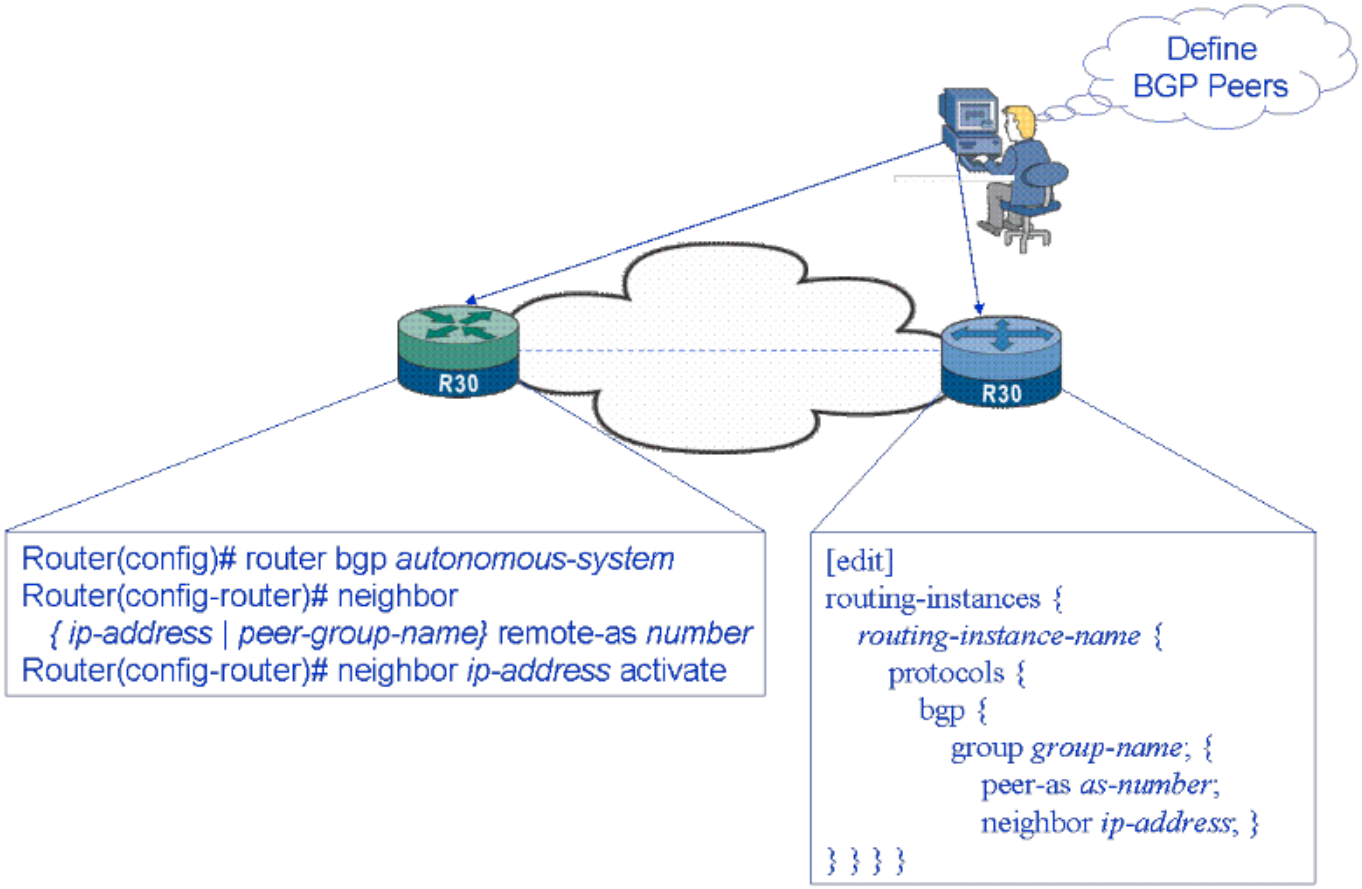

Figure SO.24 - Multiple Command Syntaxes and Programming Models

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Figure SO.25 - The Concept of a ServicePackage

A ServicePackageSpec defines the concept of bundling a set of different CustomerFacingServiceSpecs to meet the functionality specified by one or more ProductSpecifications. This enables the common characteristics of these CustomerFacingServices to be specified, so that multiple Products can be built from their associated ProductSpecification.

Treating this set of CustomerFacingServiceSpecs as a single object is very important for building complex Services, such as a VPN. This enables a single Product to be offered to the Customer (or another type of PartyRole), even though in reality the Product consists of a set of different CustomerFacingServices that must work together to provide the functionality that the Customer needs.

The composite pattern is used to make the ServicePackageSpec object extensible. A ServicePackageSpecAtomic object models different ServicePackageSpecs as a set of different instances of individual, independent CustomerFacingServiceSpecs. This is fundamentally different than the ServicePackageSpecComposite object, which is used to model one ServicePackageSpec as the combination of other existing ServicePackageSpecs (as well as providing its own extensions).

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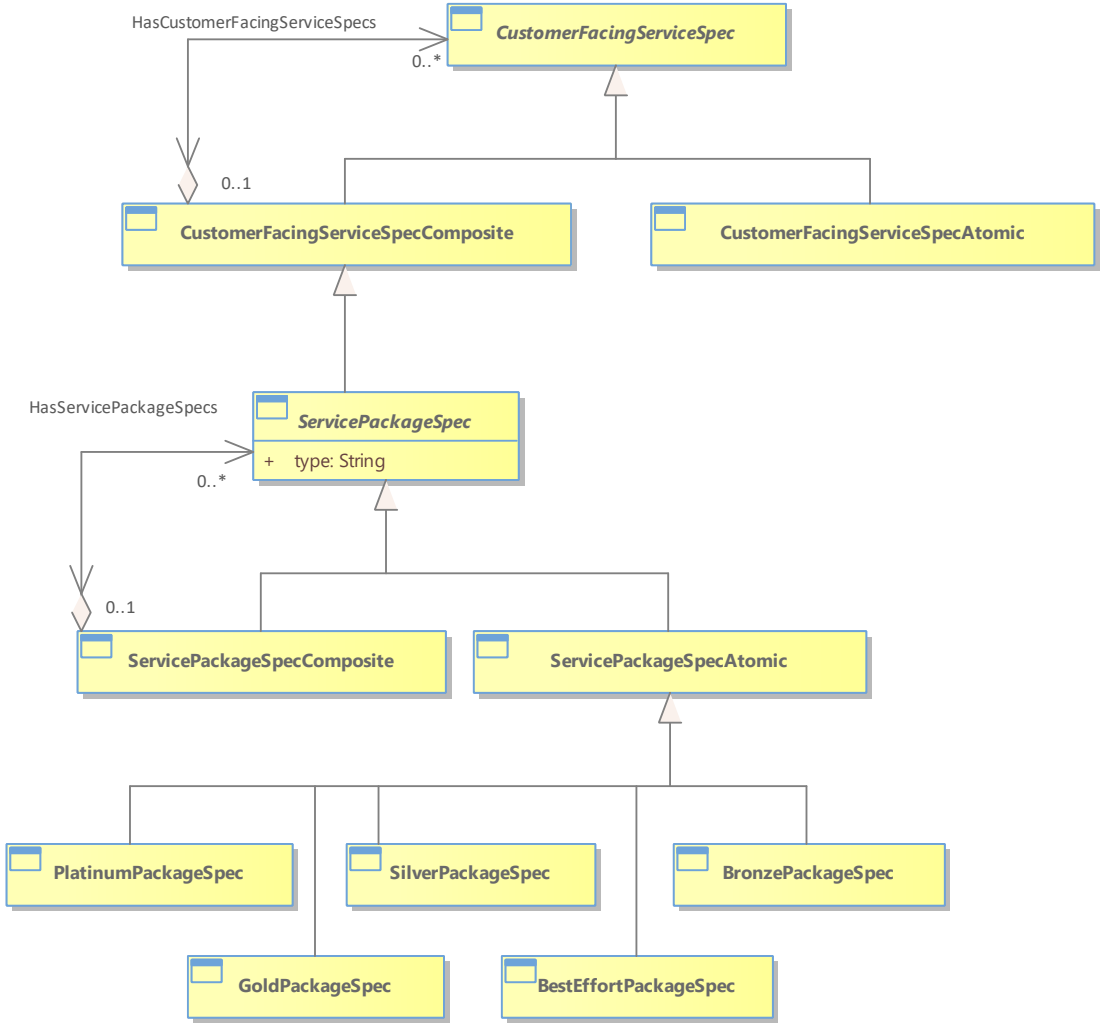

Figure SO.25 - The Concept of a ServicePackage

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Figure SO.26 - Grouping Together CustomerFacingServiceSpecifications into a ServicePackage

For example, GoldPackageSpec is an individual packaging of services, and is therefore an instance of the ServicePackageSpecAtomic class. If there was a service offering that combined the services defined by the GoldPackageSpec with those defined by the PlatinumPackageSpec, then that would be an instance of the ServicePackageSpecComposite class.

Figure SO.25 shows five exemplary subclasses of the ServicePackageSpecAtomic class – PlatinumPackageSpec, GoldPackageSpec, SilverPackageSpec, BronzePackageSpec, and BestEffortPackageSpec. Ignoring the latter for the moment, each of these classes represents a collection of CustomerFacingServiceSpecs that work together to collectively offer a set of Services. This enables these Services to be compared as a group to (for example) a ServiceLevelAgreement. Conceptually, this can be represented as shown in Figure SO.26.

There are two important differences between Platinum, Gold, Silver, and Bronze Service. The first is obvious from Figure SO.26 – the better the Service Level, the more CustomerFacingServices that a user has access to. The second is more subtle, and involves the quality of CustomerFacingService. For example, all four ServicePackages above provide Web and Data access. The difference between these applications is that the quality of Web and Data access for PlatinumService is better from an end-user's point-of-view (i.e., faster downloads, less waiting, etc.) than that for GoldService, and so forth.

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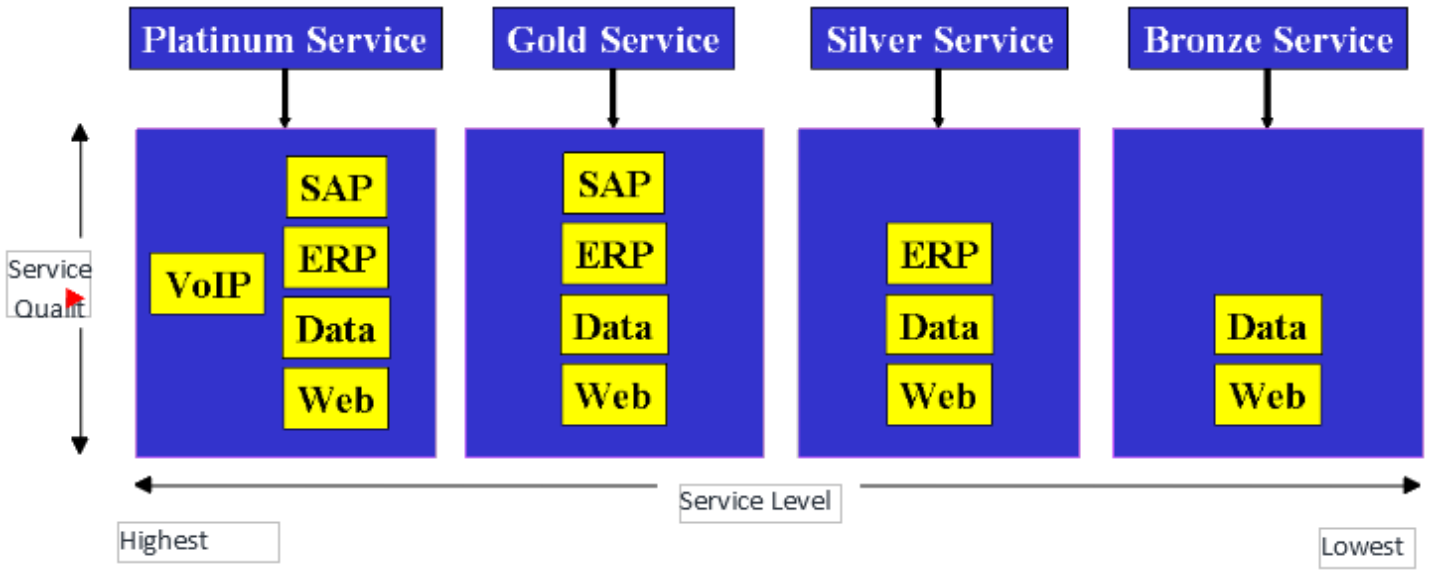

Figure SO.26 - Grouping Together CustomerFacingServiceSpecifications into a ServicePackage

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Figure SO.27 - ServicePackageSpecs and ServicePackages

The link between a ServicePackageSpec and a ServicePackage is shown in the Figure below.

A ServicePackage is derived from an associated ServicePackageSpec. The ServicePackageSpec defines the invariant attributes, methods, relationships, and constraints for all ServicePackage instances that are derived from it. This enables each individual ServicePackage to add its own application-specific changeable characteristics.

Note that there is no specific association used to relate a particular ServicePackage to the ServicePackageSpec that it is derived from. This is because the ServicePackageSpec and ServicePackage both inherit the SpecifiesService association.

Finally, while the composite pattern could be applied to ServicePackage, there is no perceived need to do so. Multiple ServicePackages will simply be related to a ProductBundle, and appear as separate ProductComponents.

Thus, we see that ServicePackage is a way to define the set of CustomerFacingServiceSpecifications that are to be used by a particular ProductSpecification.

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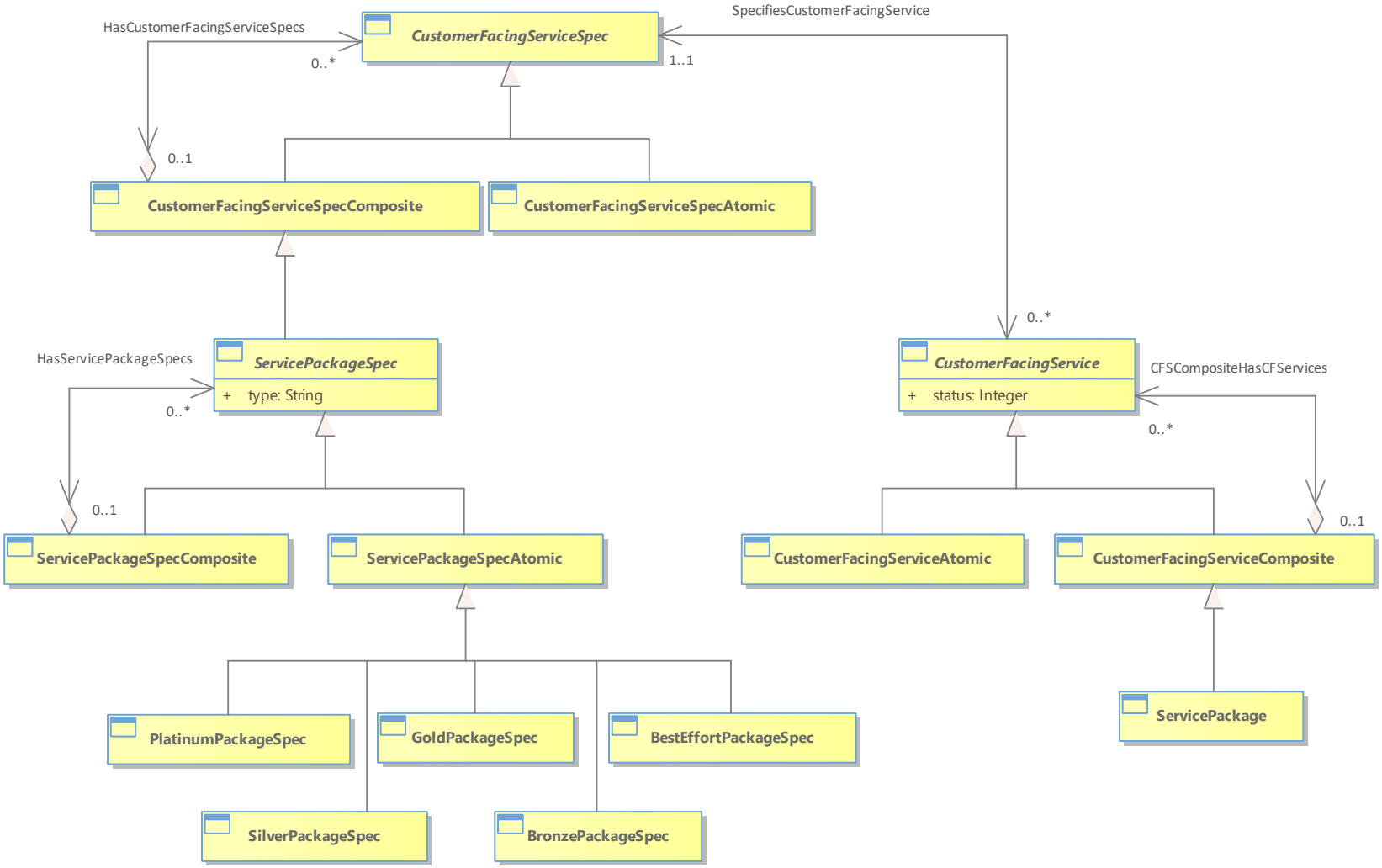

Figure SO.27 - ServicePackageSpecs and ServicePackages

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Figure SO.28 - Relationship Between ServiceBundleSpec and ServicePackageSpec

The next concept to introduce is that of ServiceBundles.

Just as ServicePackageSpecs and ServicePackages define the templates and instances of a set of CustomerFacingServices, ServiceBundleSpecs and ServiceBundles define the templates and instances of a set of ResourceFacingServices. Note the symmetry – this is once again indicative of an extensible, robust information model. The extensibility is, of course, in the use of the ServicePackageSpec and ServiceBundleSpec entities. Once the desired specifications have been defined, it is relatively easy to instantiate those templates (thereby producing ServicePackage and ServiceBundle entities). Furthermore, this enables different instances to be easily differentiated while ensuring that their important characteristics remain constant. This also greatly simplifies their definition and management.

The following Figure shows the relationship between a ServicePackageSpec and a ServiceBundleSpec.

A ServiceBundleSpecification is the base class for defining the different classes of bundled ResourceFacingServiceSpecifications that a Customer can obtain via a Product. The preferred way to represent a Customer subscription to a Product of this nature is by using the association class ServicePackageBundleDetails to define the set of specific ServiceBundleSpecifications that are used by a particular ServicePackage.

Conceptually, a ServiceBundle is thought of as a collection to enable the needs of different sets of ResourceFacingServices to be grouped together. The “bundle” conveys the concept of grouped Services that are related. Since these are ResoureFacingSpecifications, they define reusable templates for implementing the ResourceFacingServices that are required by a particular CustomerFacingService (as represented by a ServicePackage).

The following Figure shows four exemplary subclasses of ServiceBundle for defining four different groups of Class of Service (CoS). The idea is to use each CoSBundle class to represent the Class of Service settings for a specific ServiceBundle. This enables different CoS settings to be associated with the different CustomerFacingServices that are defined by a particular ServicePackage, which is what the Customer obtains

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via a Product. For example, CoS1 could be used for VoIP, CoS2 could be used for mission-critical applications, CoS3 could be used for all other applications that are not mission-critical but require better than best-effort delivery, and CoS4 could define best effort service delivery.

The combination of ServicePackage and ServiceBundle enables a Service Provider to abstract the notion of different classes of service that are sold to subscribers. For example, the troublesome use case of provisioning a Customer's request to upgrade to a higher-level Service (or downgrade to a lower-level Service) can now be much more easily and efficiently handled. More importantly, this enables a single link to be defined between the low-level QoS model (which controls the mechanisms used in the devices to condition traffic appropriately) and this higher-level, business-oriented model.

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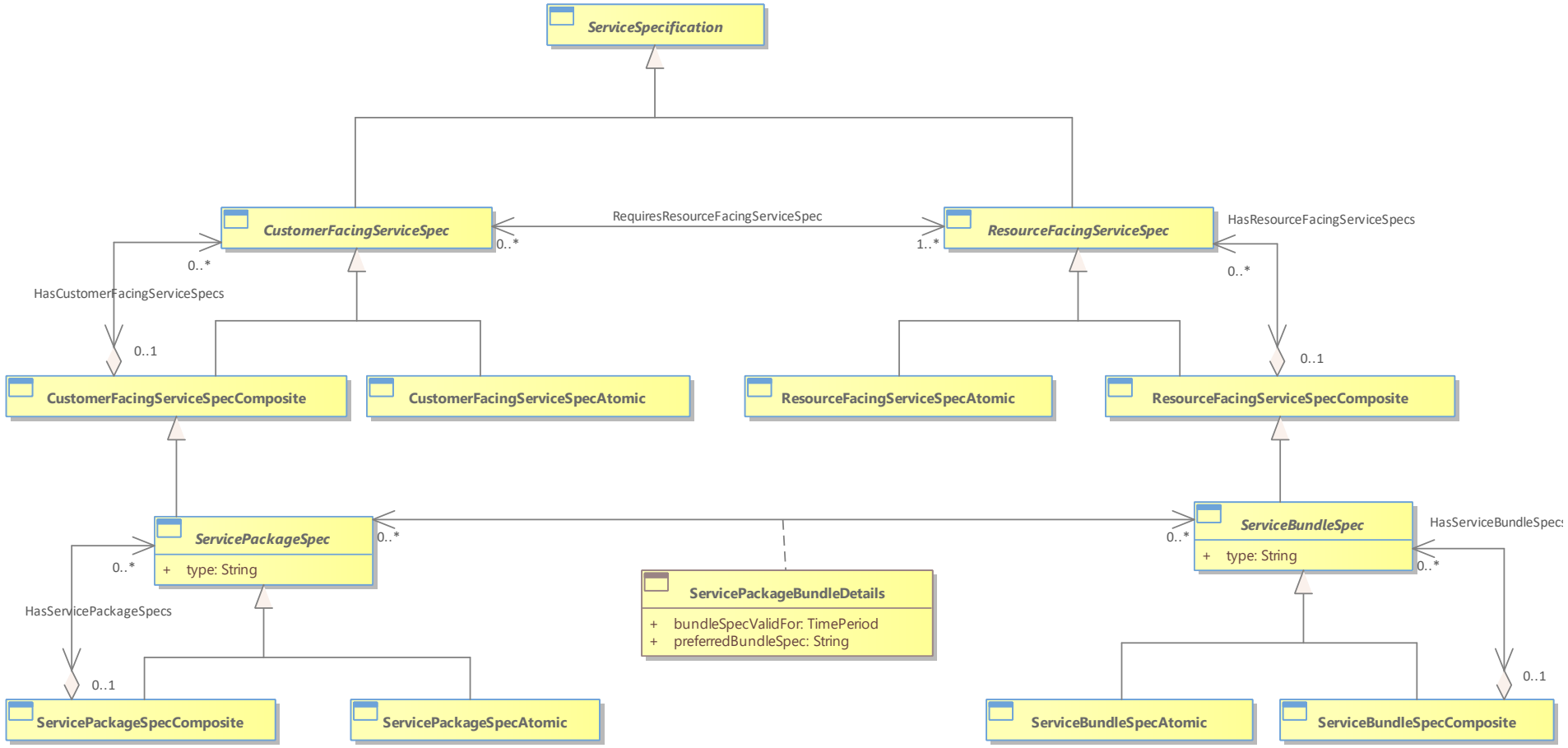

Figure SO.28 - Relationship Between ServiceBundleSpec and ServicePackageSpec

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Figure SO.29 - ServiceBundles and ServiceBundleSpecifications

Specific ResourceFacingServices are related to a ServiceBundleSpecification as shown in the Figure below.

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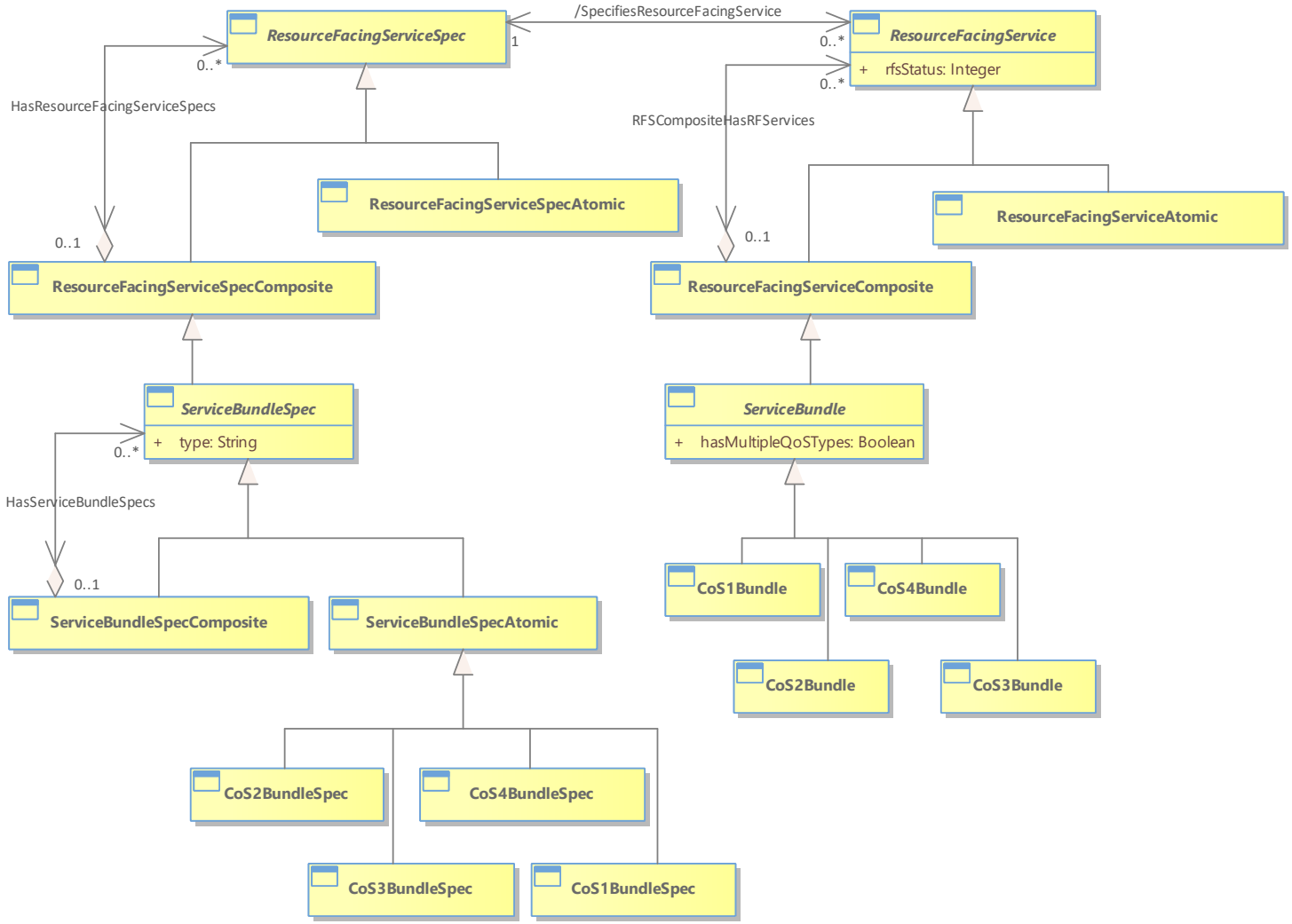

Figure SO.29 - ServiceBundles and ServiceBundleSpecifications

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Figure SO.30 - Summary Relationships Between ServicePackage and ServiceBundle

A summary view of the important (not all!) relationships between ServicePackageSpec, ServicePackage, ServiceBundleSpec, and ServiceBundle, and their parent classes (again, not all!) is shown in the Figure below. This figure was drawn in order to emphasize the symmetry that exists between ServiceSpecifications and Services.

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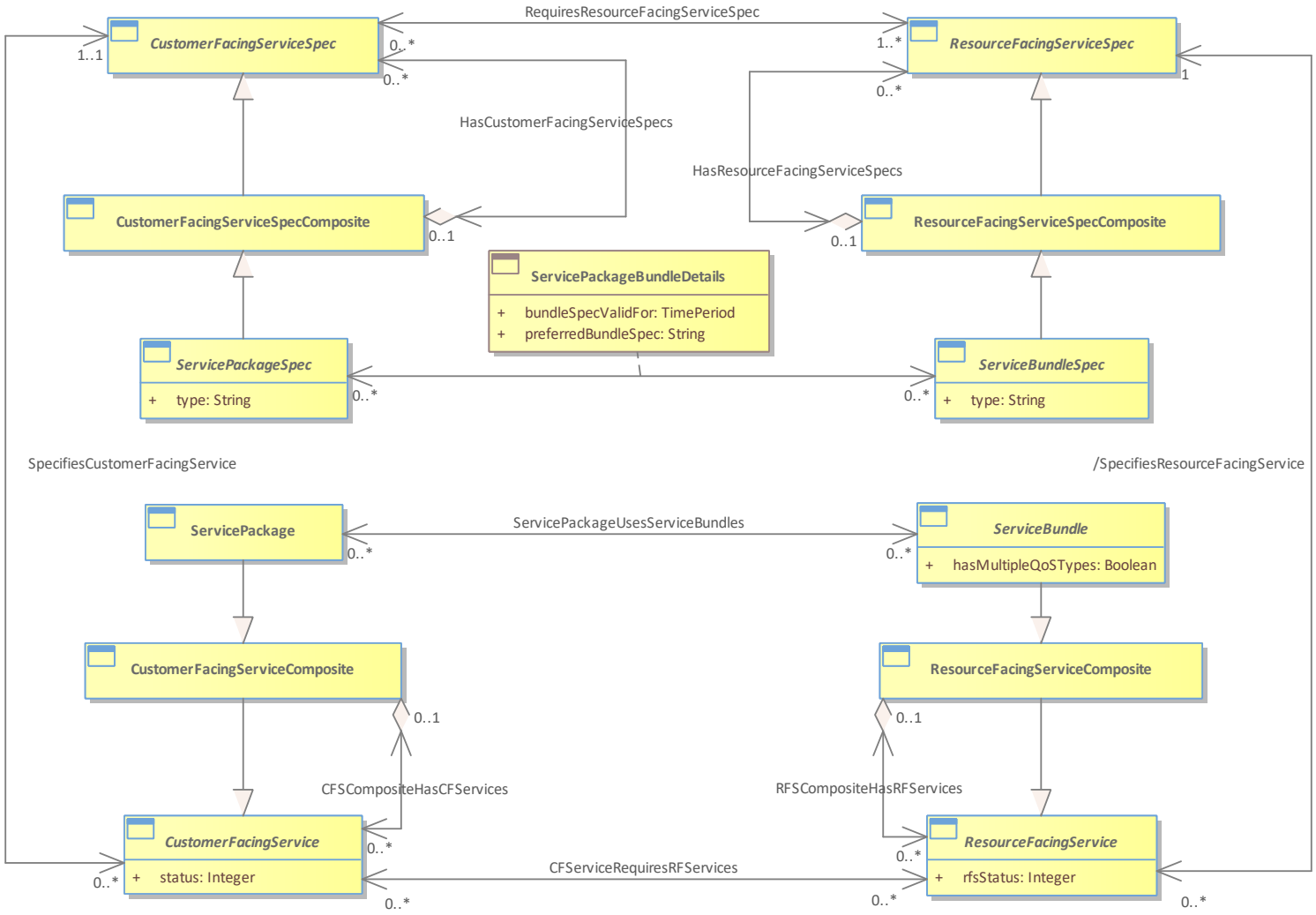

Figure SO.30 - Summary Relationships Between ServicePackage and ServiceBundle

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Figure SO.31 - Resource Facing Service Specification Pattern

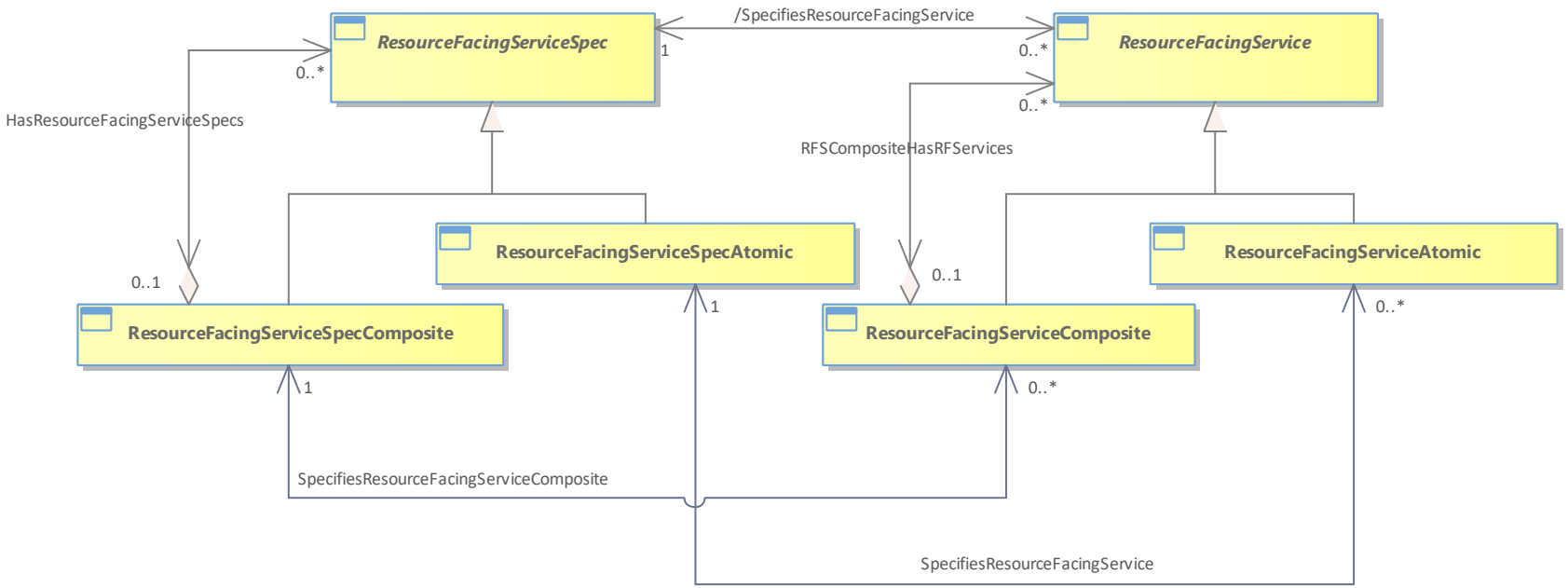

Figure SO.31 - Resource Facing Service Specification Pattern

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Figure SO.32 - Usage Volume Service Specification

NetworkServiceSpec is a type of CustomerFacingServiceSpec. It represents the common behavior and description of an installed network service that will be provisioned in the network to enables usages, like a Mobile Line Service. NetworkProductSpec is derived from NetworkServiceSpec.

NetworkServiceSpec generates 1 to many types of usage described by ServiceUsageSpecification.

UsageVolumeServiceSpec is a specialization of CustomerFacingServiceSpecAtomic. UsageVolumeServiceSpec is the CustomerFacingServiceSpec used for the creation of UsageVolumeProductSpec.

A UsageVolumeServiceSpec defines buckets of usage, such as hours of national calls, hours of international calls to a specific region, volume of data, as examples.

A UsageVolumeServiceSpec requires one or many NetworkServiceSpecs from which Usages will impact the buckets. It uses 1 to many types of usage described by ServiceUsageSpecification. ServiceUsageSpecificaton can impact 0 to many UsageVolumeServiceSpec.

UsageVolumeServiceSpec can be associated to many rules to Reserve, Debit or Credit an amount on a balance. These rules can also be associated to many different UsageVolumeServiceSpec. They inherit from PolicyRule.

UsageVolumeServiceBalanceReserveRule inherits from PolicyRule. It specifies the general rules to Reserve the UsageVolumeServiceBalance depending on ServiceUsage. It also describes to rules associated to quota. It allows the determination of quota to reserve on the UsageVolumeServiceBalance according to conditions.

UsageVolumeServiceBalanceDebitRule inherits from PolicyRule. It specifies rules to debit the UsageVolumeServiceBalance depending on ServiceUsage. It specifies the event triggering it (Ex: Call to a given destination exceeding a reserved quota, end of call...), potential conditions to be checked and the action to apply if the condition is true (Ex: debit quota or call duration on the UsageVolumeServiceBalance).

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UsageVolumeServiceBalanceCreditRule specifies rules to apply to credit the UsageVolumeServiceBalance (loyalty, bonus...). It inherits from PolicyRule for specifying the event triggering it (Ex: Sum of calls to a given destination over the period exceeding a threshold...), potential conditions to be checked (Ex: if the user bought a specific bundle during the period) and the action to apply if the condition is true (Ex: credit 30mn calls on the UsageVolumeServiceBalance).

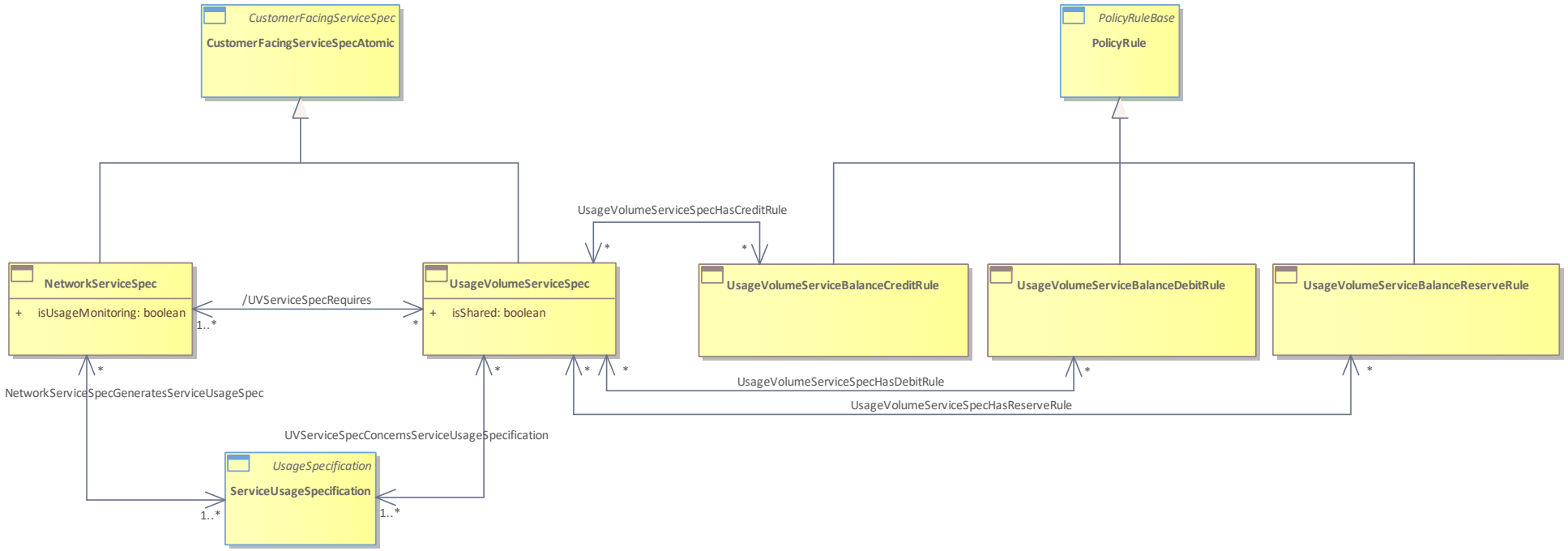

Figure SO.32 - Usage Volume Service Specification

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Figure SO.33 NetworkProductSpec / NetworkServiceSpec relationships

This figure shows the relationship between entities at Product and Service levels.

A UsageVolumeProductSpec is the restriction of one and only one UsageVolumeServiceSpec.

A NetworkProductSpec is a restriction of one and only one NetworkServiceSpec.

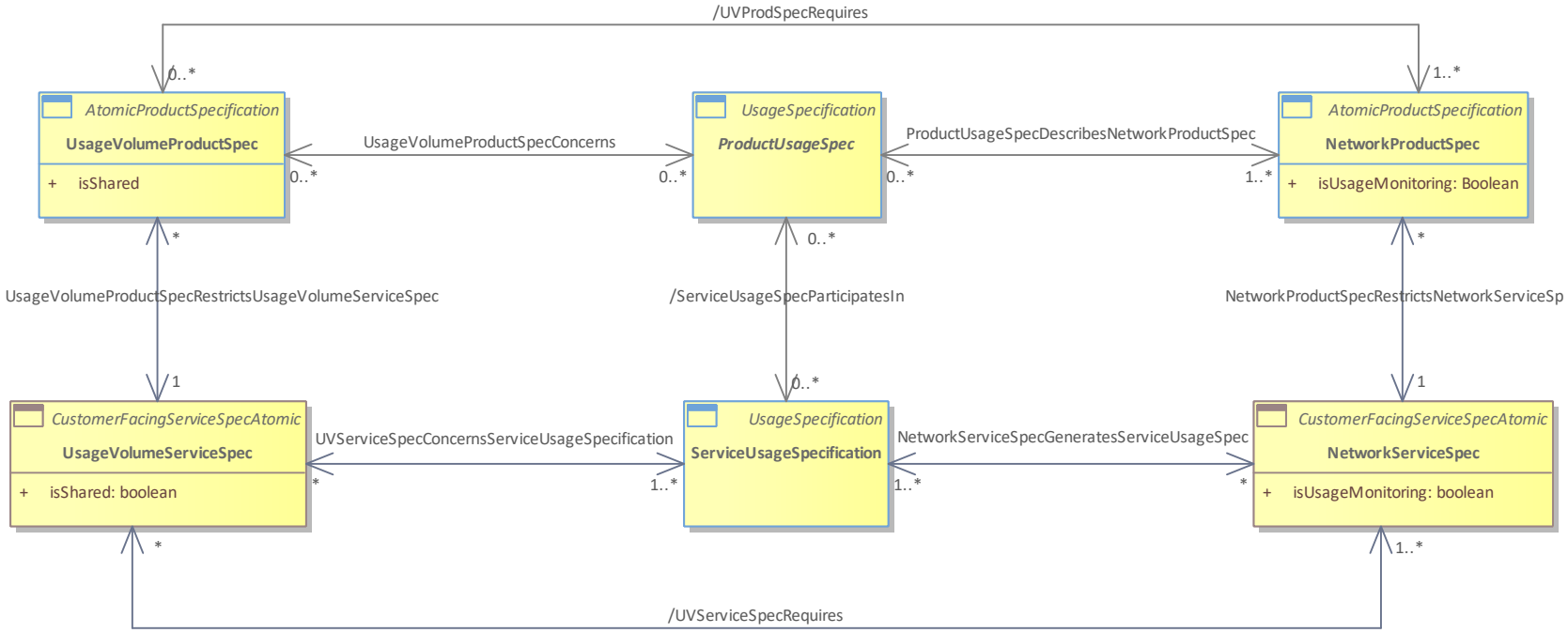

Figure SO.33 NetworkProductSpec / NetworkServiceSpec relationships

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4.3.1 Customer Facing Service Spec ABE

The Customer Facing Service Spec ABE defines the common characteristics for a set of CustomerFacingServices. This scales by enabling multiple CustomerFacingServiceSpecs to be defined to model the common characteristics of different groups of CustomerFacingServices.

Figure CFSS.01 - Customer Facing Service Spec ABE Related Entities

Following are the business entities aggregated under the **Customer Facing Service Spec** ABE.

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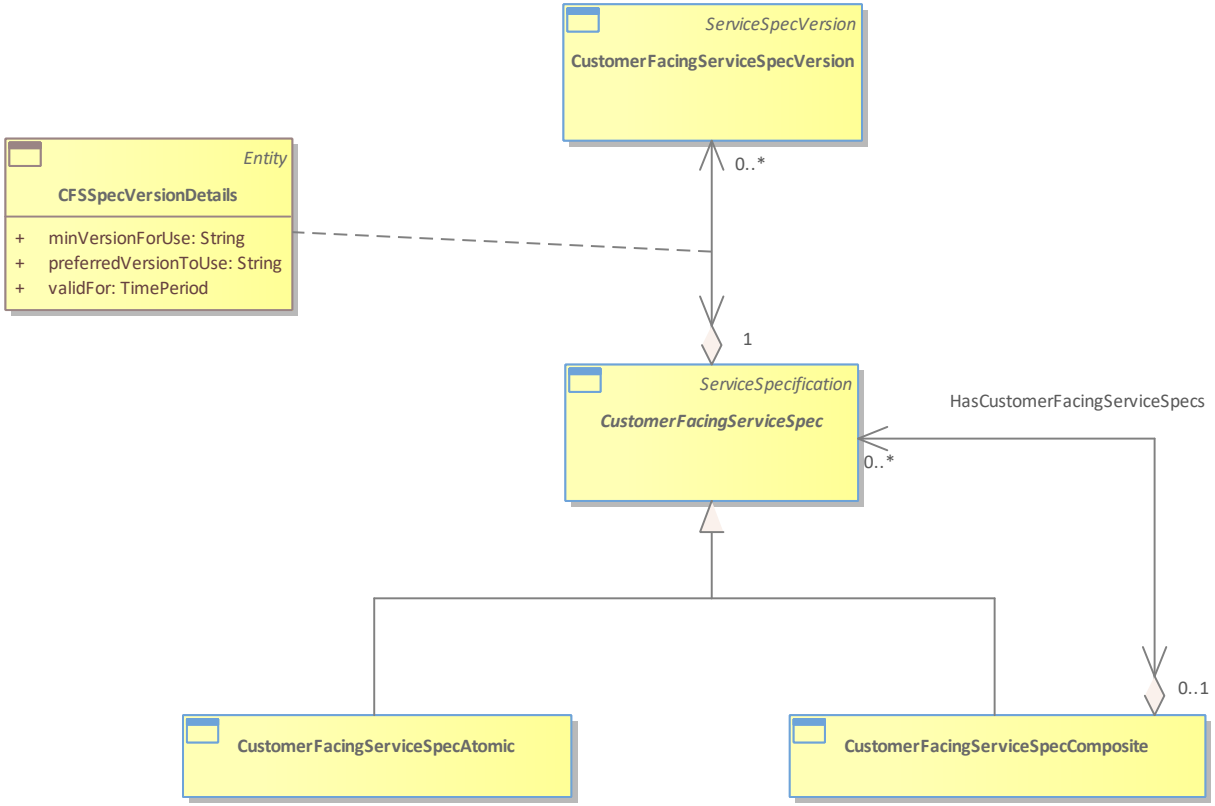

Figure CFSS.01 - Customer Facing Service Spec ABE Related Entities

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4.3.1.1 Customer Facing Service Spec Examples ABE

The Customer Facing Service Spec Examples ABE contains example subclasses of CustomerFacingServiceSpecs.

Figure CFSSE.01 - Customer Facing Service Spec Examples ABE Related Entities

Following are the business entities aggregated under the **Customer Facing Service Spec Examples** ABE.

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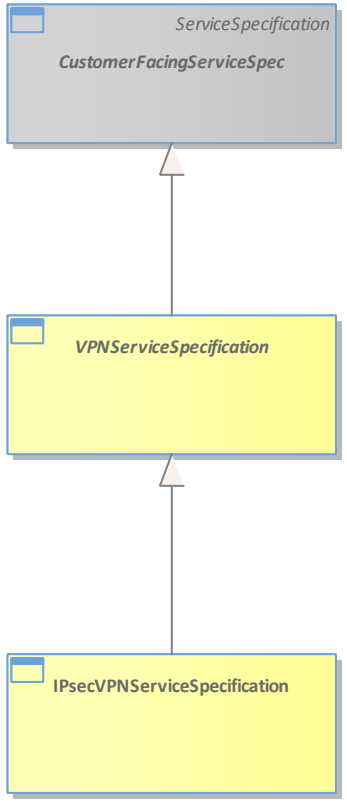

Figure CFSSE.01 - Customer Facing Service Spec Examples ABE Related Entities

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4.3.1.2 Customer Facing Service Spec Role ABE

The Customer Facing Service Spec Role ABE defines a ServiceSpecification, in terms of a set of ServiceSpecificationRoles, for a CustomerFacingService. It is used to represent the shared characteristics of a CustomerFacingService. This enables the CustomerFacingService to be managed abstractly using ServiceSpecificationRoles. It also helps define a CustomerFacingServiceSpecification in terms of the functions that it has or provides.

Figure CFSSR.01 Customer Facing Service Spec Role ABE Related Entities

Following are the business entities aggregated under the **Customer Facing Service Spec Role** ABE**.**

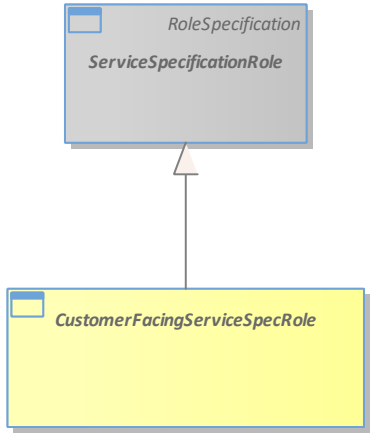

Figure CFSSR.01 Customer Facing Service Spec Role ABE Related Entities

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4.3.1.3 Network Service Spec ABE «Preliminary»

The Network Service Spec ABE contains the NetworkServiceSpec description and relationships.

Figure NSS.01 - Network Service Spec ABE Related Entities

Following are the business entities aggregated under the Network Service Spec Aggregate Business Entity.

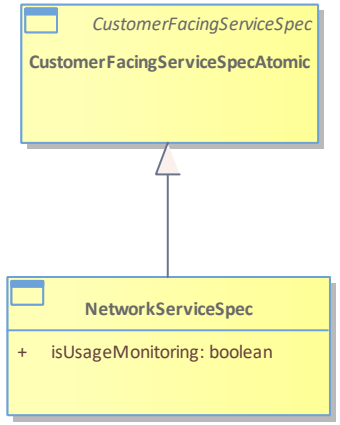

Figure NSS.01 - Network Service Spec ABE Related Entities

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4.3.1.4 Service Package ABE

The Service Package ABE defines the concept of bundling a set of different CustomerFacingServiceSpecs to meet the functionality specified by one or more ProductSpecifications. This enables the shared characteristics of these CustomerFacingServices to be specified, so that multiple Products can be built from their associated ProductSpecification.

Figure SP.01 - Service Package ABE Related Entities

Following are the business entities aggregated under the **Customer Facing Service Spec Examples** ABE.

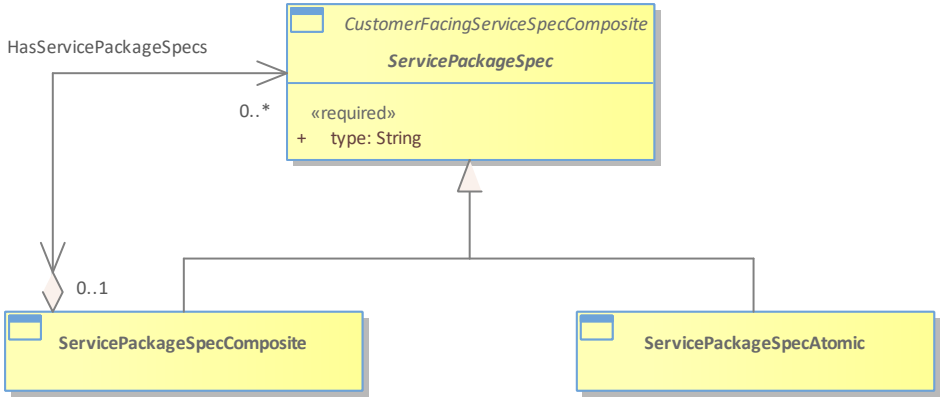

Figure SP.01 - Service Package ABE Related Entities

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4.3.1.4.1 Service Package Spec Examples ABE

The Service Package Spec Examples ABE contains example subclasses of ServicePackageSpecs.

Figure SPSE.01 - Service Package Spec Examples ABE Related Entities

Following are the business entities aggregated under the **Service Package Spec Examples** ABE.

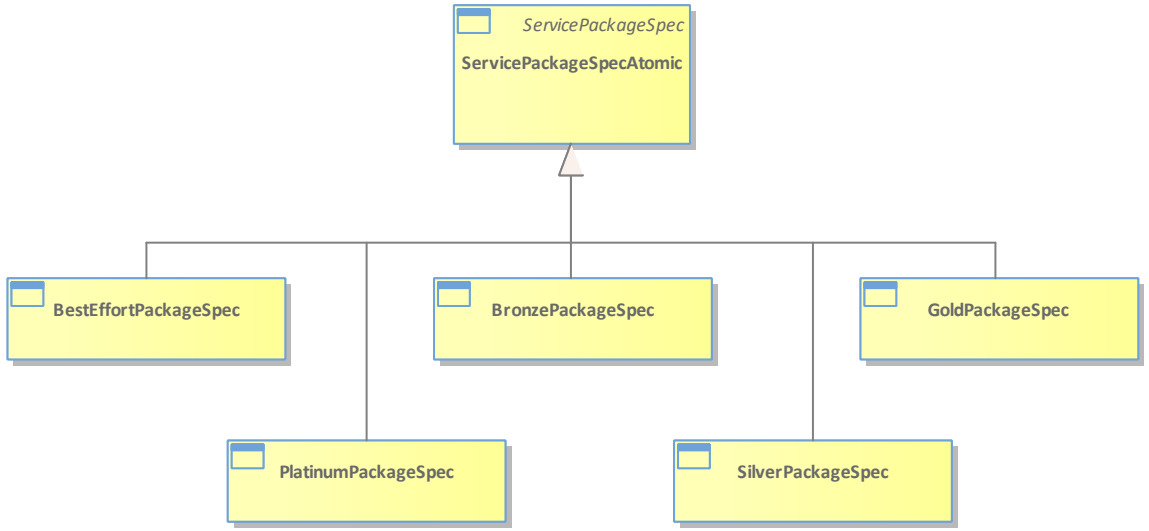

Figure SPSE.01 - Service Package Spec Examples ABE Related Entities

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4.3.1.5 Usage Volume Service Spec ABE «Preliminary»

The Usage Volume Service Spec ABE contains the UsageVolumeServiceSpec and all the entities specific to this type of service.

Figure UVSS.01 - Usage Volume Service Spec ABE Related Entities

Following are the business entities aggregated under the Usage Volume Service Spec Aggregate Business Entity.

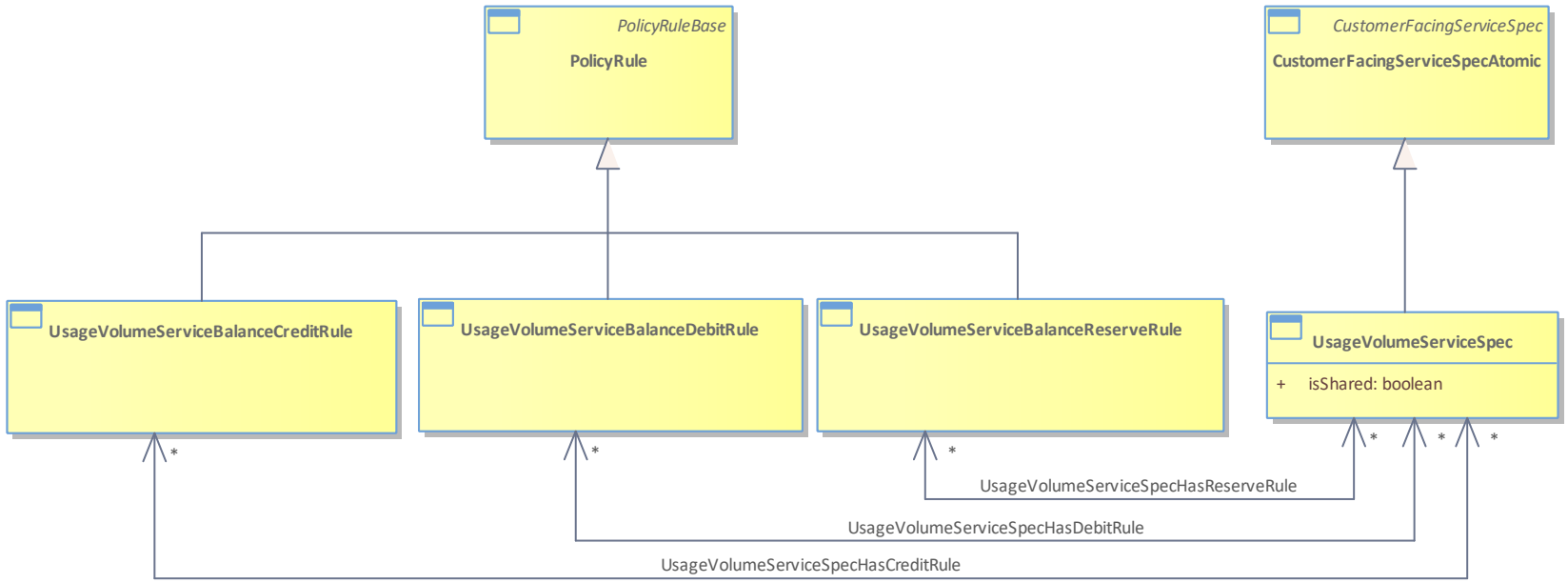

Figure UVSS.01 - Usage Volume Service Spec ABE Related Entities

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4.3.2 Resource Facing Service Spec ABE

The Resource Facing Service Spec ABE defines the common characteristics for a set of ResourceFacingServices. This scales by enabling multiple ResourceFacingServiceSpecs to be defined to model the common characteristics of different groups of ResourceFacingServices.

Resource Facing Service Spec ABE Related Entities

Following are the business entities aggregated under the **Resource Facing Service Spec** ABE.

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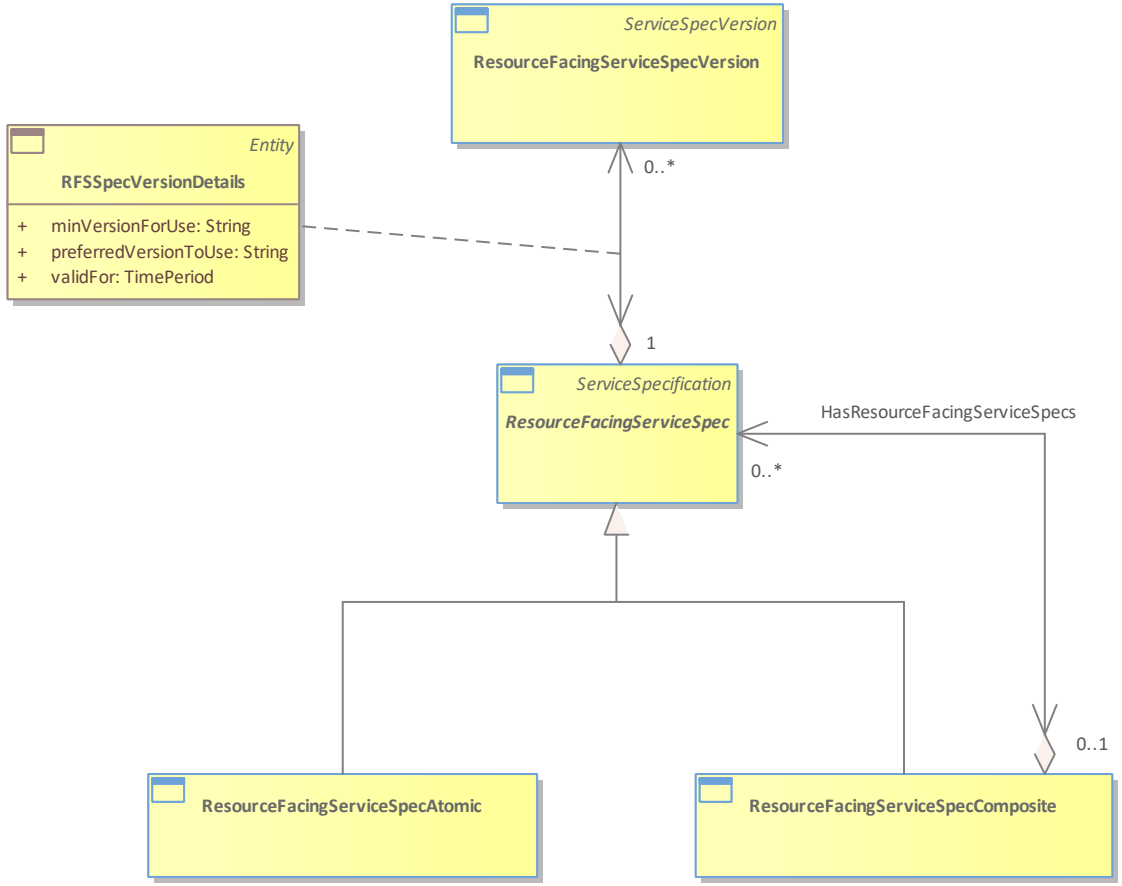

Resource Facing Service Spec ABE Related Entities

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4.3.2.1 Resource Facing Service Spec Role ABE

The Resource Facing Service Spec Role ABE defines a ServiceSpecification, in terms of a set of ServiceSpecificationRoles, for a ResourceFacingService. It is used to represent the shared characteristics of a ResourceFacingService. This enables the ResourceFacingService to be managed abstractly using ServiceSpecificationRoles. It also helps define a ResourceFacingServiceSpecification in terms of the functions that it has or provides.

Figure RFSSR.01 - Resource Facing Service Spec Role ABE Related Entities

Following are the business entities aggregated under the **Resource Facing Service Spec Role** ABE.

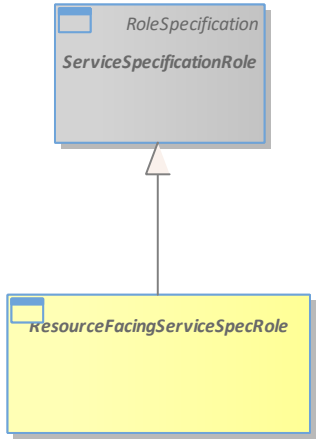

Figure RFSSR.01 - Resource Facing Service Spec Role ABE Related Entities

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4.3.2.2 Service Bundle ABE

The Service Bundle ABE is a collection of ResourceFacingServiceSpecifications. This enables the needs of different sets of ResourceFacingServiceSpecifications to be grouped together, hence the name “bundle”. Since these are ResoureFacingSpecifications, they define reusable templates for implementing the ResourceFacingServices that are required by a particular CustomerFacingService (as represented by a ServicePackage).

Figure SB.01 - Service Bundle ABE Related Entities

Following are the business entities aggregated under the **Service Bundle Examples** ABE.

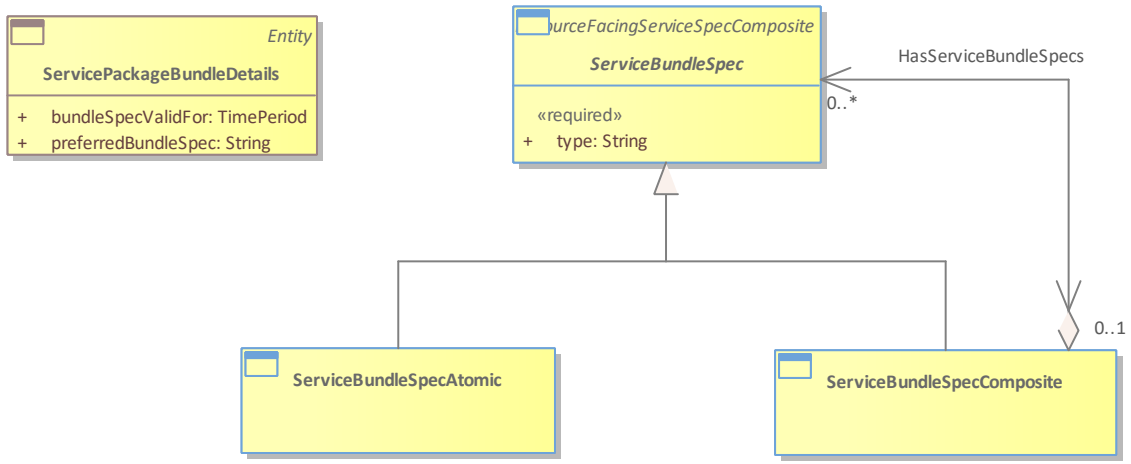

Figure SB.01 - Service Bundle ABE Related Entities

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4.3.2.2.1 Service Bundle Examples ABE

The Service Bundle Examples ABE contains example subclasses of ServiceBundleSpec.

Figure SBE.01 - Service Bundle Examples ABE Related Entities

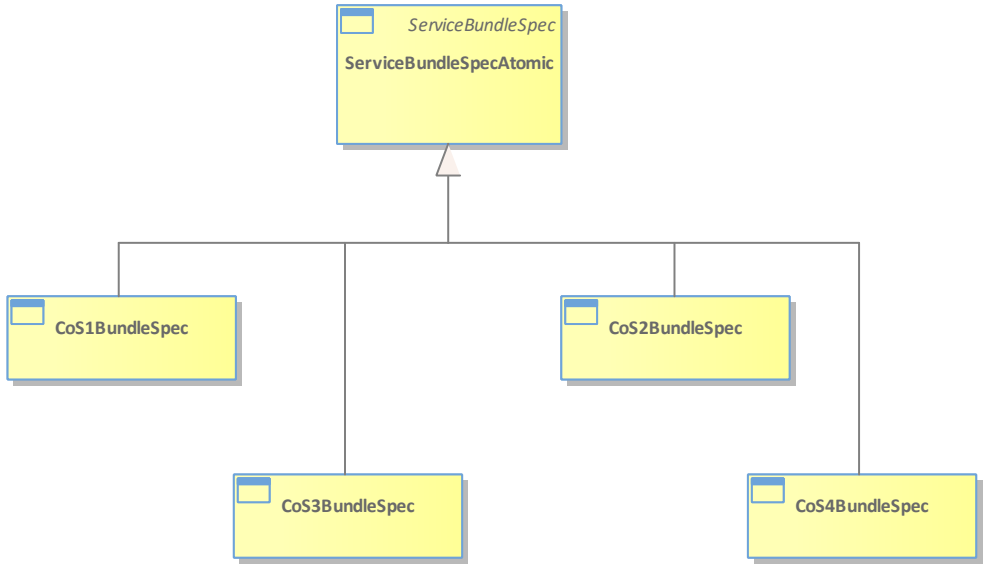

Figure SBE.01 - Service Bundle Examples ABE Related Entities

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4.3.3 Service Catalog ABE

A collection of entities that represent the specification of ServiceCatalogs and catalogs described by them that contain ServiceSpecifications.

Figure SC.01 - Service Catalog ABE Related Entities

Following are the business entities aggregated under the **Service Catalog** ABE.

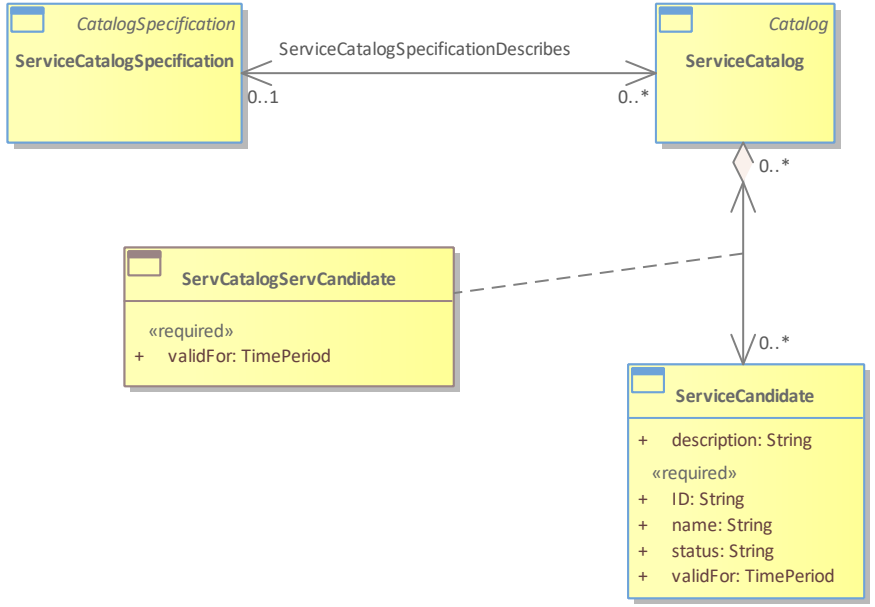

Figure SC.01 - Service Catalog ABE Related Entities

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4.4 Service ABE

A Service represents an instance of ServiceSpecification. The **Service ABE** contains entities that are used to represent both types of Services: CSP Know-Hows instances (a.k.a. Customer Facing Services or CFS) and Technical solution (a.k.a. Resource Facing Services or RFS).

The Service carries the place where the Service is in use if relevant, as well as configuration characteristics, such as assigned telephone numbers and internet addresses. The Service also tracks the links to Product realized and/or Resources used.

Figure SO.06 - Basic Service Model – A Starting Point

A ComprisedOf recursive association on Service is tempting to add, because it defines the ability to aggregate subordinate Services into a higher-level Service. However, this aggregation is a bad idea, as it enables ResourceFacingServices to aggregate CustomerFacingServices, which destroys the point of separating them in the first place. Hence, it will not be defined in this model.

Specifically, a CustomerFacingService is what is bound to a Product, not a Service. ResourceFacingService is not linked directly to Product; rather, it is linked to Resource.

The Figure shows that zero or more CustomerFacingServices are optionally used to realize a Product. The cardinality is zero or more in order to allow for the development of Services in a Service Provider or Enterprise before they are offered as a Product or ProductOffering.

The CFServiceRequiresRFServices association defines the set of ResourceFacingServices that are required by a particular CustomerFacingService in order for that CustomerFacingService to operate correctly.

A ResourceFacingService is related to LogicalResources and PhysicalResources through two associations classes, ServiceLRDependency and ServicePRDependency. These association classes define the set of LogicalResources and PhysicalResources, respectively, which are required for this ResourceFacingService to function correctly. The ServiceLRDependency is used to define the set of LogicalResources, such as memory and DeviceInterfaces, which are required for this particular ResourceFacingService to function correctly. The ServicePRDependency have a slightly different connotation. This defines the set of PhysicalResources, such as PhysicalDevices and

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PhysicalPorts, which are required for this particular ResourceFacingService to function correctly. Thus, the former association class defines the operational aspects of the ResourceFacingService, while the latter defines its hosting aspects. This separation is, of course, facilitated by the design of the Service (CustomerFacingServices vs. ResourceFacingServices) and Resource (PhysicalResource vs. LogicalResource) models.

Note that a Product has two subclasses, ProductComponent and ProductBundle. A ProductComponent represents a stand-alone Product, while a ProductBundle represents Products formed from other Products. Clearly, either of these can serve as the instance that is being realized through a combination of CustomerFacingServices and Resources.

The model shown was built to distinguish between two types of Services – those that are used to realize a Product, and those that are required to implement the functionality provided by a Product. These are called CustomerFacingServices and ResourceFacingServices, respectively.

In addition, note the strong link between infrastructure entities and business entities, such as a Product. In particular, this model ensures that Customers access Resources and Services through their packaging in a Product. This particular modeling of the complex abstractions and relationships between Customer, Product, Service and Resource provides some of the key underpinnings that support what the SID is all about.

The important point of this example is that the VPN Product requires a set of Services. Specifically, our VPN Product requires a combination of the BGP ResourceFacingService (and others, but let's keep it simple for the time being) and the VPN CustomerFacingService. The utility of this Figure is that it provides the ability to "package" multiple heterogeneous Services together to realize a Product. This models a CustomerFacingService requiring a set of ResourceFacingServices through the CFServiceRequiresRFServices association.

Continuing the example, the VPN Product that the Customer bought may already have a router that the Customer can use to access the VPN, or the Customer may need to purchase one. This is represented using the ProductHasPhysicalResources association.

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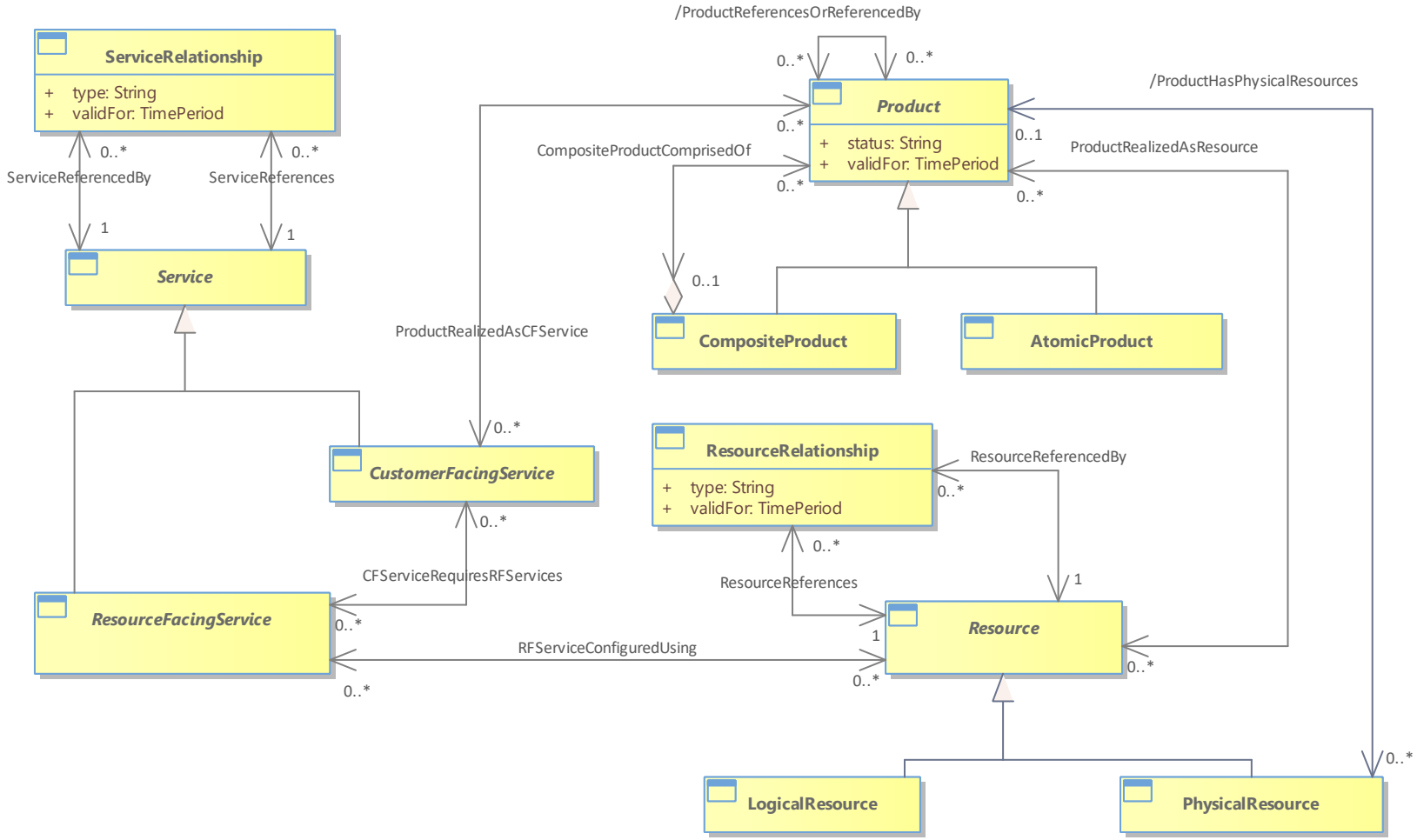

Figure SO.06 - Basic Service Model – A Starting Point

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Figure SO.07 - Representing BGP as a LogicalResource

Let's consider the simple example of a Customer purchasing a VPN. The VPN itself may be both a Product and a CustomerFacingService, since the VPN provides a number of functions (e.g., connectivity) that people commonly refer to as Services. Suppose the VPN is a particular type of VPN, defined by RFC2547bis. This specification mandates that BGP (the Border Gateway Protocol, which is an inter-domain routing protocol that is used to exchange reachability information with other BGP systems) is used for route advertisement. The Customer doesn't buy BGP services, yet BGP is required for this type of VPN to function. Therefore, BGP is a ResourceFacingService.

The following Figure takes an extract from the LogicalResource model and shows that BGP is a particular type of RoutingProtocol. This model defines several different types of Protocols. A RoutingProtocol is a particular type of protocol that accomplishes routing through the implementation of a specific algorithm. Protocols are modeled as a type of LogicalResource.

Note: Routing is defined as the process of finding a path to a destination.

To connect this properly, we also need the concept of a LogicalDevice. This is the logical analog to the PhysicalDevice class. A LogicalDevice is an abstract base class for all logical resources that are inherently manageable and make up a Product. It is used to describe different types of logical features and services that constitute a Product. It has two main purposes: (1) to collect common attributes and relationships for all logical resources, and (2) to provide a convenient, single point where relationships with other managed objects can be defined.)

Thus, BGP is associated with a ResourceFacingService through the RFServiceUsesProtocol aggregation.

The ProtocolServiceDetails association class defines the Protocols that are required by a particular ResourceFacingService.

Thus, BGP is associated with a ResourceFacingService through the ProtocolServiceDetails association class. The ProtocolServiceDetails association class defines the Protocols that are required by a particular ResourceFacingService.

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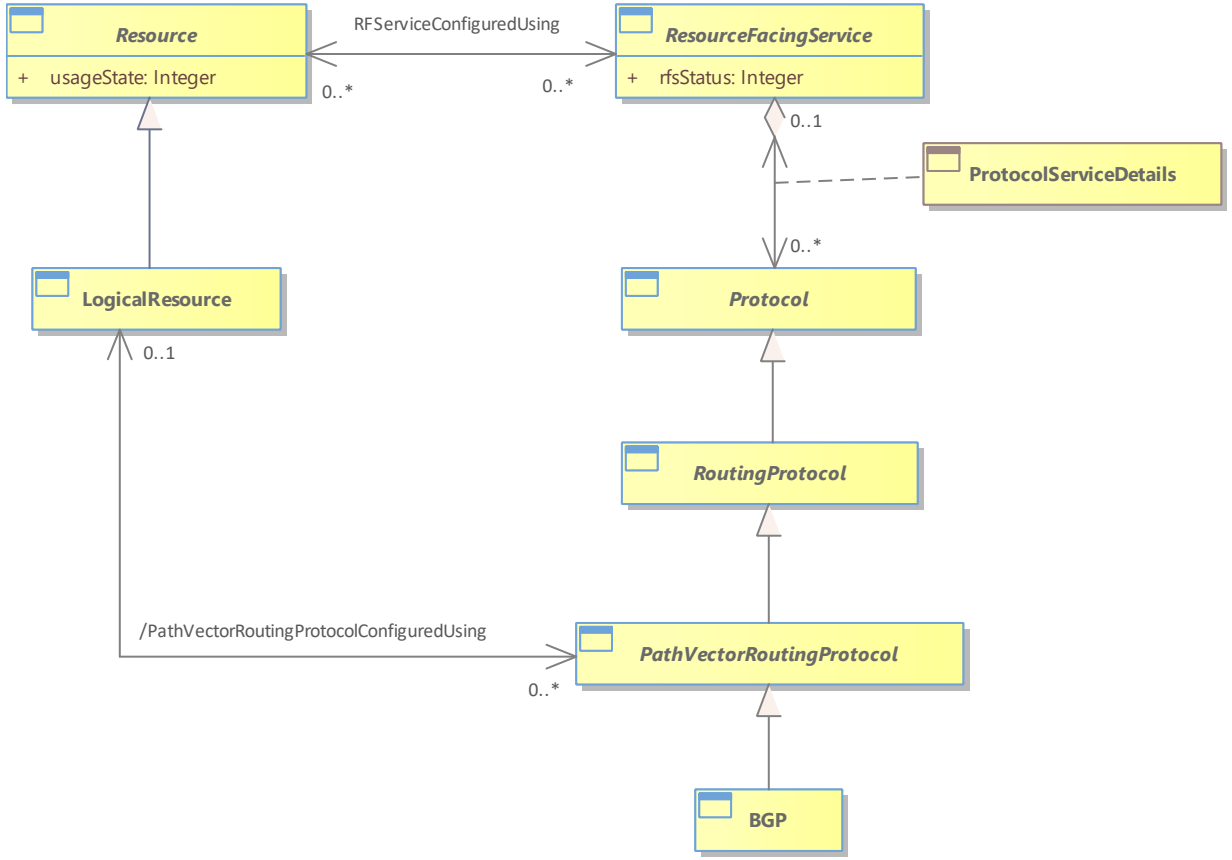

Figure SO.07 - Representing BGP as a LogicalResource

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Figure SO.12 - Modeling Stand-Alone and Aggregate Services

Similarly as specification part, CustomerFacingService and ResourceFacingService both use the same composite pattern. The difference is that since Services are in reality instances defined by a ServiceSpecification, the atomic and composite services are melded into CustomerFacingServices and ResourceFacingServices. In other words, generic Services are CustomerFacingServices and ResourceFacingServices themselves. While other types of Services could certainly be defined, their associated ServiceSpecification needs to be defined first. The following Figure shows the composite pattern applied to CustomerFacingServices and ResourceFacingServices.

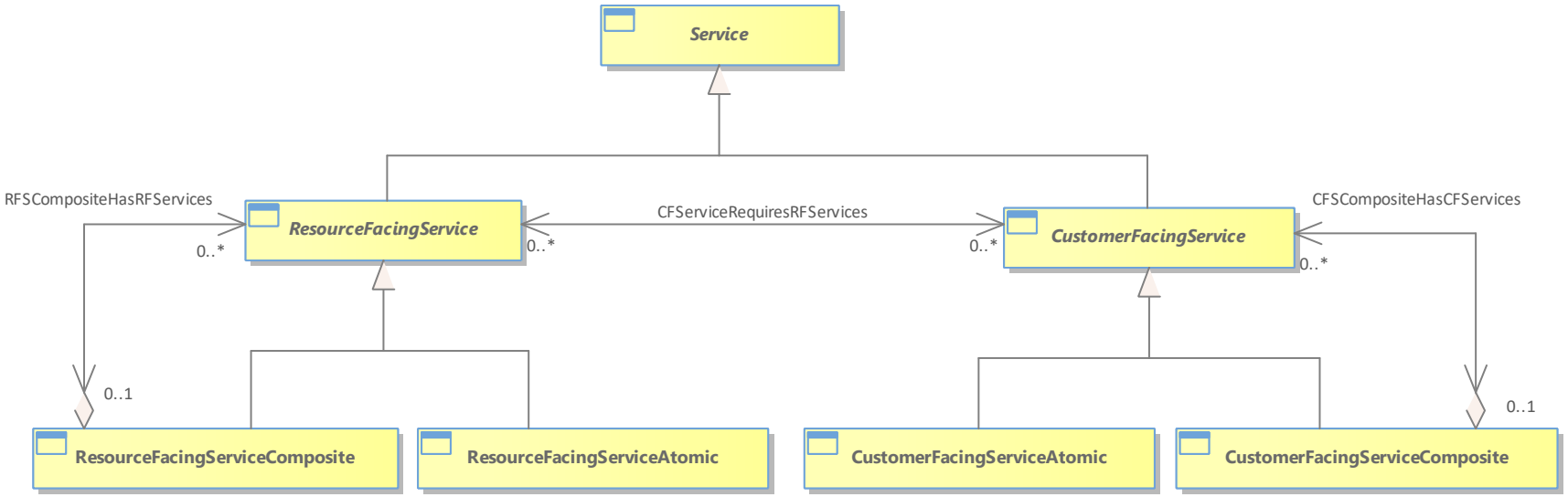

Figure SO.12 - Modeling Stand-Alone and Aggregate Services

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Figure SO.20 - The Concept of ServiceRoles

The ServiceRole class defines a Service in terms of a set of roles. The roles are then associated for both CustomerFacing and ResourceFacing services through the CustomerFacingServiceRole and ResourceFacingServiceRole subclasses. The utility of the ServiceRole class is to present a single point for accumulating common relationships that its subclasses can inherit.

The CFSRoleRequiresRFSRoles aggregation is used to define the set of ResourceFacingServiceRoles that are required by a particular CustomerFacingServiceRole. Conceptually, this enables roles to be used to define the common characteristics of a set of ResourceFacingServices that are required by a particular CustomerFacingService.

Note that the cardinality of the CFSRoleRequiresRFSRoles is * on the CustomerFacingServiceRole (aggregate) side and 1..* on the ResourceFacingServiceRole (component) side. This is because a ResourceFacingServiceRole can exist without being bound into a CustomerFacingServiceRole (e.g., in testing the network), but a CustomerFacingServiceRole requires at least one ResourceFacingServiceRole to function. Note also that this cardinality is symmetric with the cardinality defined in the CFServiceRequiresRFServices association.

The InvolvedServiceRoles association is used to define a set of ServiceRoles that are involved with, or related to, each other in order to build a particular type of Service. This association applies equally to CustomerFacingServices and ResourceFacingServices. For example, consider the definition of a set of “QoS packages” that each consist of a set of different Classes of Service. Each Class of Service definition is associated with different types of traffic, and defines how each type of traffic should be conditioned independently of other types of traffic. The higher-level QoS package is clearly a CustomerFacingService, since it is obtained via a Product by a Customer. Equally clearly, the lower-level Class of Service definitions are each ResourceFacingServices, since the Customer has obtained a higher-level package and doesn’t have explicit knowledge of these sub-services. Now, assume that there is a dependency that exists between a sub-service and either another sub-service or the QoS package itself, such as a limit as to how many instances of one type of Service can be created, or that each sub-service must first be instantiated before a QoS package is completed. These are examples of the InvolvedServiceRoles association.

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Continuing the example, the CustomerFacingServiceRole (e.g., “Gold Service”) is built, or constructed from, a set of ResourceFacingServiceRoles (e.g., “Rate Limiting Service” and “Remarking Service”). The former ensures that the Customer sends only the amount of traffic that he or she is contractually allowed to send, and the latter prioritizes different types of traffic. The Gold Service simply won’t work unless these two Services are built and active, but the Customer doesn’t need to explicitly know about them. These and other QoS functions are discussed in the QoS Service guide book.

Conceptually, ServiceRoles represent the variable characteristics of a Service from the viewpoint of functions (i.e., roles) that the Service provides to the rest of the environment. One example is that VPNs need to control how routing information is exchanged between the CE and PE routers at the edges of the Service Provider’s backbone and between the PE routers across the Service Provider’s backbone. ServiceRoles can be defined that represent how routing is constrained, depending if the router is a CE, PE, or P router.

This enables the Service to be managed abstractly using ServiceRoles. It also helps define the Service in terms of the functions that it has or provides. This is important, as it makes the management independent of the particular type of VPN and, more importantly, type of device and protocol used.

The CustomerFacingServiceRole and ResourceFacingServiceRole subclasses are used to represent the variable characteristics of CustomerFacingServices and ResourceFacingServices, respectively, in terms of roles that these services play. The set of roles define the Service in terms of the functions that it performs. This enables it to be modeled independently of the protocols, devices, and other dependent entities, as well as the environment in general.

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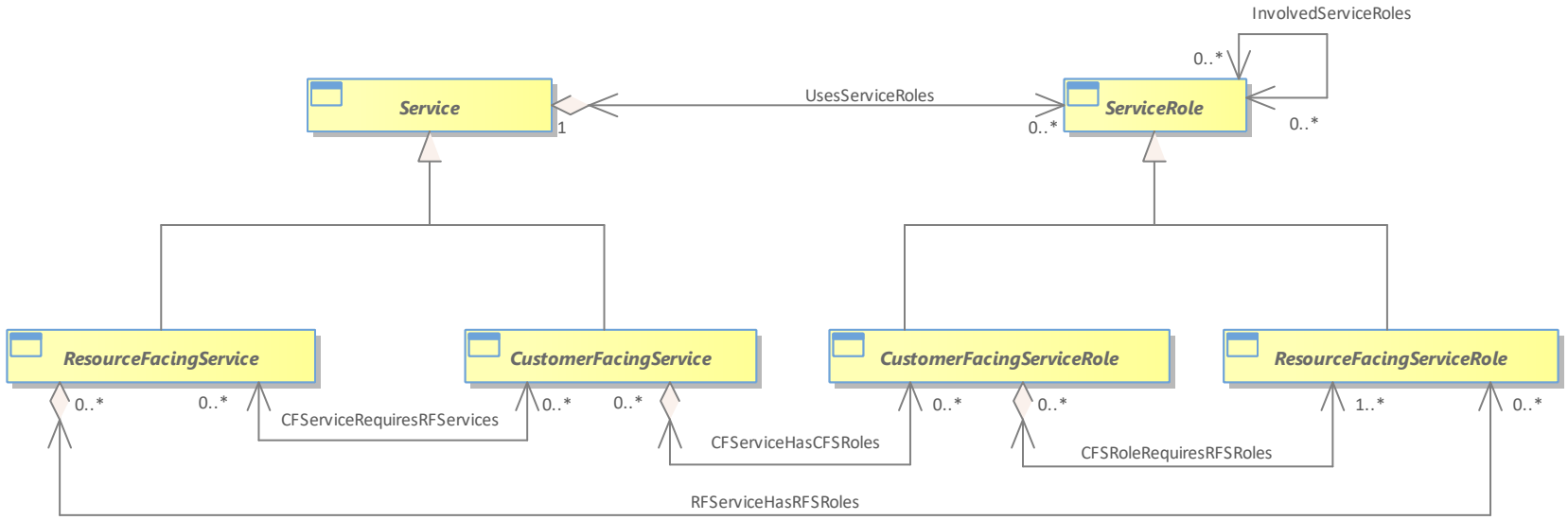

Figure SO.20 - The Concept of ServiceRoles

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Figure SO.35 - Interaction Between PartyRole and Service

A PartyRole defines the function that a Party takes on. PartyRoles can represent the ability to manage, configure, use, and perform other types of interactions with Services. A simple starting model is shown in the Figure below.

One can see that the Service and Resource management methods are symmetrical. Therefore, this section will only discuss the Service ownership and administration classes.

The OwnsServiceDetails association class defines a Service (CustomerFacing or ResourceFacing) that a particular PartyRole owns. Note that this is a very different relationship than the AdministersServiceDetails association class. The OwnsServiceDetails association class defines the PartyRole that is the owner of the Service. Here, “owner” is the person or group that is responsible (e.g., organizational, financial, and so forth) for the Service. Note that the cardinality of the OwnsServiceDetails association class on the PartyRole side is 1 – this means that a Service MUST have at least one PartyRole to manage it. Furthermore, the cardinality of this association on the Service side is *, which ensures that a PartyRole doesn’t have to own a Service.

The AdministersServiceDetails association class defines a Service (CustomerFacing or ResourceFacing) that a particular Party, playing a PartyRole, administers. From a business perspective, the Owner has to either appoint or give permission to the Administrator to administer the Service that is owned. This is done using the GrantsServiceAdminRights association class.

The GrantsServiceAdminRights association defines which PartyRoles are allowed to administer a particular owned Service. Different PartyRoles will in general have different administration rights for the same Service. For example, different types of Administrators will in general each have the ability to make more changes than different types of Technicians.

Policies can be defined for controlling the specific rights of administering a Service. This is the function of the ServiceManagementPolicy class. Only a simple single attribute is shown.

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The ServiceManagementPolicy class defines the particular policies that are used to define how different aspects of the Service are managed and maintained. For example, assume that a Service is made up of several sub-Services. One of these sub-Services might require multiple PartyRoles to approve changes to its configuration, whereas another sub-Service might not require any approvals to change its configuration. Since Policy also uses the composite pattern, if a set of Policies are needed, these can be aggregated into a single Policy).

Note the power of this pattern – the ManagementPolicyForService association, along with its ServiceManagementPolicy class, provide a generic template for defining which set of PartyRoles can perform which type of management function for which type of Service. Figure SO.35 shows that Services and Resources can be managed using the same pattern, even though the contents of the classes will be different.

There is a lot more to the model that may be developed further.

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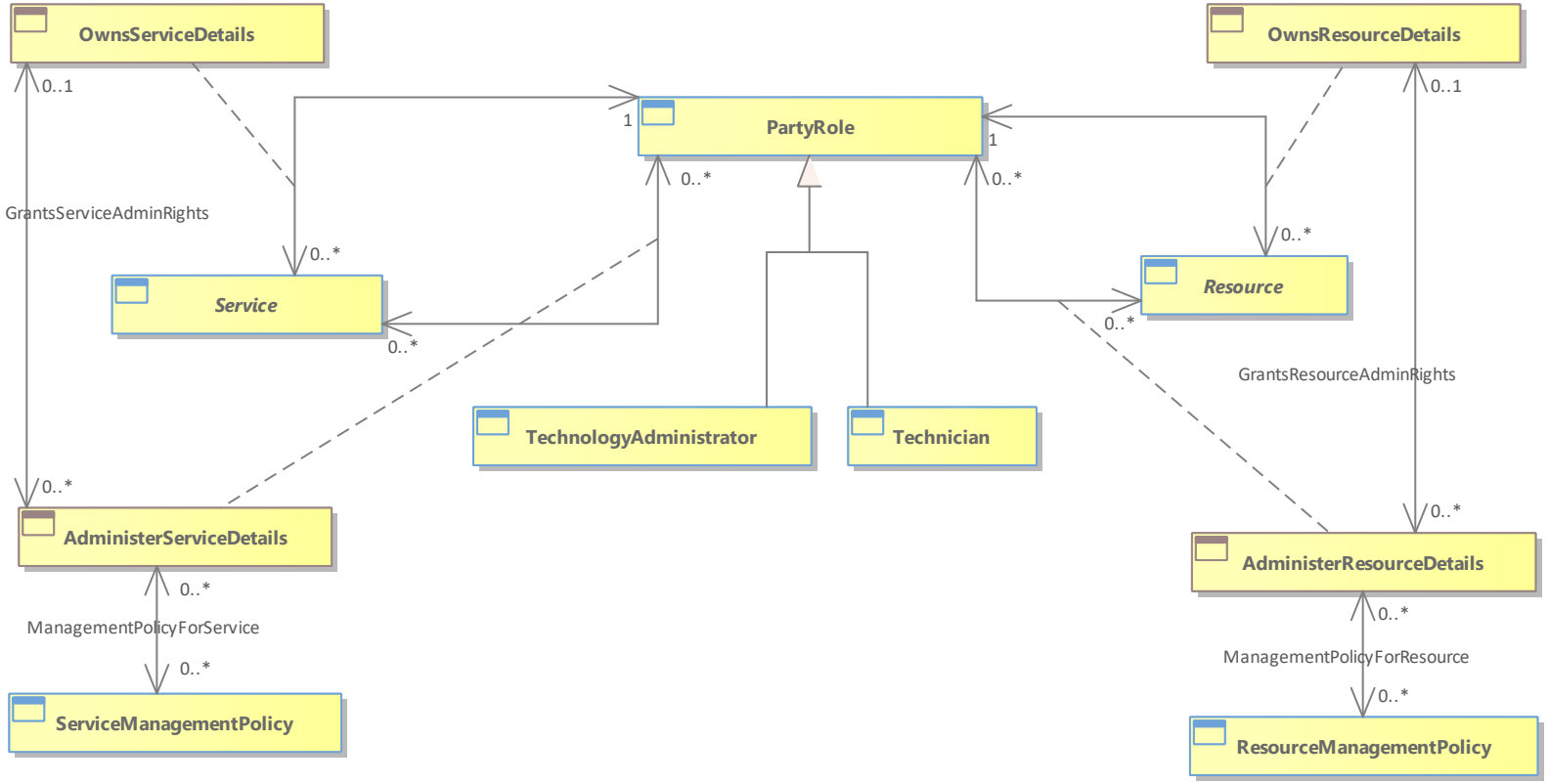

Figure SO.35 - Interaction Between PartyRole and Service

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Figure SO.36 - Interaction between Service and Place

In the SID, different types of Places have different types of semantics associated with them. For example, a Wireless Access Point has an associated coverage area. Only applications within that coverage area can use the services associated with the Wireless Access Point. As another example, subscribers will access different POPs (Points of Presence, the Service Provider term for an access network) as they travel nationally and internationally. Services may vary, and additional peering relationships may come into effect. A simple starting model relating Service to Place is shown in Figure below.

Note that the ServicePlaceDetails association is implemented as an association class. Its purpose is to represent the semantics of this particular association, such as which specific Services are associated with this Place. Clearly, policies can be used to more fully define the semantics of a particular Place.

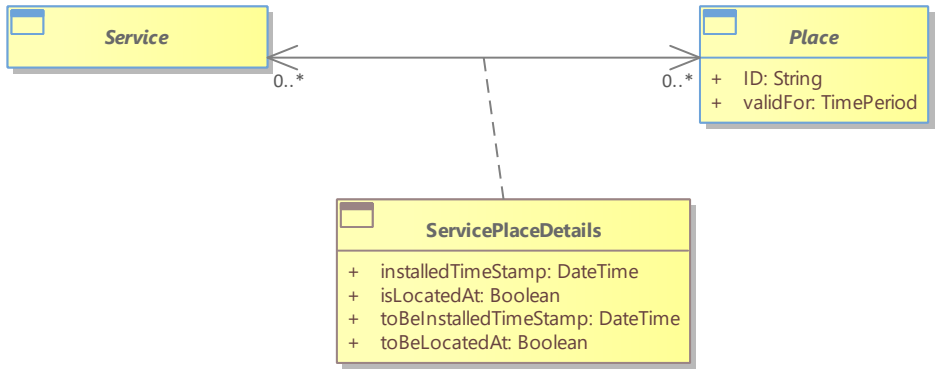

Figure SO.36 - Interaction between Service and Place

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Figure SO.37 - NetworkServiceSpec / UsageVolumeServiceSpec instantiation

A NetworkService is described by one and only one NetworkServiceSpec. A NetworkServiceSpec can describe several NetworkService.

NetworkService is a type of CustomerFacingServiceAtomic.

A UsageVolumeService is described by one and only one UsageVolumeServiceSpec. A UsageVolumeServiceSpec can describe several UsageVolumeService.

UsageVolumeService is a type of CustomerFacingServiceAtomic.

A NetworkService generates 1 to many ServiceUsage.

A UsageVolumeService is concerned by 1 to many ServiceUsage.

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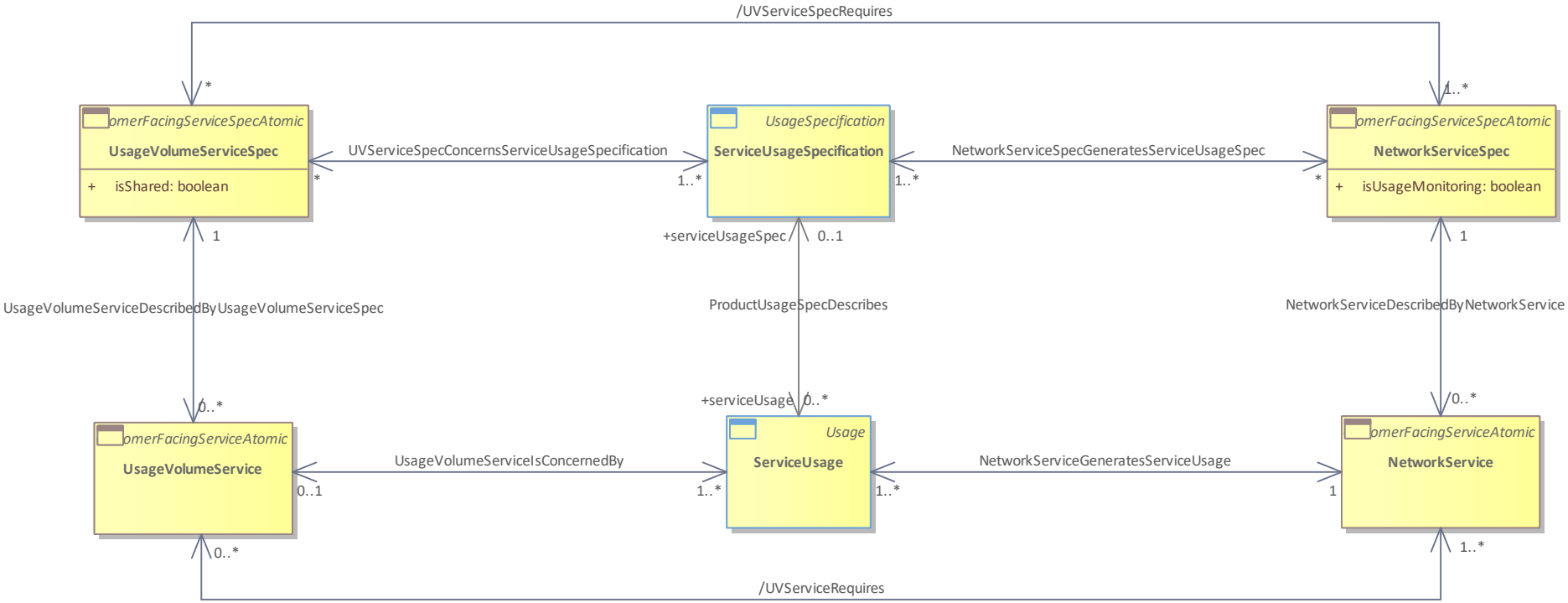

Figure SO.37 - NetworkServiceSpec / UsageVolumeServiceSpec instantiation

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Figure SO.38 - UsageVolumeService detailed view

NetworkService generates usage when Services are used. The usage managed by associated UsageVolumeService is analyzed according to the rules associated to the UsageVolumeService.

These rules determine, according to events and conditions, the action to execute that can correspond to:

- UsageVolumeServiceBalanceReserve (used only in case of supervized NetworkService): Reservation of a Quota on the UsageVolumeServiceBalance according to the evaluation of the usage event,
- UsageVolumeServiceBalanceDebit: Debit of an amount that may correspond to:

- In case of supervized NetworkService: the previously reserved Quota or the amount of the Quota that was really consumed according to the evaluation of the usage event,

- In case of non supervized NetworkService: The usage performed.

- UsageVolumeServiceBalanceCredit: Credit of an amount or a bonus on a UsageVolumeServiceBalance according to the evaluation of the usage event.

A ServiceUsage can be associated to a UsageVolumeService depending on its type. The proposed model supports standard usage cases and also some more complex. As examples:

- In case of calls to toll free numbers UsageVolumeServiceBalanceReserveRule and UsageVolumeServiceBalanceDebitRule can be defined to consider that usage related to toll free numbers do not have any impact on the UsageVolumeServiceBalance.

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- In case of usage that can be done several times during a period (example: a VOD that can be viewed as many times as user wants during 48 hours), UsageVolumeServiceReserveRule and UsageVolumeServiceBalanceDebitRule can be defined to consider that for this specific type of event, only the first event should require a reservation and a debit during a configured period.

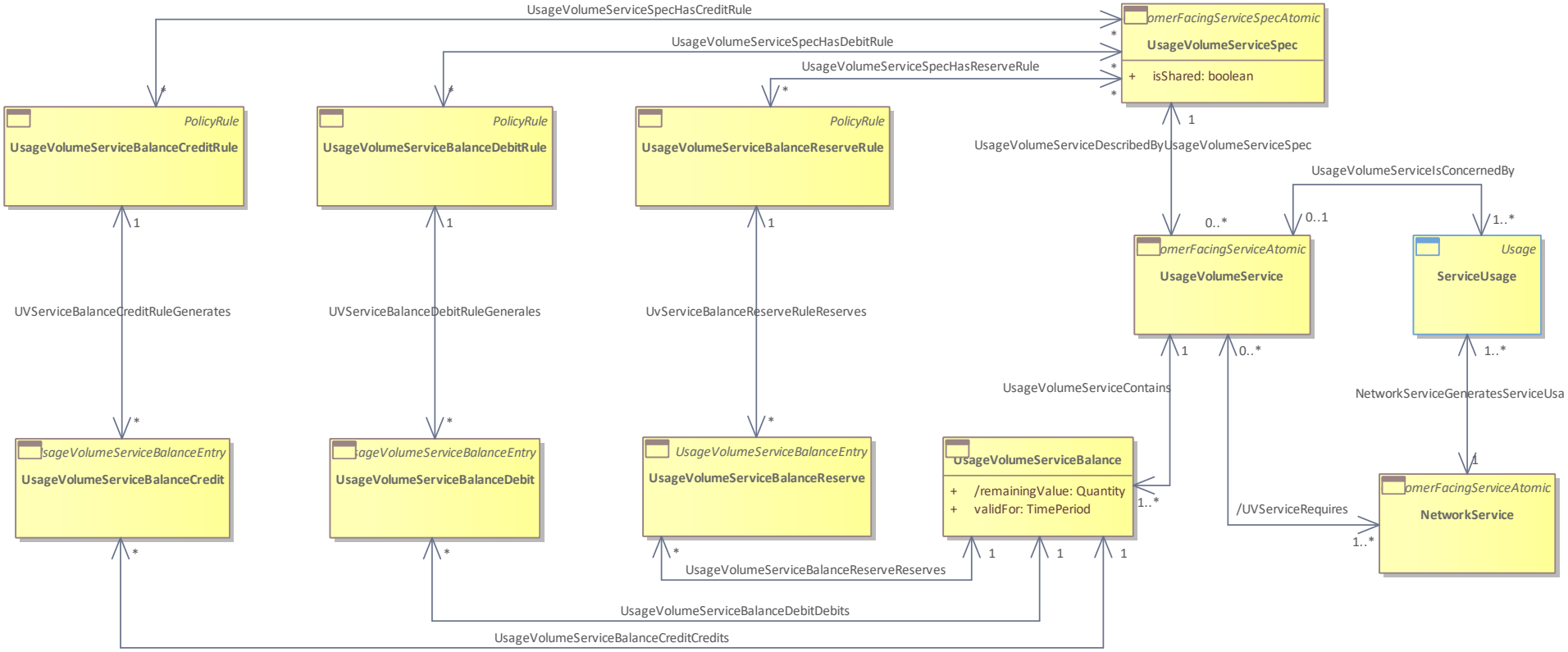

Figure SO.38 - UsageVolumeService detailed view

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4.4.1 Customer Facing Service ABE

The Customer Facing Service ABE defines the characteristics of a particular CustomerFacingService that represents a realization of a Product within an organization's infrastructure. In contrast, the Resource Facing Service ABE is an abstraction that defines the characteristics of a particular ResourceFacingService which support the network/infrastructure facing part of CustomerFacingService. ResourceFacingServices are "internal" Services that are required to support a CustomerFacingService.

Entities in this ABE provide different views to examine, analyze, configure, monitor and repair CustomerFacingServices of all types. Entities in this ABE are derived from Customer Service Specification entities.

Figure CFS.01 - Customer Facing Service ABE Related Entities

Following are the business entities aggregated under the **Customer Facing Service Spec Examples** ABE

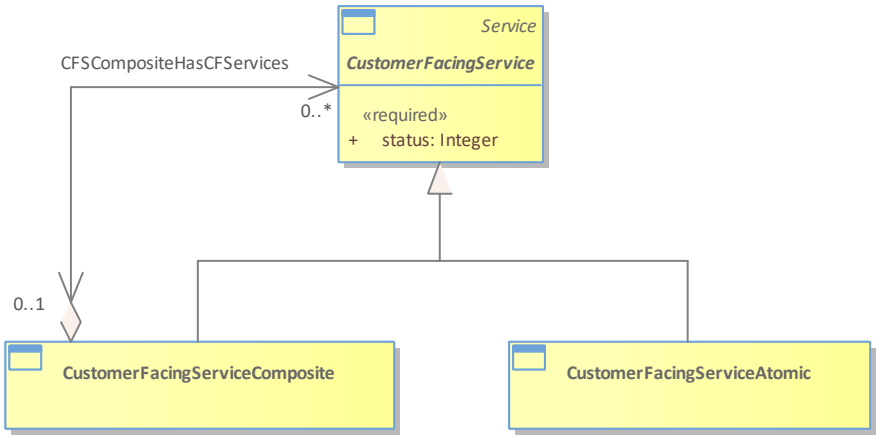

Figure CFS.01 - Customer Facing Service ABE Related Entities

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4.4.1.1 Customer Facing Service Example ABE

The Customer Facing Service Example ABE contains an example CustomerFacingService entity.

Figure CFSE.01 - Customer Facing Service Example ABE Related Entities

Following are the business entities aggregated under the **Customer Facing Service Example** ABE.

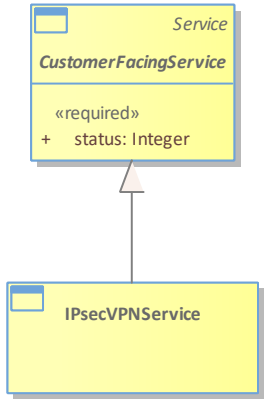

Figure CFSE.01 - Customer Facing Service Example ABE Related Entities

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4.4.1.2 CustomerFacing Service Role ABE

The Customer Facing Service Role ABE defines CustomerServiceRoles that are used to represent the variable characteristics of a CustomerFacingService in terms of roles that this type of Service plays.

4.4.1.3 Network Service ABE «Preliminary»

The Network Service ABE contains the NetworkService description and relationships.

Figure NS.01 - Network Service ABE Related Entities

Following are the business entities aggregated under the Network Service Aggregate Business Entity.

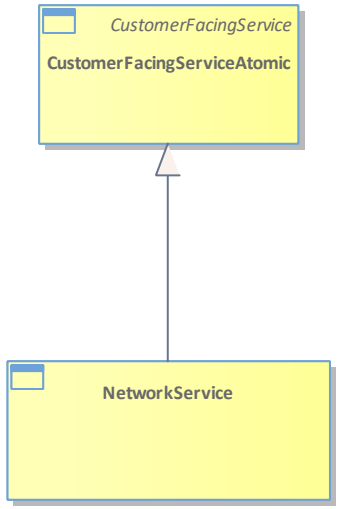

Figure NS.01 - Network Service ABE Related Entities

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4.4.1.4 Usage Volume Service ABE «Preliminary»

The Usage Volume Service ABE contains the UsageVolumeService and all the entities specific to this type of service.

Figure UVS.01 - Usage Volume Service ABE Related Entities

Following are the business entities aggregated under the **Usage Volume Service** Aggregate Business Entity.

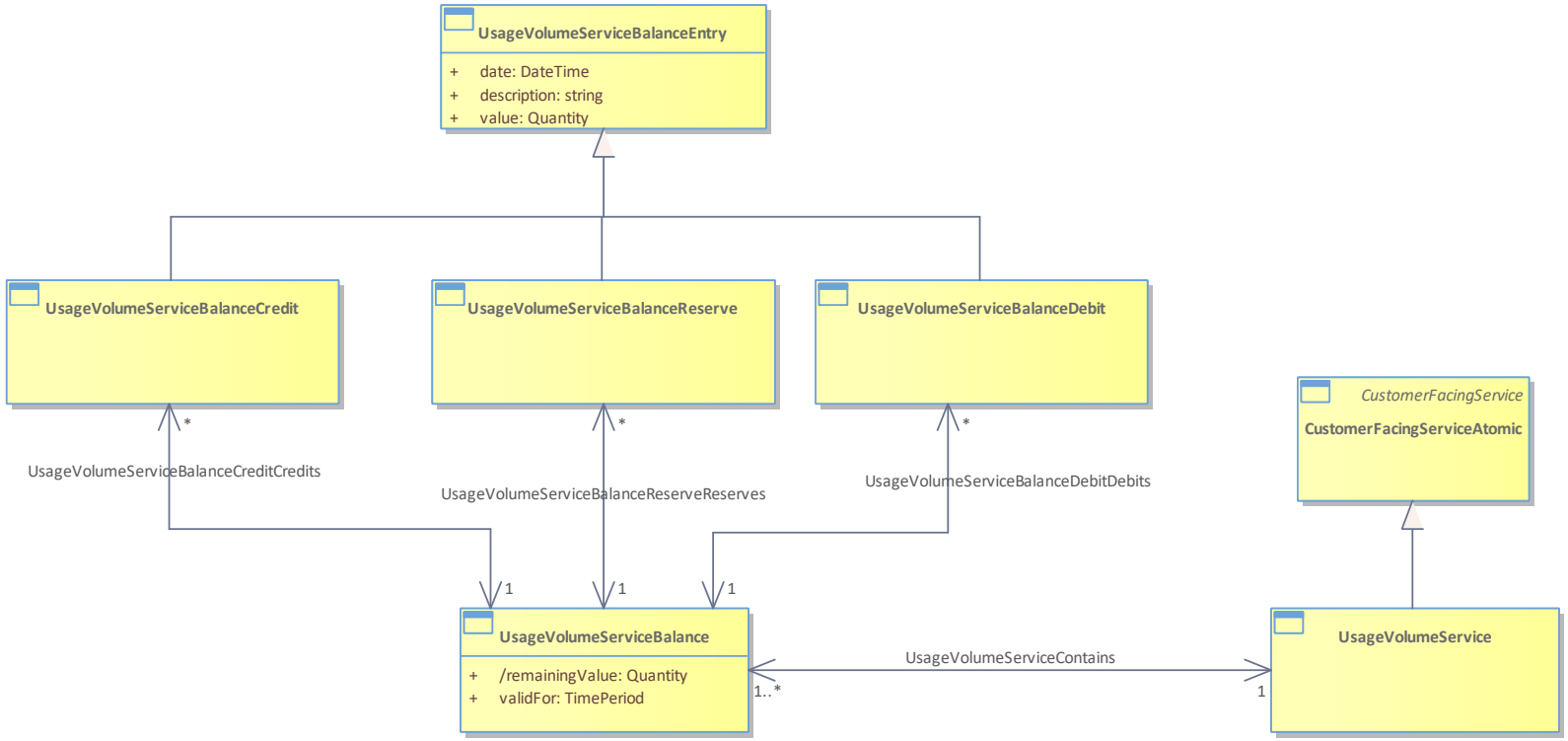

Figure UVS.01 - Usage Volume Service ABE Related Entities

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4.4.2 Resource Facing Service ABE

The Service ABE contains entities that are used to represent two fundamentally different types of Services, called CustomerFacingServices and ResourceFacingServices.

A CustomerFacingService defines the characteristics of a particular Service that represents a realization of a Product within an organization's infrastructure. In contrast, a ResourceFacingService is an abstraction that defines the characteristics of a particular Service which support the network/infrastructure facing part of the service. For example, a VPN is an example of a CustomerFacingService, while the sub-services that perform different types of routing between network devices making up the VPN are examples of ResourceFacingServices. ResourceFacingServices are "internal" Services that are required to support a CustomerFacingService.

Entities in this ABE provide different views to examine, analyze, configure, monitor and repair Services of all types. Entities in this ABE are derived from Service Specification entities.

Figure RFS.01 - Resource Facing Service ABE Related Entities

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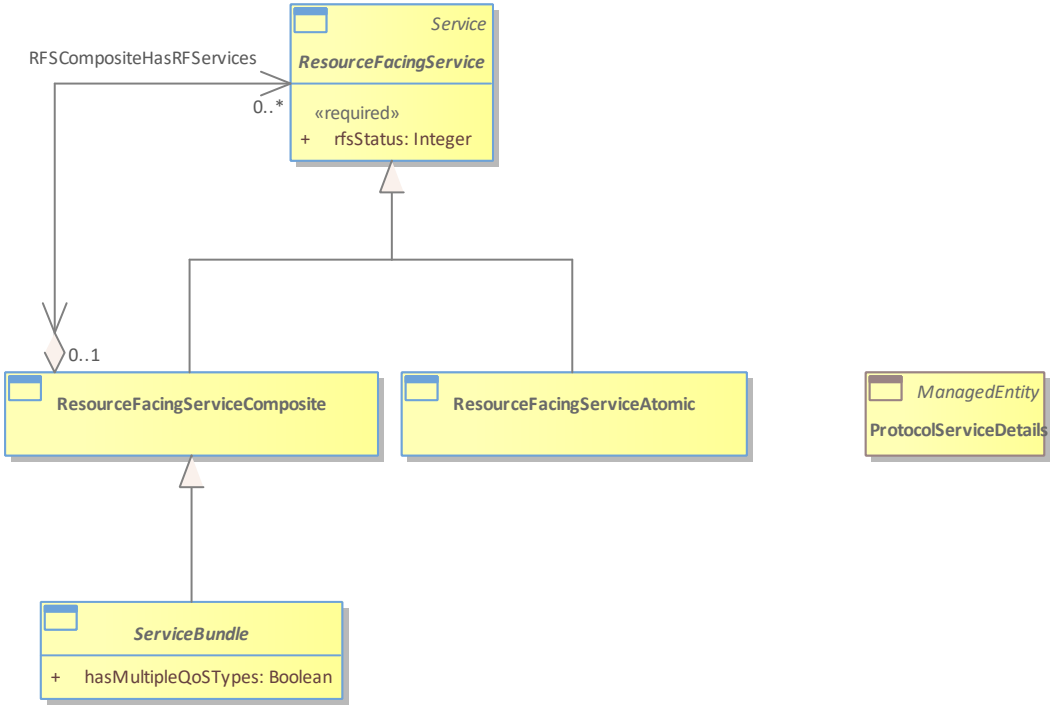

Figure RFS.01 - Resource Facing Service ABE Related Entities

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4.4.2.1 Resource Facing Service Examples ABE

The Resource Facing Service Examples ABE contains examples of ResourceFacingServices.

Figure RFSE.01 - Resource Facing Service Examples ABE Related Entities

Following are the business entities aggregated under the **Resource Facing Service Examples** ABE.

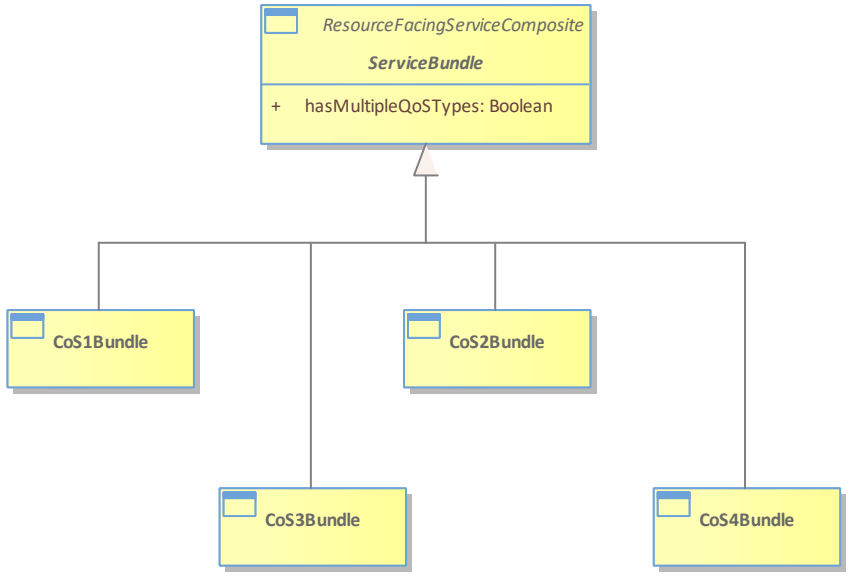

Figure RFSE.01 - Resource Facing Service Examples ABE Related Entities

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4.4.2.2 Resource Facing Service Role ABE

The Resource Facing Service Role ABE defines ResourceServiceRoles that are used to represent the variable characteristics of a ResourceFacingService in terms of roles that this type of Service plays.

Figure RFSR.01 - Resource Facing Service Role ABE Related Entities

Following are the business entities aggregated under the **Resource Facing Service Role** ABE.

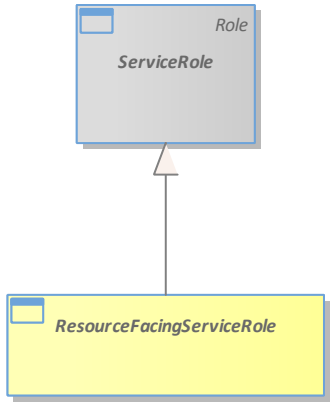

Figure RFSR.01 - Resource Facing Service Role ABE Related Entities

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4.5 Service Configuration ABE

The **Service Configuration ABE** represents the specification of the different possible Service configurations as well as the Service Configurations instantiated for a Service.

Figure CP.01 - Configuration Spec and Configuration (copy from Common)

A Configuration (also referred to as a Profile) defines how a Resource, Service, or Product operates or functions. A Configuration may contain one or more parts (which is realized by using the Atomic/Composite pattern, but it is represented as a single entity - ConfigurationRelationship), and each part may contain zero or more fields. Each field may have attributes that are statically or dynamically defined. Some of these fields have fixed values, while others provide values from which a choice or choices can be made (e.g., using the EntitySpec/Entity and/or CharacteristicSpec/CharacteristicValue patterns).

Starting from the right-side of the figure:

- The Resource, Service and Product classes are as defined in the SID model.
- The associated configuration objects are used to define additional properties (attributes) of the associated entity. For example, consider an Equipment instance (a Resource). The intrinsic properties of the Equipment instance (perhaps but not necessarily common to all Equipment instances) are defined in the Resource itself. Additional properties (which may not be common to all Equipment instances) are provided in one or one Resource Configuration instances.
- It is possible to relate one instance of Configuration to several others via the ConfigRelationship. This is useful when configuring complex (aggregate) entities. For example, to configure a circuit pack, one may also need to configure the various ports supported by the circuit pack and in turn, the connection termination points supported by the ports.

On the left-side of the figure, there are various Configuration Specifications that correspond to the various Configuration classes on the right-side of the figure. This is the typical EntitySpecification/Entity pattern already present in the SID model but now applied to a particular kind of Entity, i.e., a Configuration. The EntitySpecification represents the definition of an entity. The idea is that the various ResourceConfigSpec instances are reusable across many instances of Resource.

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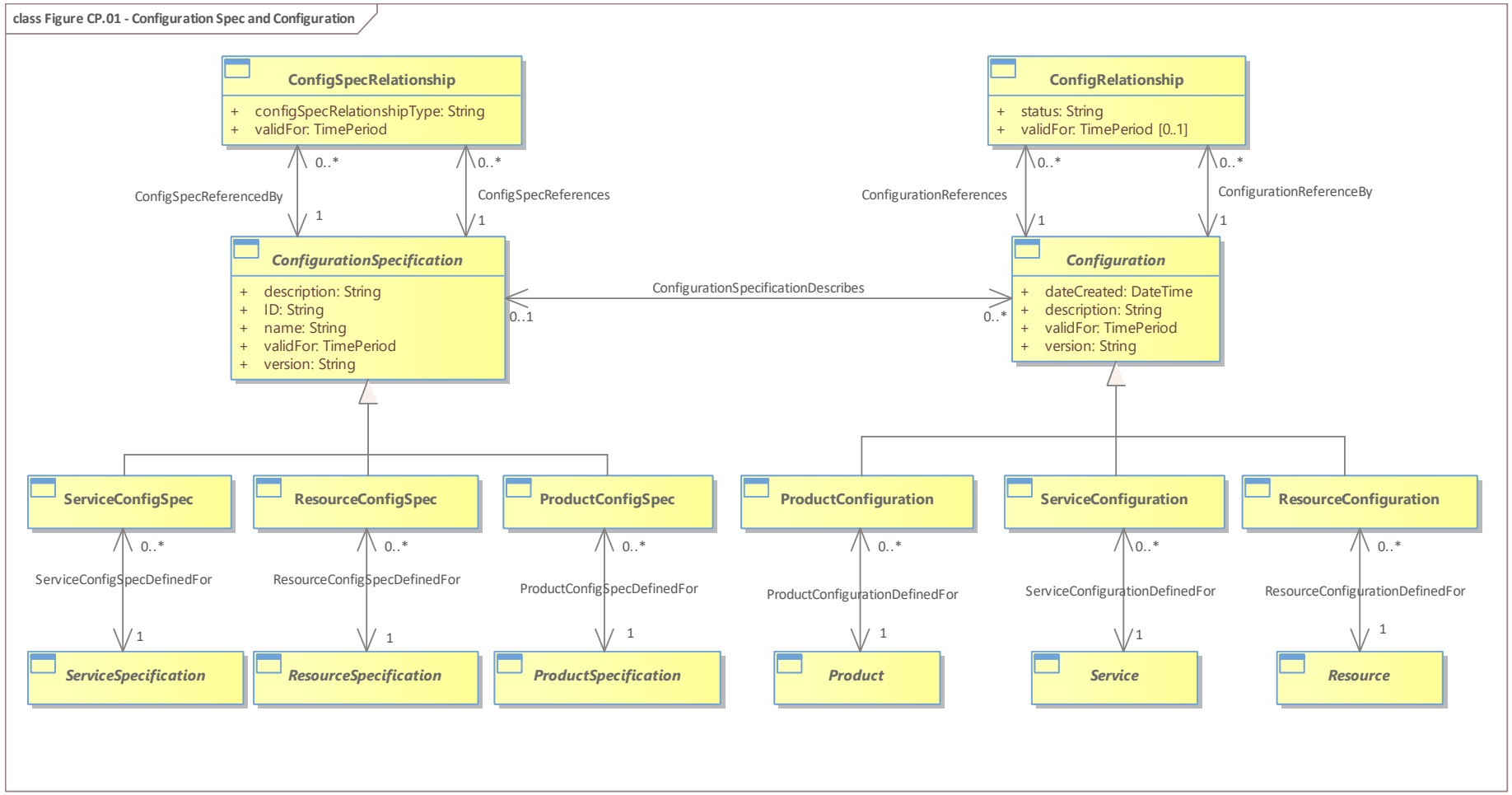

Figure CP.01 - Configuration Spec and Configuration (copy from Common)

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Figure CP.02 - Configuration Specs Associations (copy from Common)

The following Figure emphasizes the reusability of the various Configuration Specifications. So, for example, a single instance of ProductConfigSpec could be applies to many ProductConfiguration instances.

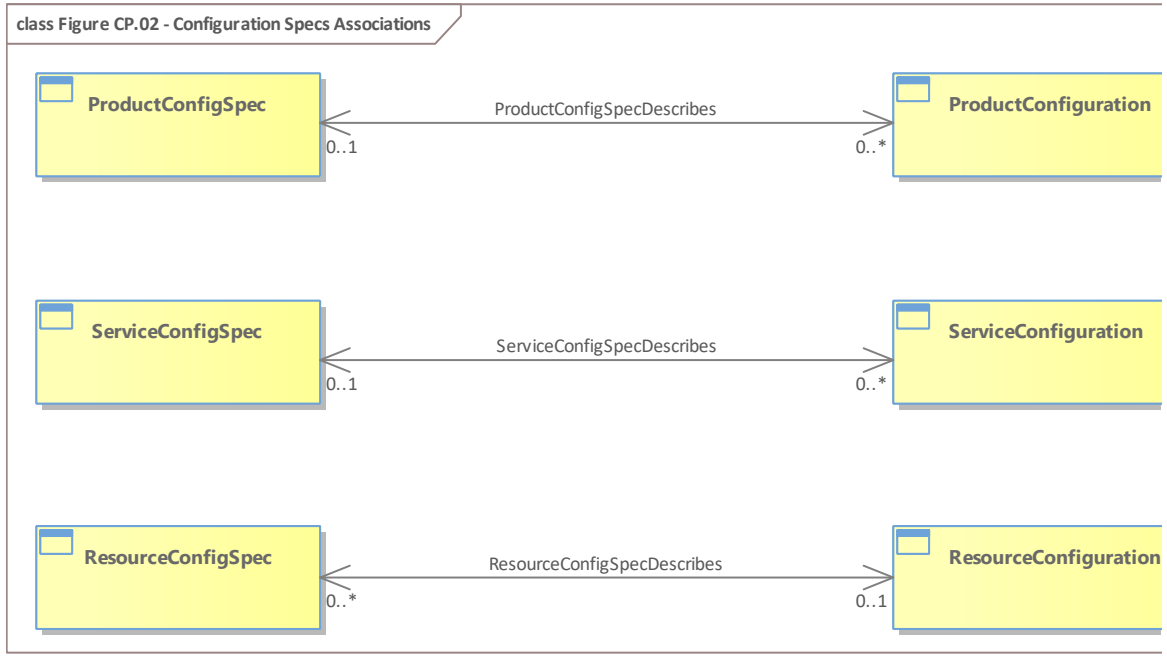

Figure CP.02 - Configuration Specs Associations (copy from Common)

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Figure CP.03a - Entities Serve as Configuration Specifications (Copy from Common)

In some cases a Product, Service, or Resource entity may play the role of a Configuration Specification as shown in the following Figure.

The entity's attributes (with the entity playing the role of a Configuration Specification) are used to set the values for other entities. In this case, the attributes can be "discovered" by code. If they have a value, it is used to set the value for another entity of the same type. One restriction applies here, i.e., attributes of an entity (playing the role of a Configuration Specification) must be able to be set to null (if not needed for a given situation) even if the attribute value is normally required to be set in the entity (when the entity is not playing the role of a Configuration Specification). This approach (of allowing entities to play the role of a Configuration Specification) is needed to avoid turning fixed (intrinsic) attributes of an entity into flexible, dynamic characteristics.

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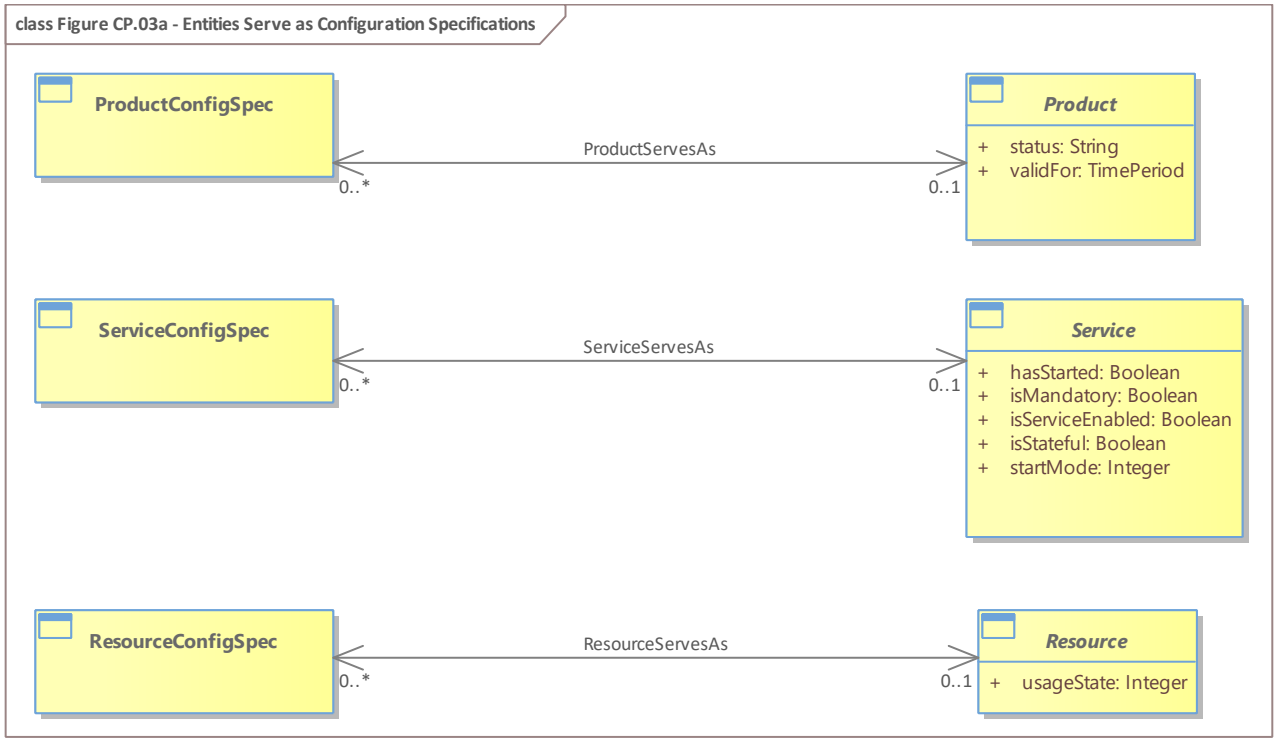

Figure CP.03a - Entities Serve as Configuration Specifications (Copy from Common)

4.6 Service Order ABE

The **Service Order ABE** contains all entities that represent a type of Request of one or several services for any type of Service Specification. A Service Order decomposes a Product Order's products into the services associated with a ServiceOrder through which the products are realized.

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Figure SOr.01 - Service Order Entities

ServiceOrders, which are used to provision services, can be initiated in a number of ways, such as by a CustomerOrder to provision CustomerFacingServices that realized the Product(s) on the order or to reconfigure/reprovision Services as a result of a problem with them.

The following Figure shows the basic model and how ServiceOrder and ServiceOrder item inherit from entities in the Business Interaction ABE.

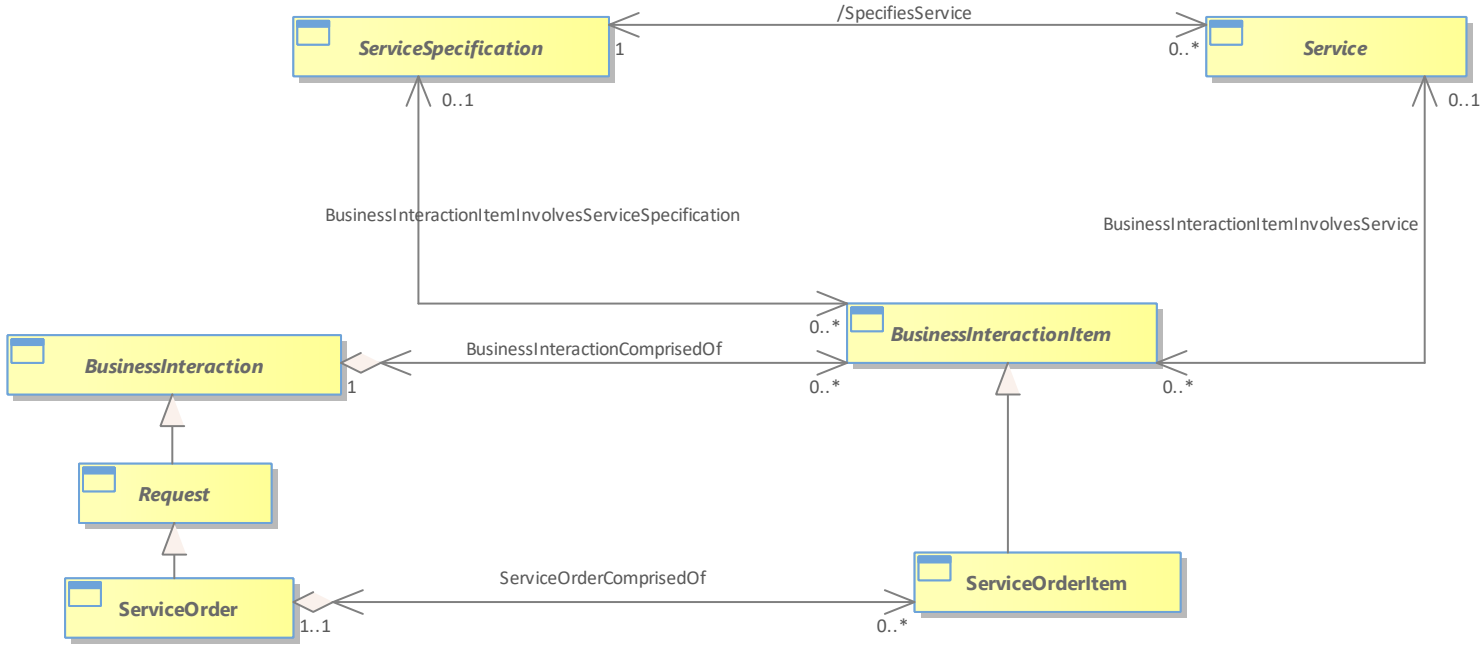

Figure SOr.01 - Service Order Entities

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Figure SOr.02 - Relationships Inherited From Business Interaction

ServiceOrder inherits a number of associations that entities in the Business Interaction ABE have with entities in other ABEs as shown in the following Figure.

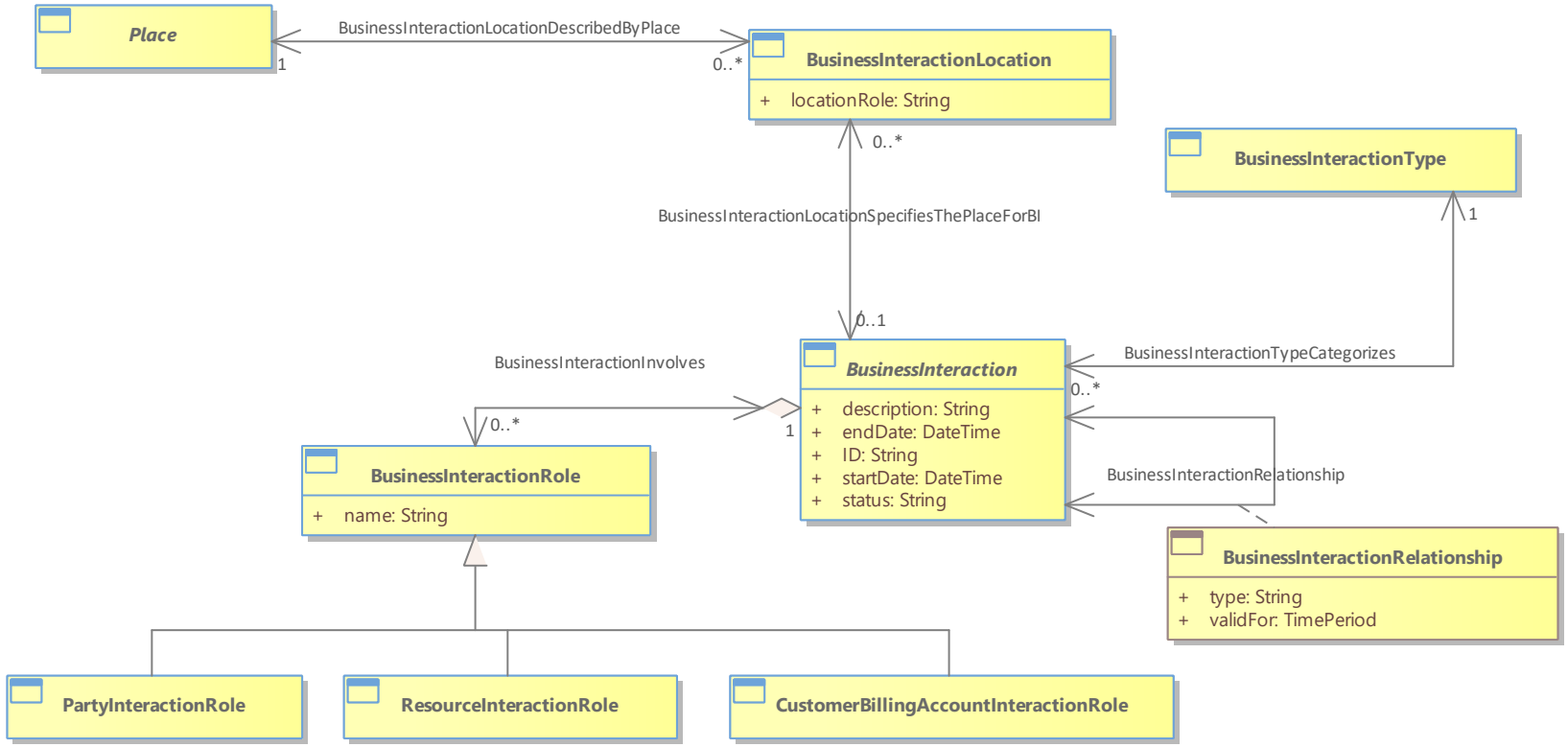

Figure SOr.02 - Relationships Inherited From Business Interaction

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Figure SOr.03 - Relationships Inherited From BI Item

ServiceOrderItem also inherits a number of associations that entities in the Business Interaction ABE have with entities in other ABEs as shown in the following Figure.

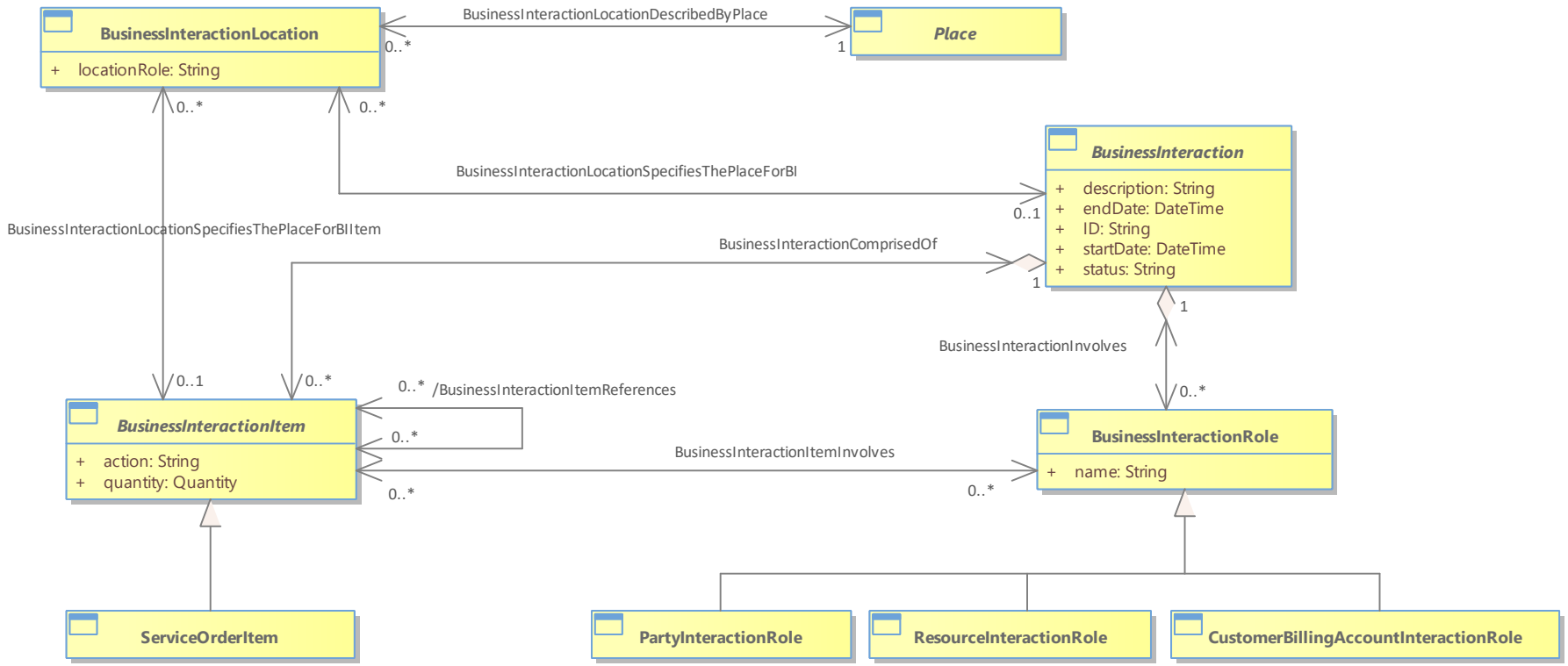

Figure SOr.03 - Relationships Inherited From BI Item

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4.7 Service Usage ABE

The **Service Usage ABE** represents the specification of Service Usage types and collects Service Usage records with their characteristics (a.k.a. consumption data).

Figure U.02 - Usage Specification (copy from Common)

The following Figure shows abstract business entities that are used to model usage. The model shown below is generic.

The UsageSpecification class describes a type of usage. In order to achieve a flexible structure of the UsageSpecification all its attributes are stored as characteristics. Alternatively, different types of usage could be modeled as sub-classes of the Usage entities (Usage, ProductUsage, ServiceUsage, ResourceUsage).

The UsageSpecification is comprised of UsageSpecCharacteristics that define all characteristics (attributes) known by a particular type of usage. Each characteristic is described by its name, category, presence and a set of allowed values.

- Name defines a unique name that identifies the characteristic.
- Category defines one of the high-level aspects of the usage information described by the characteristic. These categories are commonly referred to as: Who, When, What, Where, and Why.
- Min and Max cardinalities that defines whether the characteristic is required or optional.

Additional details about the Characteristic pattern can be found in GB922-1R Root Business Entities addendum.

The set of allowed values is modeled with UsageSpecCharacteristicValue. This class describes all allowed values or value ranges.

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The Usage class represents a usage event. It is described by the UsageSpecification and comprised of UsageCharacteristicValues that hold data attributes of a usage specified by UsageSpecCharacteristics. The UsageCharacteristic can either store its value in the usageCharValue property or reference predefined UsageSpecCharacteristicValue.

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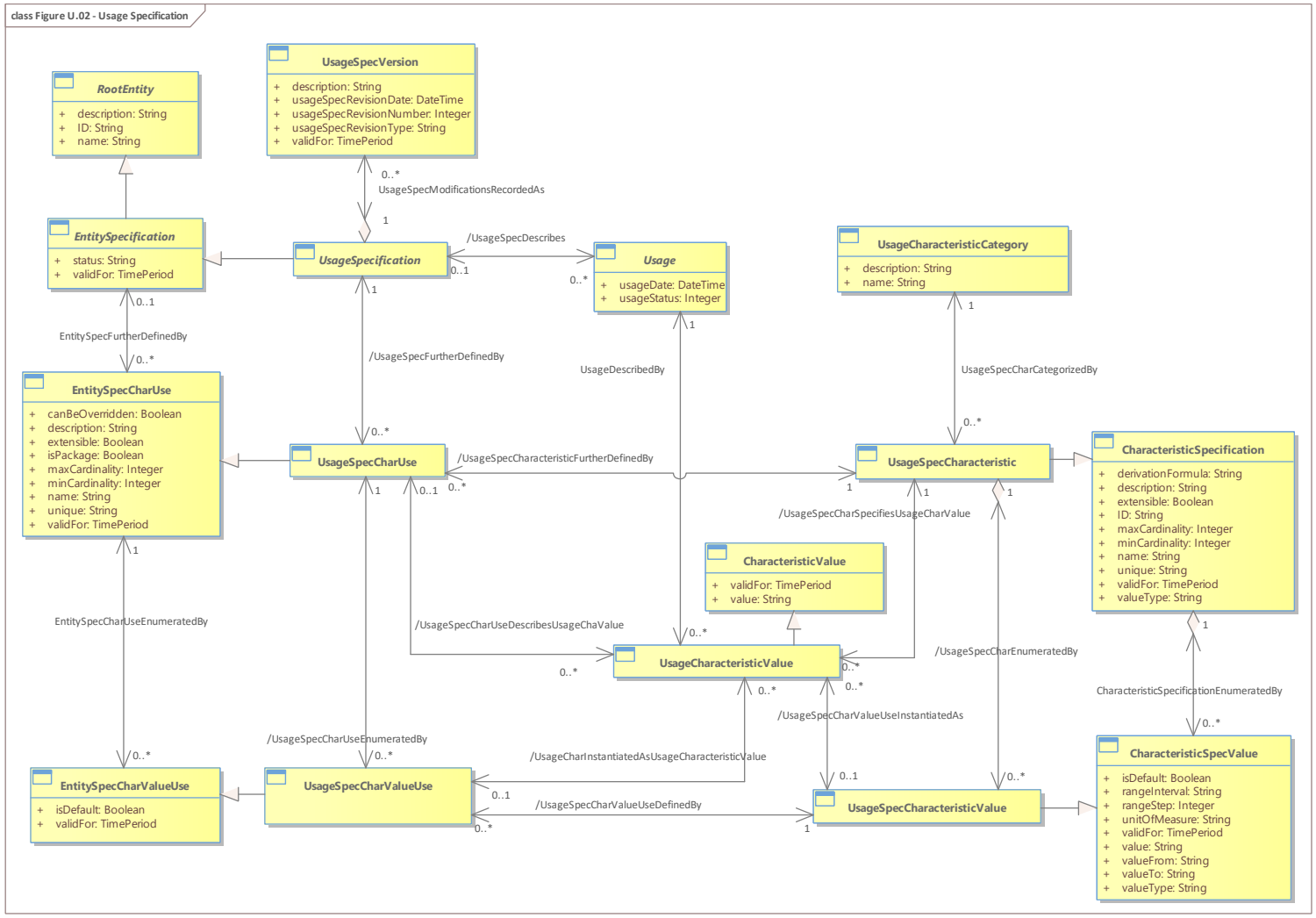

Figure U.02 - Usage Specification (copy from Common)

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Figure U.04 - Resource and Service Usage (copy from Common)

A resource usage represents any usage of resources in its broadest meaning, for example a purchase or lease of a product realized by a resource. A service usage represents any usage of service in its broadest meaning, for example a purchase of a product realized/implemented by a service or a usage of a service that realizes a product. Classes that are used to model resource and service usages are subclasses of abstract usage classes as shown in the following Figure, the ResourceUsageSpec is a subclass of the UsageSpecification and represents a type of resource usage, while the ServiceUsageSpec represents a type of service usage. A typical source of resource or service usage is a mediation application. Other sources may be application servers and other applications (for example, an online shopping application) that are aware of the usage format and can generate usage directly without mediation.

An example of the ServiceUsage class is a voice call that is characterized by:

An origination and a destination address (Who), typically complete information is available only for ProductUsage

- A time of a day the call was connected and a duration of the call (When),
- A call type and its details (What),
- Connection locations of the call (Where) and
- A cause of event recording (Why).

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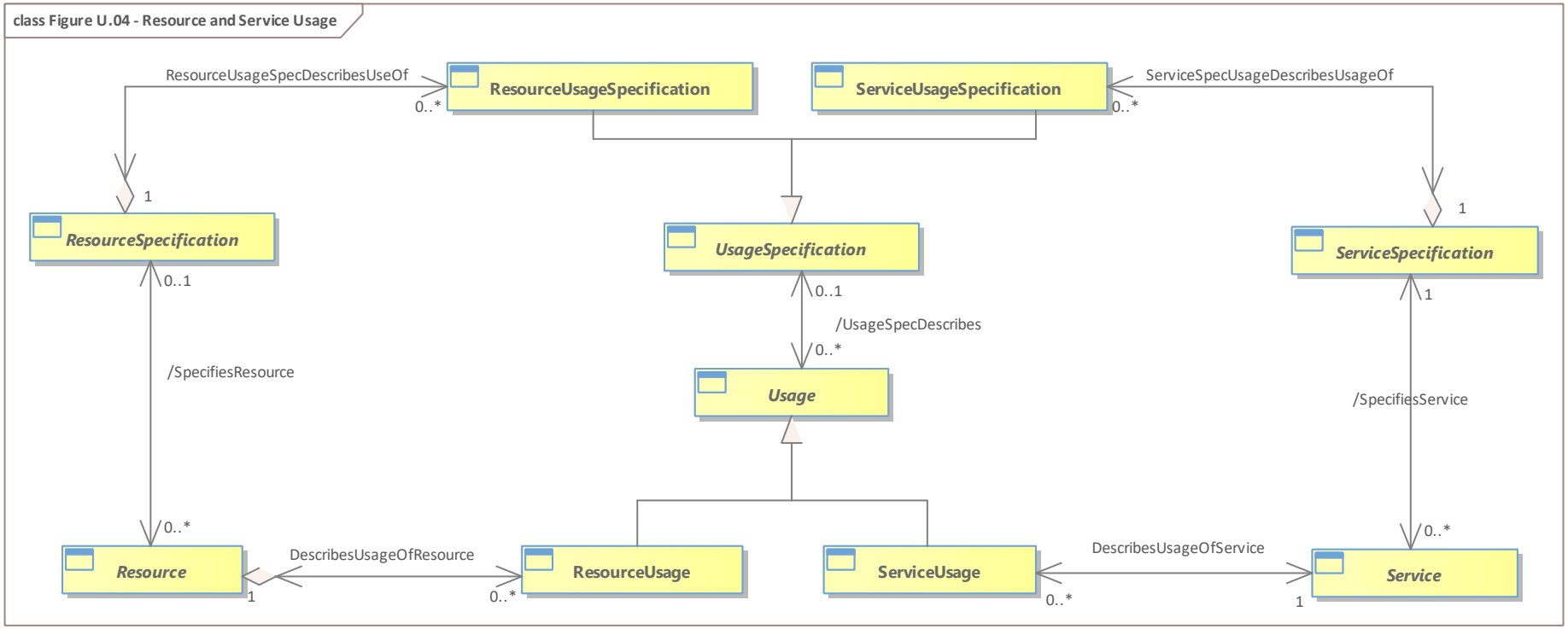

Figure U.04 - Resource and Service Usage (copy from Common)

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Figure U.06 - Product Usage, Part 2 (copy from Common)

As mentioned above, a product is realized as one or more services and/or resources. Consequently, the ProductUsageSpec business entity has to be associated with the ResourceUsageSpec and the ServiceUsageSpec business entities. The ProductUsage, the ServiceUsage and the ResourceUsage business entities have to reflect these associations as well. The ProductComponentUsage can be associated with either ServiceUsage or ResourceUsage and included in a ProductBundleUsage. These entities and associations are shown on the Figure bellow.

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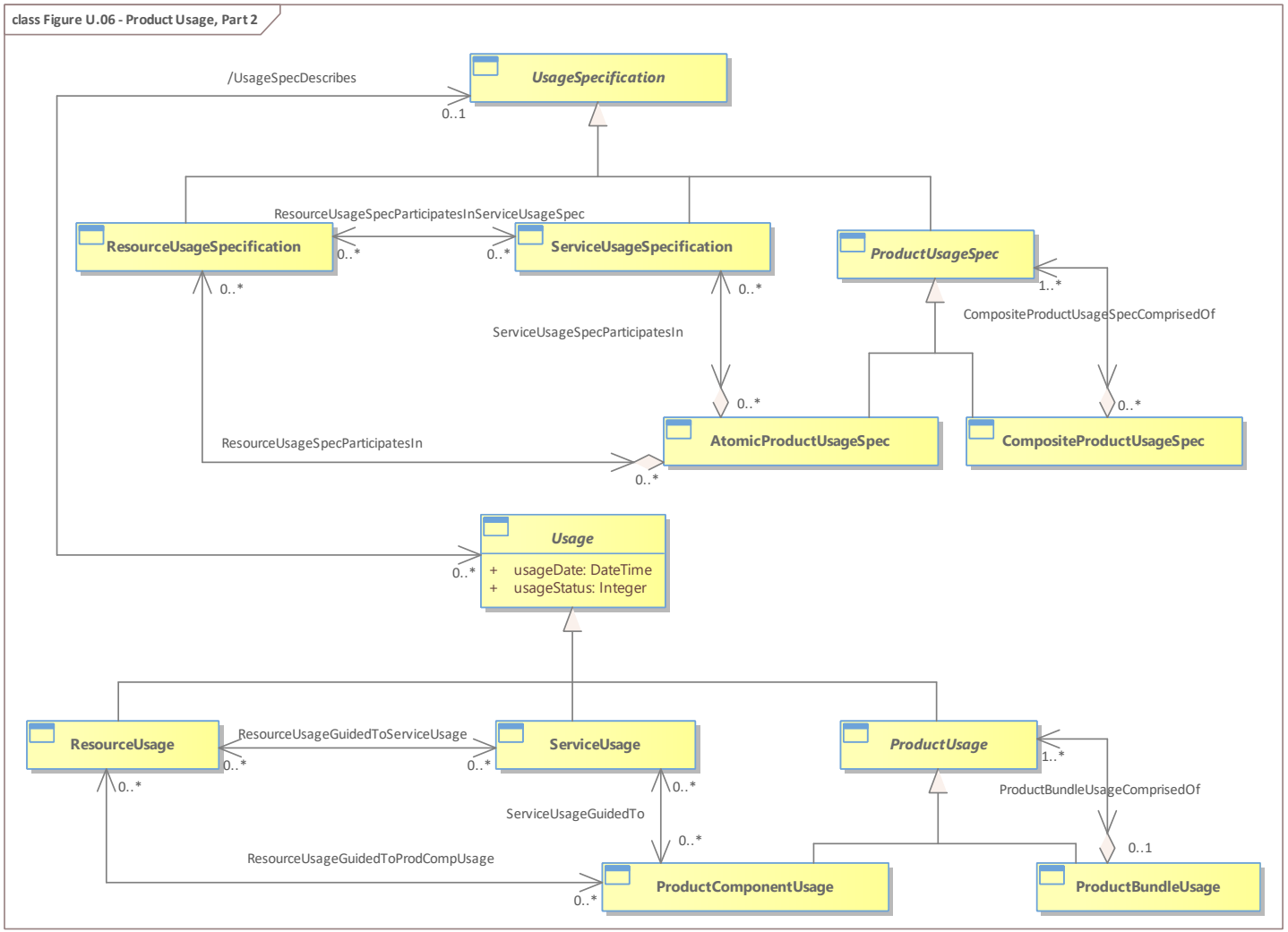

Figure U.06 - Product Usage, Part 2 (copy from Common)

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4.7.1 Service Usage Specification ABE

The Service Usage Specification ABE specifies types and characteristics of usage of ServiceSpecifications.

Figure SUS.01 - Service Usage Specification ABE Related Entities

Following are the business entities aggregated under the **Service Usage Specification** Aggregate Business Entity.

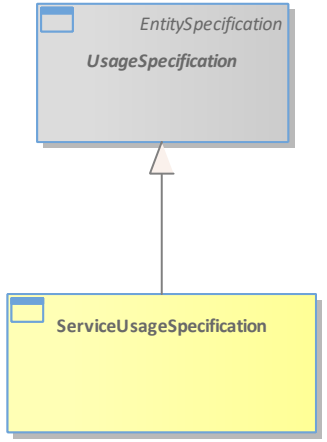

Figure SUS.01 - Service Usage Specification ABE Related Entities

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4.7.2 Service Usage Entity ABE

The Service Usage Entity ABE collects service usage information, and generates ServiceUsage according to ServiceUsageSpecification for use by other business entities.

Figure SUE.01 - Service Usage Entity ABE Related Entities

Following are the business entities aggregated under the **Service Usage Entity** Aggregate Business Entity.

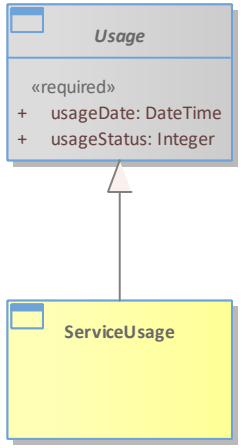

Figure SUE.01 - Service Usage Entity ABE Related Entities

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4.8 Service Test ABE

The **Service Test ABE** defines measurement including units and values and how the values are determined. It also includes thresholds used to evaluate the measure and the consequences of violating the thresholds as well as the Service Test measures that reflect the functioning of the Service under test.

Figure T.04 - Product, Service and Resource Test Entities (Copy from Common)

The following Figure shows one way to support tests for Product, Services, and Resources by applying the Entity/EntityRole pattern. This allows the same "Test Specification" to be used for many "Product TestSpecifications", "ServiceTestSpecifications", and "Resource TestSpecifications".

Product, Service, and Resource Tests are defined to be generalizations of the abstract "Test" class. In this way, the "Test" model may be used for the specification of all three types of tests. Each test, such as "ServiceTest" is associated to its relative entity under test, which in this case is "Service". The relationships therefore between products, services, and resources, are maintained to the associated tests. As an example, a service test measure such as "Frame Loss Ratio" may translate to resource test measures "Number of frames sent" and "Number of frames received". The metric "Number of Frames Lost" is derived from the resource test measures, and the "Frame Loss Ratio" is further calculated by dividing the number of frames lost by the number of frames sent.

Refer to Common Domain guide book section Test ABE for more details

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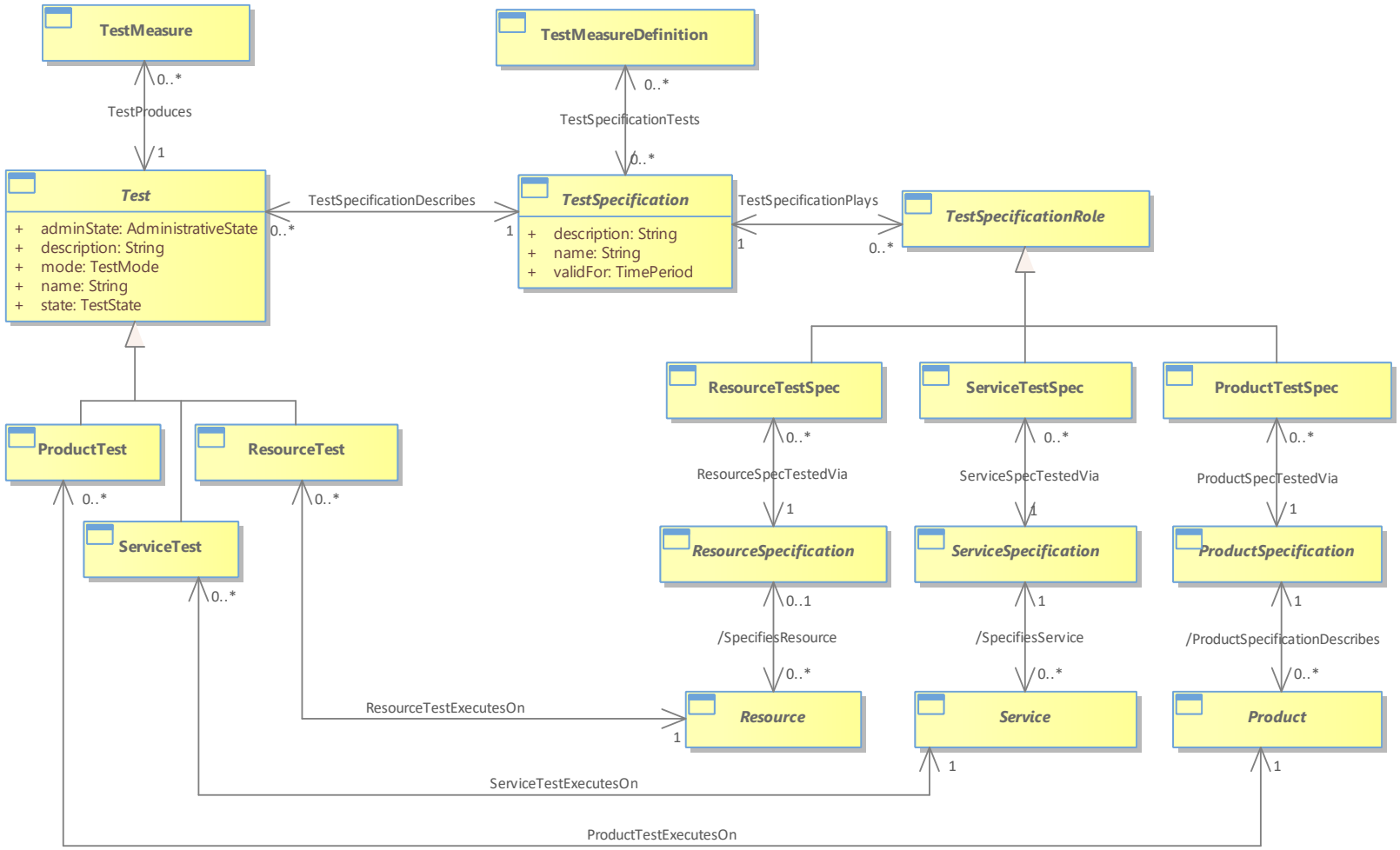

Figure T.04 - Product, Service and Resource Test Entities (Copy from Common)

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4.9 Service Performance ABE

The **Service Performance ABE** collects information used to correlate, consolidate, and validate various performance statistics and other operational characteristics of Know-Hows (a.k.a. CFS) and Technical Solutions (a.k.a. RFS). It provides a set of entities that enables performance monitoring and reporting. Each of these entities also enables network performance assessment against planned goals, performs various aspects of trend analysis, including error rate and cause analysis and Service degradation.

Entities in this ABE also enable the traffic trend analysis.

Figure SO.37 - Service Level Specification Overview

It should be noted that this section is still a work in progress, and is subject to change.

A business participant, such as a Customer (or other type of PartyRole), often expects a certain level of Service to be associated with a Product or Service procured from another business participant, such as a Service Provider or some other Enterprise. Such expectations are defined by a Service Level Specification, which is associated to a Product, Service or their respective Specifications. Sometimes referred to as a Service Level Agreement Template, a Service Level Specification represents a pre-defined or negotiated set of Service Level Objectives. In addition, certain consequences are associated with not meeting the Service Level Objectives. Service Level Agreements are expressed in terms of Service Level Specifications.

This figure shows a simple, un-attributed view of a Service Level Specification. Note: the two subclasses of ServiceLevelSpecification are shown merely to emphasize the two different types of specifications.

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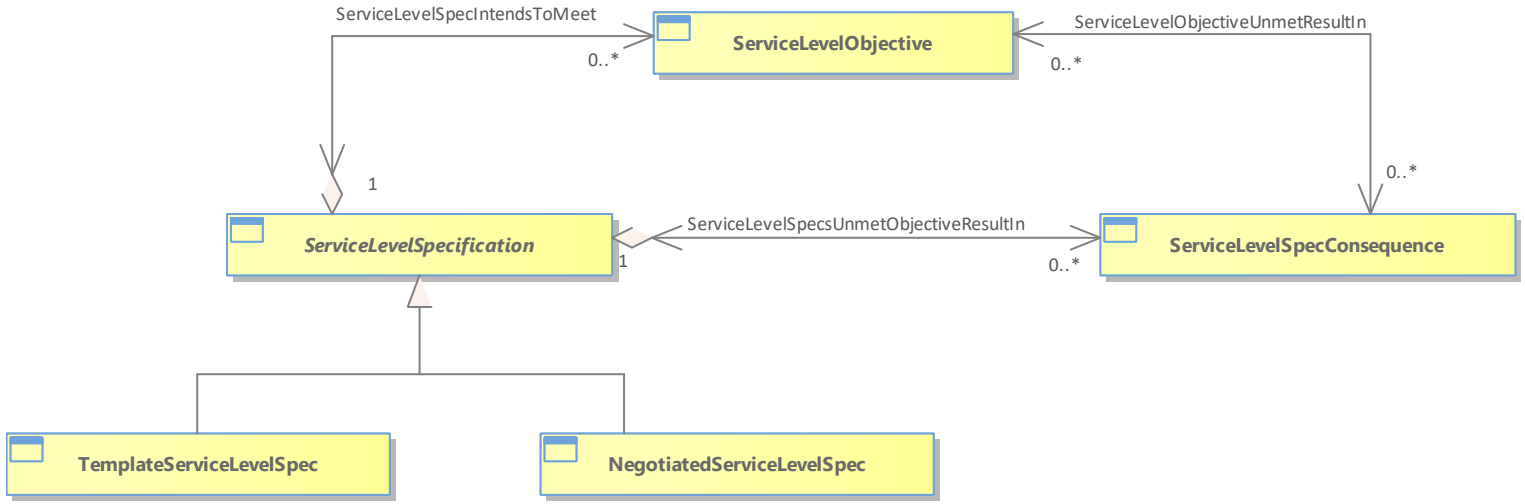

Figure SO.37 - Service Level Specification Overview

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Figure SO.38 - Products, Services, and ServiceLevelAgreements

Examples of products and services, along with two different associated SLAs, are depicted in the diagram below. The diagram is a modified extract from the Service Level Agreement Handbook (GB917).

- Note: the above uses terminology from GB917. An Internal SLA is also known as a Service Provider SLA.

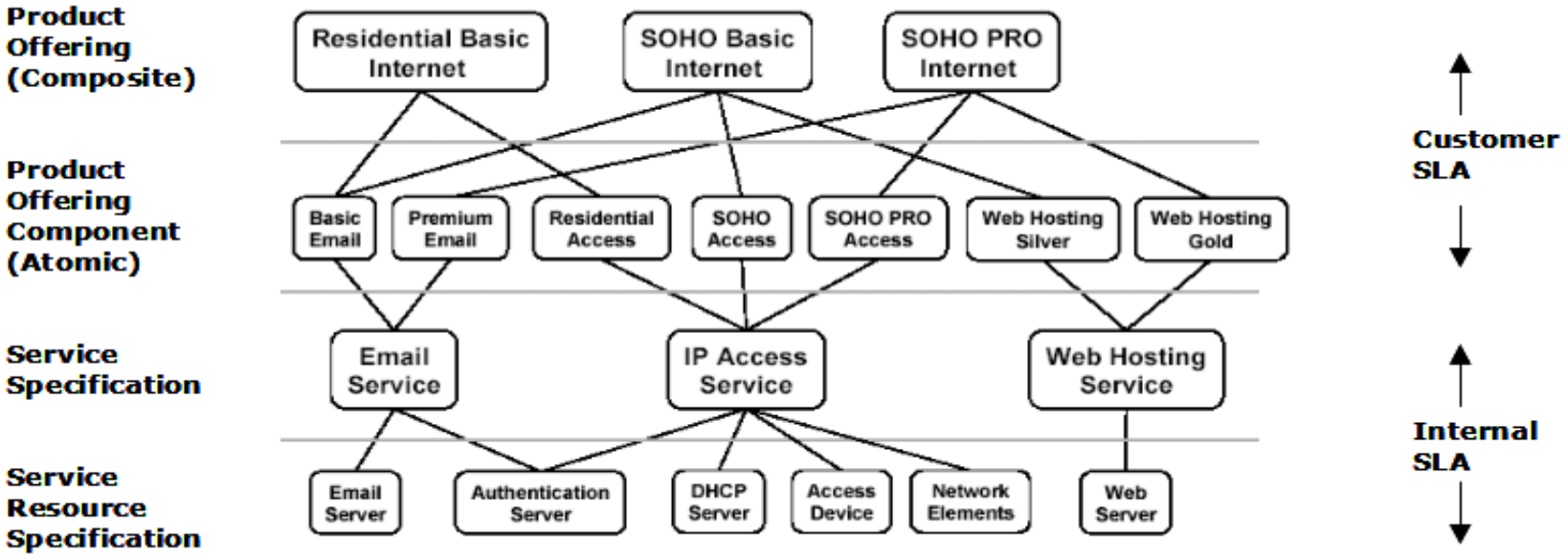

Figure SO.38 - Products, Services, and ServiceLevelAgreements

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Figure SO.39 - Associating Service Level Specifications with Products and Services

Customer Service Level Agreements are expressed in terms that a Customer can understand. These Service Level Agreements are defined in terms of Service Level Specifications related to ProductOfferings and ProductSpecifications. ProductSpecifications are used in order to define a set of characteristics that the set of Service Level Specification will cover.

Internal Service Level Agreements are usually defined in terms of resource-facing Service Level Specifications, which are typically too technical to be of interest to a customer. Supplier/partner Service Level Agreements can be expressed in terms of either customer-facing or resource-facing Service Level Specifications. The Figure below shows a Service Level Specification's relationships to Products, Services, and their associated Specifications.

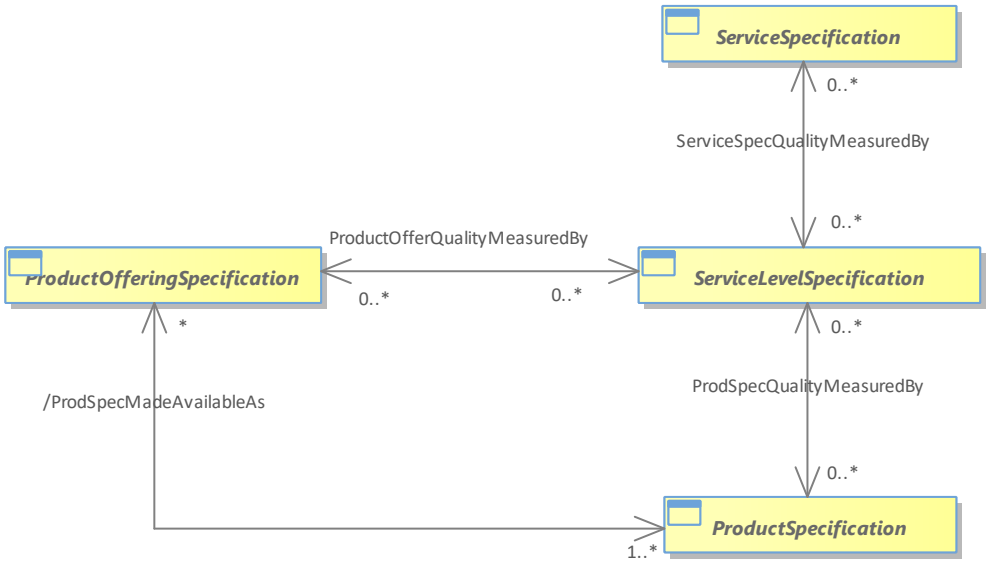

Figure SO.39 - Associating Service Level Specifications with Products and Services

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Figure SO.40 - Service Level Specification Parameters and Objectives

Service level objectives are defined in terms of parameters and metrics, thresholds, and tolerances associated with the parameters. The Figure below depicts Service Level Objectives, along with the Parameters that define them.

Service Level Specification parameters can be one of two types. A Key Quality Indicator (KQI) provides a measurement of a specific aspect of the performance of a Product (i.e., ProductSpecification, ProductOffering, or Product) or a Service (i.e., ServiceSpecification or Service). A KQI draws its data from a number of sources, including Key Performance Indicators (KPIs). A KPI provides a measurement of a specific aspect of the performance of a Service (whether it is a network- or a non-network-based Service) or a group of Services of the same type. A KPI is restricted to a specific resource type. The figure below extends Figure SO.40 by showing how KPIs and KQIs are related to products and services.

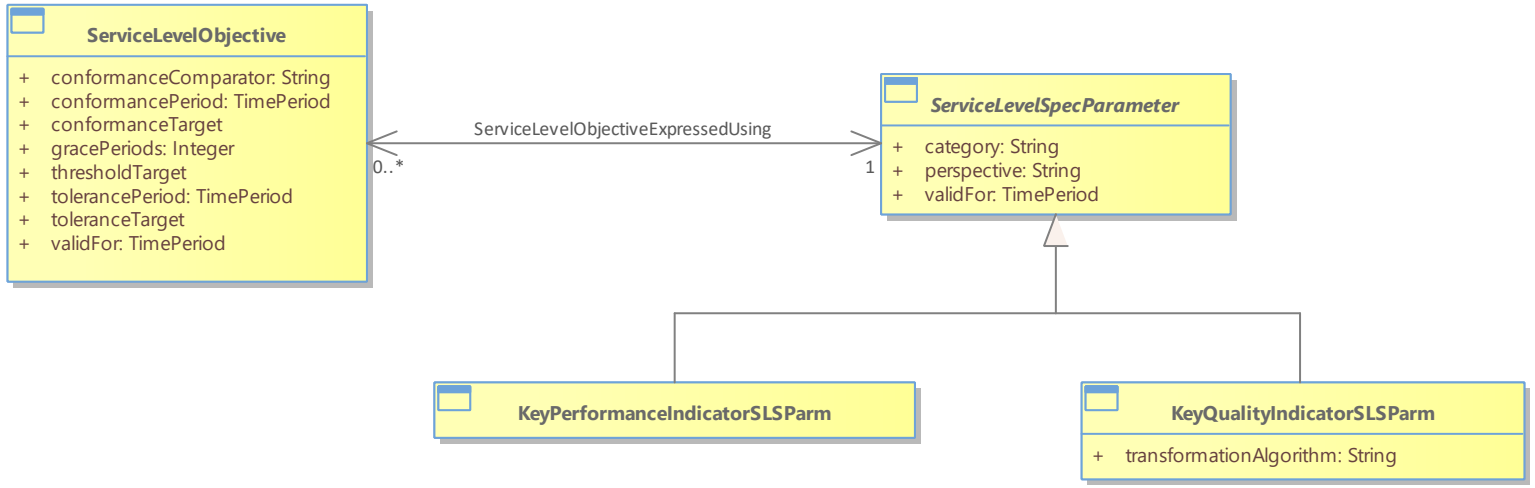

Figure SO.40 - Service Level Specification Parameters and Objectives

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Figure SO.41 - KPIs, KQIs, Products, and Services

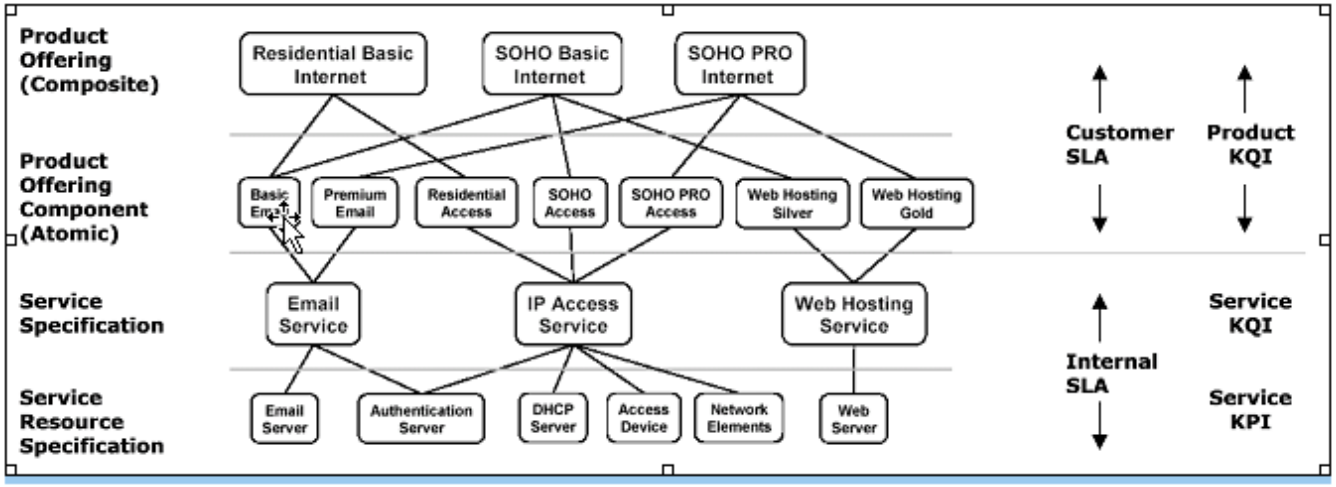

Figure SO.41 - KPIs, KQIs, Products, and Services

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Figure SO.42 - ServiceLevelSpecConsequences

Customer-facing Product KQIs, such as the number of trouble tickets opened for a Product, characterize Customer Service Level Agreements. Service KQIs, such as the maximum web server unavailability time that is permitted, represent an aggregation of Service KPIs or a simple one-to-one mapping to a Service KPI, and characterize internal and engaged party Service Level Agreements. Atomic Service KPIs, such as the maximum IP packet loss ratio, do not characterize Service Level Agreements, but form the basis for the determination of KQIs. Please see the SLA Management Handbook (GB917) for additional examples and the Wireless Services Measurement Handbook (GB923) for more details about KPIs and KQIs.

Violations to Service Level Agreements usually result in some consequence for the provider of the Service. The consequence could be a general one that applies to any objective violation, or the consequence could be associated with a specific objective. An example of a consequence is a discount applied to a customer's bill if the number of trouble tickets for a Product exceeds some number over a specified period of time. The figure below shows how consequences of violating a Service Level Agreement fit into the Service Level Specification model.

No attempt was made to build a detailed model that represents the prescribed actions. Rather, the above model is intended to be an extensible framework that enables each adopter of the model to customize their extensions for this business entity.

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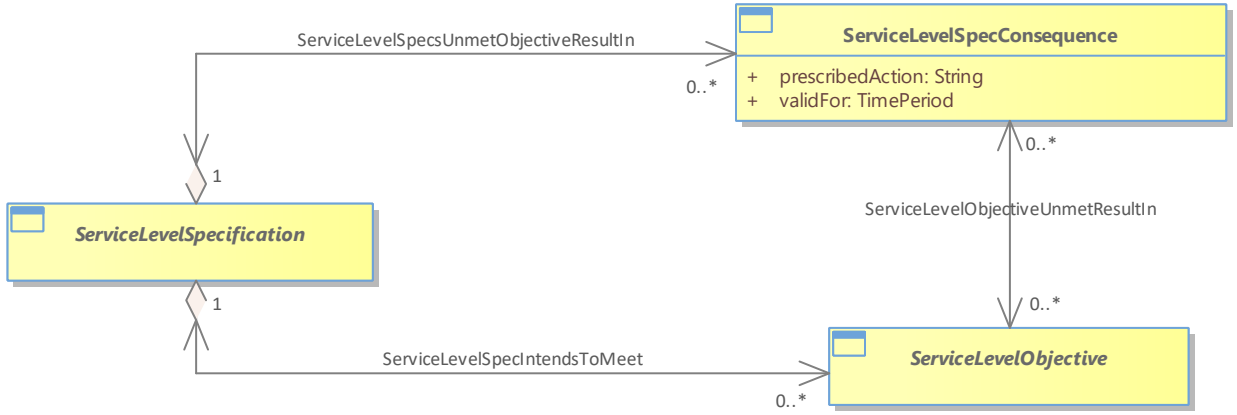

Figure SO.42 - ServiceLevelSpecConsequences

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Figure SO.43 - ServiceLevelSpecApplicability

The applicability of Service Level Specifications, Service Level Objectives, and Service Level Specification consequences may vary by day and/or time of day. For example, a Service Level Objective may be represented by one set of metrics during weekdays and by another set during weekends. Thus, we may need to define a set of consequences to associate the proper effect of violating a Service Level Objective. For example, one consequence might be the result of a Service Level Violation during normal working hours, while a second consequence might be used to represent Service Level Violations that occur during non-working hours. This is handled by introducing a Service Level Specification applicability business entity as shown in the figure below.

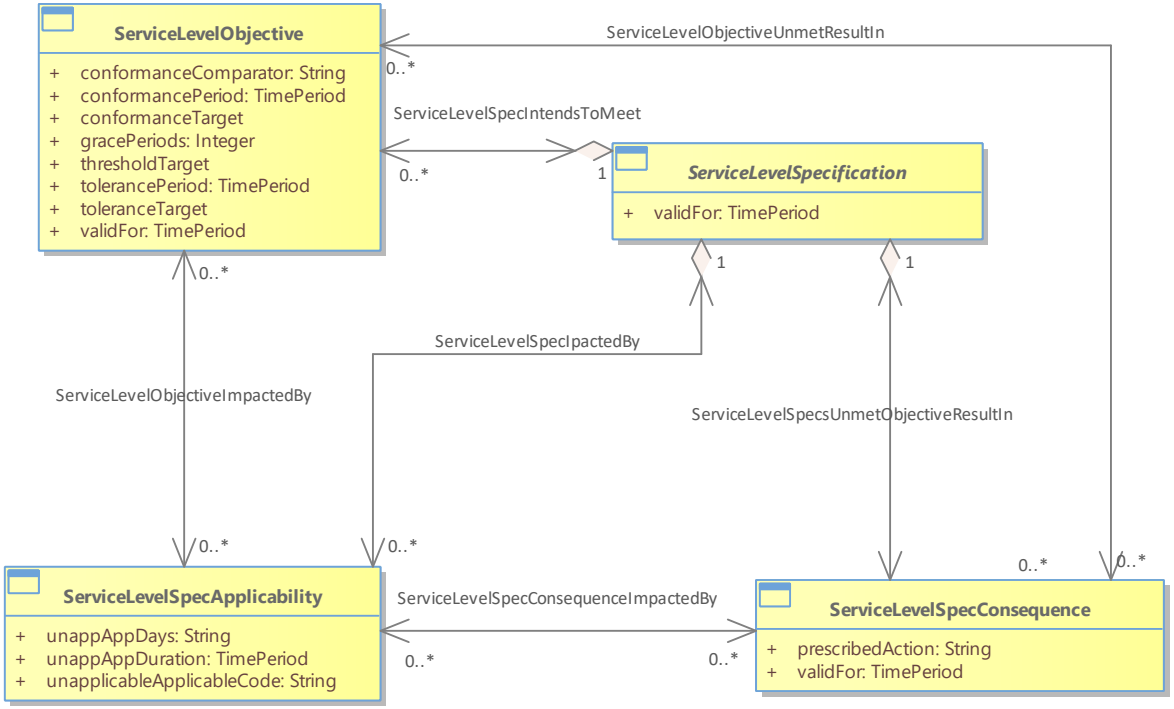

Figure SO.43 - ServiceLevelSpecApplicability

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Figure SO.43a - ServiceLevelSpecificationExpression

The ServiceLevelSpecExpression entity is intended to describe strict relations between ServiceLevelSpecification subordinates (ServiceLevelSpecParameter, ServiceLevelObjective, ServiceLevelSpecApplicability, and ServiceLevelSpecConsequence). The baseline idea of ServiceLevelSpecExpression entity resolves the multiple associations among ServiceLevelSpecification subordinate entities. So in this case it is possible to have variety of ServiceLevelSpecification subordinates combined using ServiceLevelSpecExpression.

When it is needed to calculate Time-To-Resolve of a customer's` issue and the system must notify different departments upon reaching different Time-To-Resolve values.

In this case first ServiceLevelObjective_A with thresholdTarget = 15 min and conformanceTarget = 45 min will be created and associated with ServiceLevelSpecConsequence_A (this consequence will raise notification action: Send letter to the user about existed issue when threshold target was reached and apologies letter when conformance target was reached).

On the other hand second ServiceLevelObjective_B with thresholdTarget = 5 min and conformanceTarget = 30 will be created and associated with SevericeLevelSpecConsequence_B (this consequence will raise notification action: Send letter to the engineer about existed issue when threshold target was reached and escalation letter to the management when conformance target was reached)

So in this case it is crucial to understand which ServiceLevelObjective is related to which consequence and this could be done with in Expression in following way:

- Create ServiceLevelSpecExpression_A
- where

- serviceLevelSpecApplicability = **Applicability_A**

- serviceLevelSpecConsequence = **Consequence_A**

- serviceLevelSpecParameter = **Parameter_A**

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- serviceLevelObjective = **SeviceLevelObjective_A**

  - and

- create second ServiceLevelSpecExpression_B were

- serviceLevelSpecApplicability = **Applicability_A**

- serviceLevelSpecConsequence = **Consequence_B**

- serviceLevelSpecParameter = **Parameter_A**

- serviceLevelObjective = **ServiceLevelObjective_B**

This section has defined a ServiceLevelSpecification, and its associated related objects, to be compatible with the SLA Handbook and the work done by the WSMT. The next version of this Addendum will seek to align this work with other industry work.

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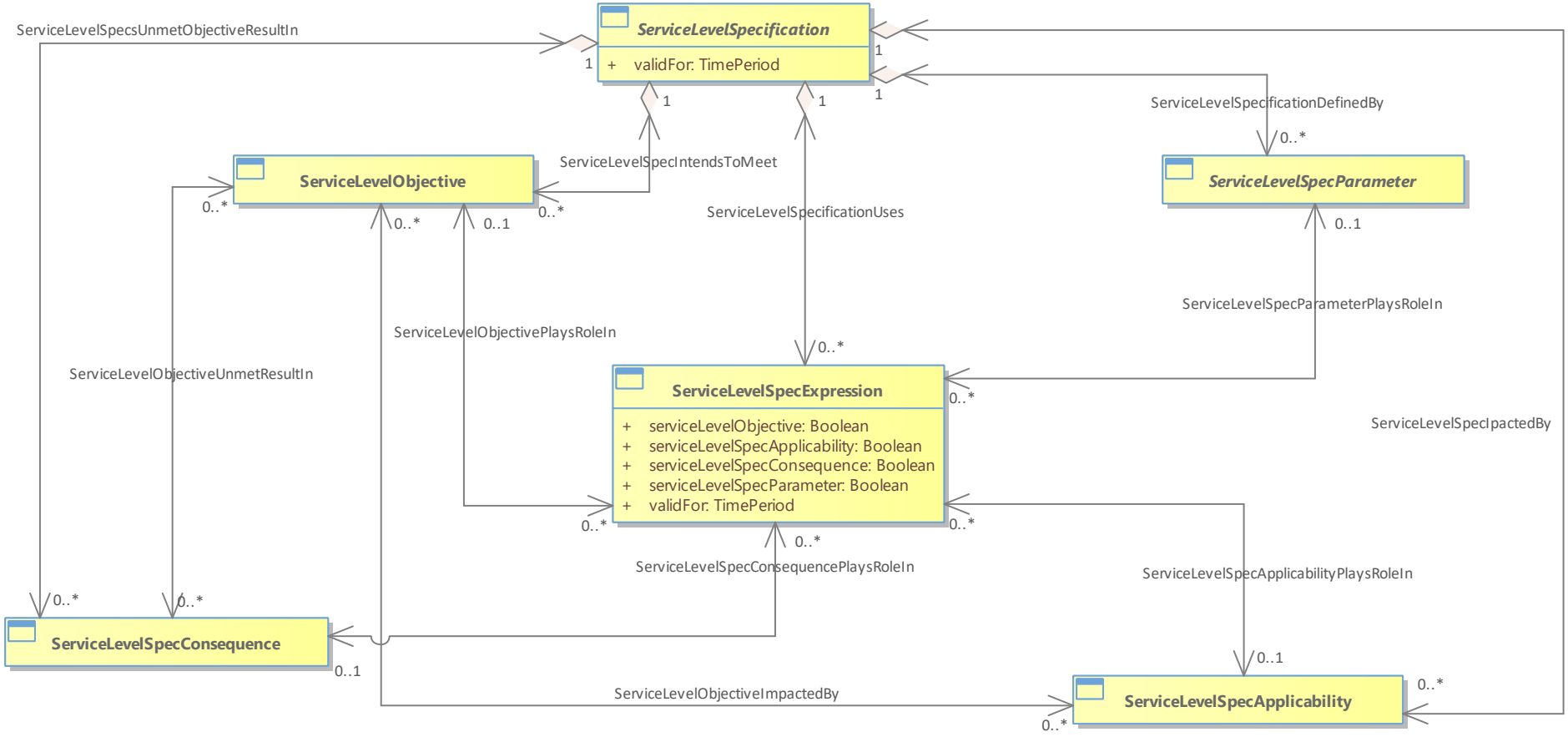

Figure SO.43a - ServiceLevelSpecificationExpression

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4.9.1 Service Level Spec ABE

The Service Specification ABE contains entities that define the invariant characteristics and behavior of both types of Service entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the invariant characteristics and behavior defined in its associated template.

This ABE includes a third type of Service Specification entity: that of a ServiceLevelSpecification. This type of specification templatizes Services that are bound to a Service Level Agreement.

Figure SLS.01 - Service Level Spec ABE Related Entities

Following are the business entities aggregated under the **Service Level Spec** ABE.

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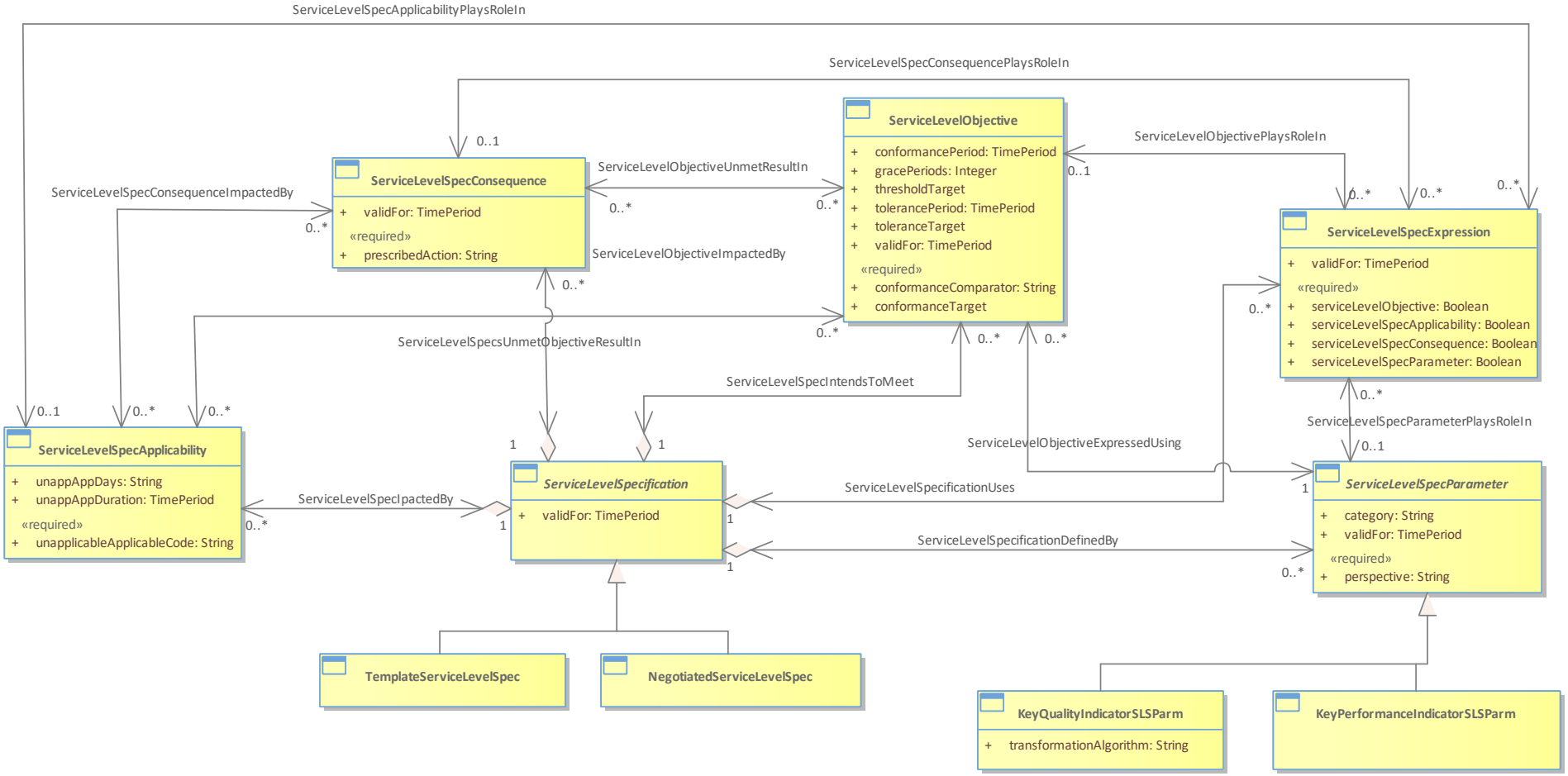

Figure SLS.01 - Service Level Spec ABE Related Entities

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4.9.2 Service Performance Specification ABE

The Service Performance Specification ABE defines measure of the manner in which a Service is functioning.

Figure SPS.01 - Service Performance Specification ABE Related Entities

Following are the business entities aggregated under the **Service Performance Specification** ABE.

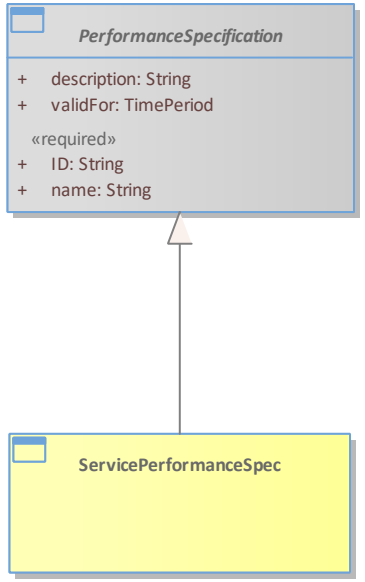

Figure SPS.01 - Service Performance Specification ABE Related Entities

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4.10 Service Capacity ABE

The **Service Capacity ABE** specifies ability of Service to provide capability measured in quantity and units of quantity over an extended period. It also manages the Service Capacity Demands.

Figure SC.01 - Basic Capacity Entities (based on CM.02 from Capacity ABE in Common Domain)

A generalized capacity and capacity demand model is shown in the following Figure.

The need to satisfy the capacity requirement during the first week in June is expressed using the fromToDateTime attribute, which is a composite of from and to dates/times, and the rangeInterval attribute of ApplicableTimePeriod. Refinements to the schedule can include days of the week as shown by the daysOfWeek attribute in the ApplicableTimePeriod entity. The rangeInterval attribute is used to handle some combination of >, <, >=, <=. It is assumed this is a planned Capacity, which would be indicated using the plannedOrActutalCapacity attribute. The CapacityRelationship entity enables a planned Capacity to be associated with an actual Capacity or for Capacities to be aggregated.

The requirement for this capacity to be available in the Northeast geographic area is expressed using the relationship to GeographicPlace. Other entities that further define/refine capacity requirements could be added as needed by extending the SID.

The ability to deliver 10000 units of Product described in the previous example is expressed via the CapacityAmount, whose datatype is Quantity. Quantity is a composite of an amount attribute and a units (unit of measure) attribute. For this instance of Capacity, the capacityAmount attribute contains the amount “10000”, and the unit of measure “units”. Other examples of units of measure could be transaction volumes, storage requirements, traffic volumes, port availabilities, and so forth. The capacityAmountFrom, capacityAmountTo, and rangeInterval enable ranges of capacity to be specified.

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Demands are placed on the capacity as defined by the CapacityDemand entity. The related ApplicableTimePeriod is used to here to express the timeframe over which the demand exists. It may or may not be the same instance of the ApplicableTimePeriod associated with the Capacity. Therefore, it may be a demand placed on more than one CapacityAmount. Similarly, the CapacityAmount may have many demands placed against it.

The attributes are similar to those in CapacityAmount with the addition of a priority attribute that defines the relative importance of this demand.

The AppliedCapacityDemand entity represents the amount of the CapacityDemand that has been applied against a CapacityAmount.

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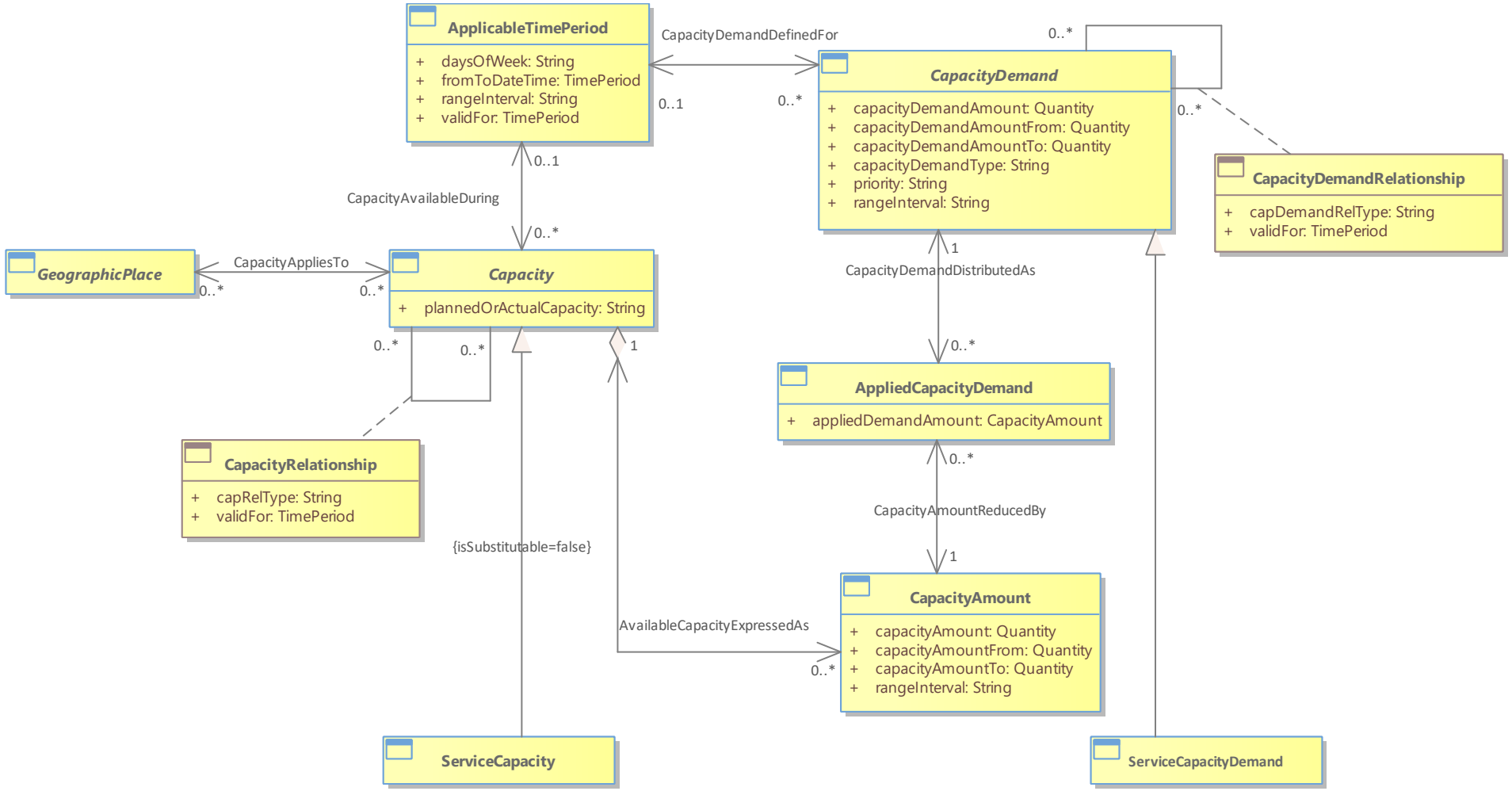

Figure SC.01 - Basic Capacity Entities (based on CM.02 from Capacity ABE in Common Domain)

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Figure SC.02 - Basic Service Capacity Entities (CM.06 Copy from Capacity ABE in Common Domain)

The following Figure shows ServiceCapacity specific relationships.

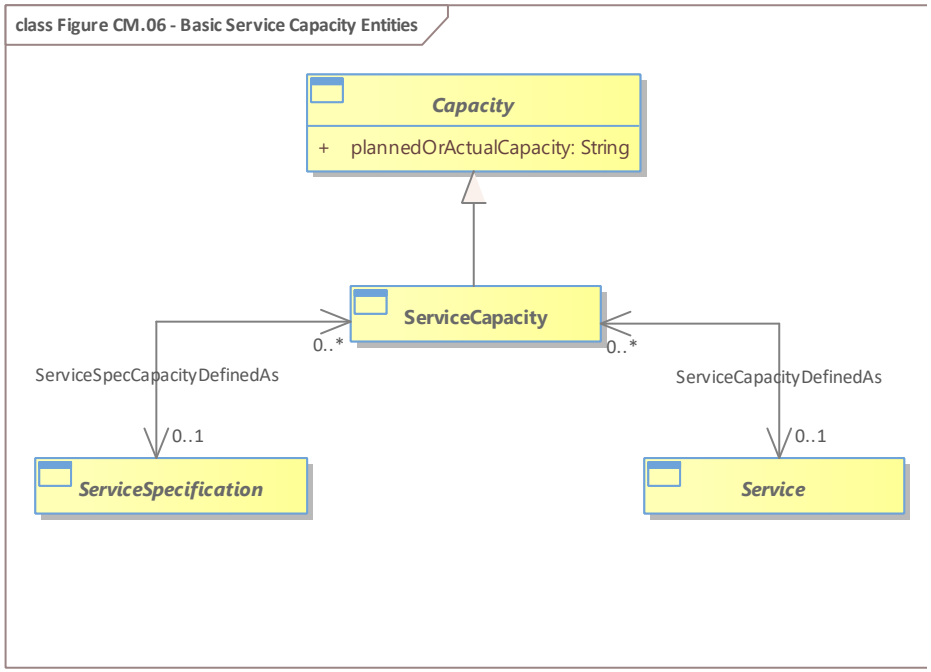

Figure SC.02 - Basic Service Capacity Entities (CM.06 Copy from Capacity ABE in Common Domain)

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Figure SC.03 - Basic Service Demand Entities (CM.09 Copy from Capacity ABE in Common Domain)

The following Figure shows ServiceCapacityDemand and its specific relationships.

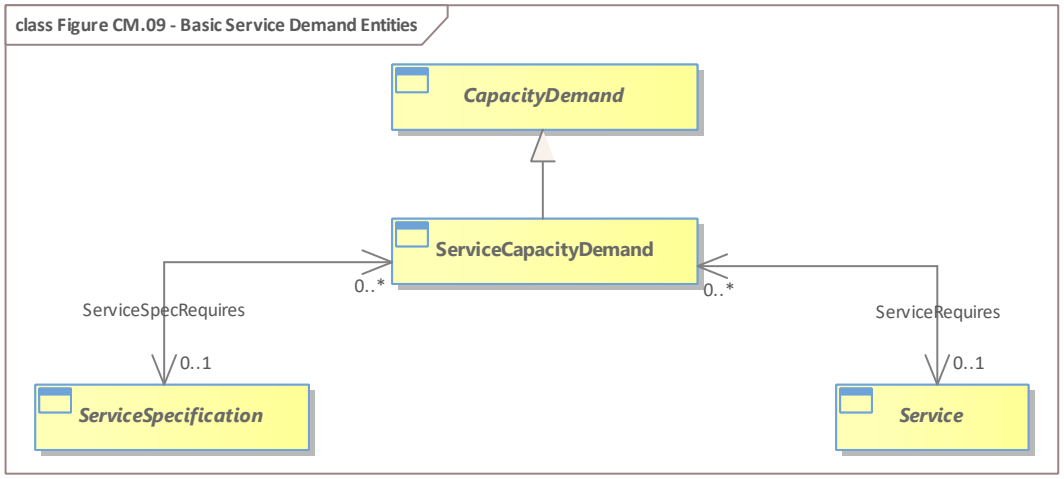

Figure SC.03 - Basic Service Demand Entities (CM.09 Copy from Capacity ABE in Common Domain)

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4.11 Service Problem ABE «NotFullyDeveloped»

The **Service Problem ABE** manages faults, alarms, and outages from a Service point-of-view. This is then correlated to trouble tickets, regardless of whether the cause is physical or logical.

Other entities in this ABE are used to direct the recovery from each of these types of problems. They provide the ability to associate Resource faults and alarms to degradation and outages of Services that run on those Resources.

Figure SP.01 - Service Problem ABE Related Entities

Following are the business entities aggregated under the **Service Problem** ABE.

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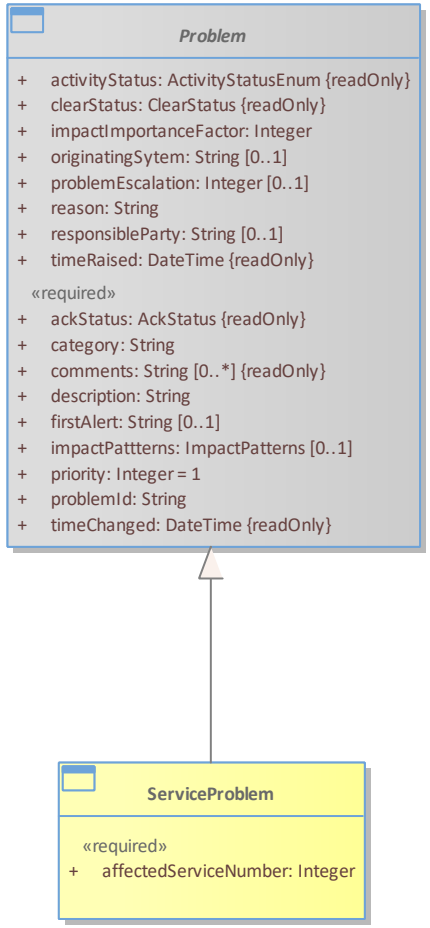

Figure SP.01 - Service Problem ABE Related Entities

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4.12 Service Strategy & Plan ABE «notFullyDeveloped»

The **Service Strategy and Plan ABE** contains entities that are used to address the need for enhanced or new Services, as well as the retirement of existing Services, by the enterprise. These entities have a strong dependency to both entities in the Resource and Product domains. Resulting efforts, such as deciding what Resources to use to host a Service, or what Services are used to support new Product Specifications, are also supported, as are service demand forecasts.

4.13 TIP Service Management ABE «likelyToBeDepreciated»

This ABE is likely to be removed in Release 25.5

The TMF Interface Program introduced in the Multi-Technology Operations System Interface (MTOSI) 2.0, Service Management interfaces based upon extensions to the SID service model in this ABE. The resulting solution set introduced two new service management activation interfaces, and a service inventory interface.

Introduction

- Note: TMF Interface Program (aka TIP) is legacy stuff to define interfaces and interfaces are now defined by the Open API Program.

The TMF Interface Program (aka TIP) introduced in the Multi-Technology Operations System Interface (MTOSI) 2.0, Service Management interfaces based upon extensions to the SID service model. The resulting solution set introduced two new service management activation interfaces, and a service inventory interface:

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- The Service Activation interface (SAI) between the Customer Relationship Management (CRM) and Service Management and Operations (SM&O) layers allows for the activation of Customer Facings Services (CFS)
- Interactive Service Activation
- *Best-effort Transaction interface* Service requests entailing multi-task transaction execution. In this type of interface, the entire transaction execution is completed despite single task failures.
- *Atomic Transaction interface* This interface has the same operations as the previous one; however, the failure of any single task will incur the termination of the transaction execution and will invoke a roll-back of the preceding task.
- *Service Order Management interface* Add-on interface to both the Best-effort Transaction or Atomic Transaction interfaces. Provides full exposure and control of the service order associated with the service request over the interface. It depends on the functions provided by the previous interfaces.
- The Service Component Activation interface (SCAI) is an internal SM&O layer interface. It allows for the activation of Service Components, also known as Resource Facing Services (RFSs), given specific service related information defined in a Service Template. The SCAI, contrary to the SAI, has no knowledge of the product domain.
- The Service Inventory interface supports the retrieval of service related entities from a Service Inventory repository.

The TIP Service Management ABE provides the extensions made to the SID model in support of the new Service Management interfaces introduced. The basic *service* and *service specification* concepts as defined in this addendum remain unchanged, and are merely enhanced to support the TIP Service Management interfaces.

Figure SO.31 TIP Service Management - Service Definition and Service Template

The abstract class ServiceSpecification has been extended to support two new concepts introduced in the TIP Service Management interfaces:

ServiceDefinition

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- used during service design time to define the characteristics of a service, their associated data types, and default values. These characteristics, known as ServiceSpecCharacteristics (SSC), may be globally set, which implies they can be defined in a service template, or otherwise only passed in real-time over an activation interface

ServiceTemplate

- defines specific ServiceSpecCharacteristicsValues (SSCVs) for the SSCs that were defined in the Service Definition as being globally settable. The Service Template is checked against its associated Service Definition by verifying the presence of the SSCs and the validity of the corresponding assigned SSCVs. Only Service Templates that contain SSCs/SSCVs that are in conformance with their specification in the Service Definition are considered valid.

Both Service Activation interfaces make use of the Service Definition and Service Template concepts in the activation of CustomerFacingServices and ResourceFacingServices.

The ServiceDefinition and ServiceTemplate further define the Service Specification through their respective associations, with the versions being identified through the ServiceSpecVersion. The association class ServiceSpecRelationship allows for Service Specifications to contain other Service Specifications, which thus holds true for Service Definitions and Service Templates. For example, there may be a type of Service Definition for Ethernet Services, and contained within that type are E-LINE, E-LAN, and E-TREE Service Definitions.

The ServiceAccessPoint (SAP) represents a unique (logical and/or physical) resource where the Service can be accessed. The Sap Specification defines a set of attributes and associated values that may be applied to one or more SAPs. The relationship between the Service Specification and the Sap Specification illustrates that for every Service Specification, there may be associated multiple Sap Specifications. For example, given an Ethernet E-LINE Service Specification, there may be multiple Sap Specifications that correspond to various Bandwidth Profile Specifications that can be applied to the Sap – one for GOLD, one for SILVER, and one for BRONZE. Likewise, there may be a single Sap

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Specification , or in this case Bandwidth Profile Specification, that is used for multiple Ethernet Service Specifications – E-LINE and E-LAN. A Bandwidth Profile Specification defines Service Level Agreement guarantees and may be applied per UNI, per EVC, or per CoS Id. They contain frame delivery per Service Level Objectives (Committed Information Rate, Committed Burst Size), and amount of excess frame delivery allowed (Excess Information Rate, Excess Burst Size).

A ServiceCatalog represents a grouping of Service Specifications and SSCs. For example, a catalog could group all internet related Service Specifications. The following figure illustrates the Service Definition, and Service Template concepts, and the relationship between the Service Catalog and the Service Specification.

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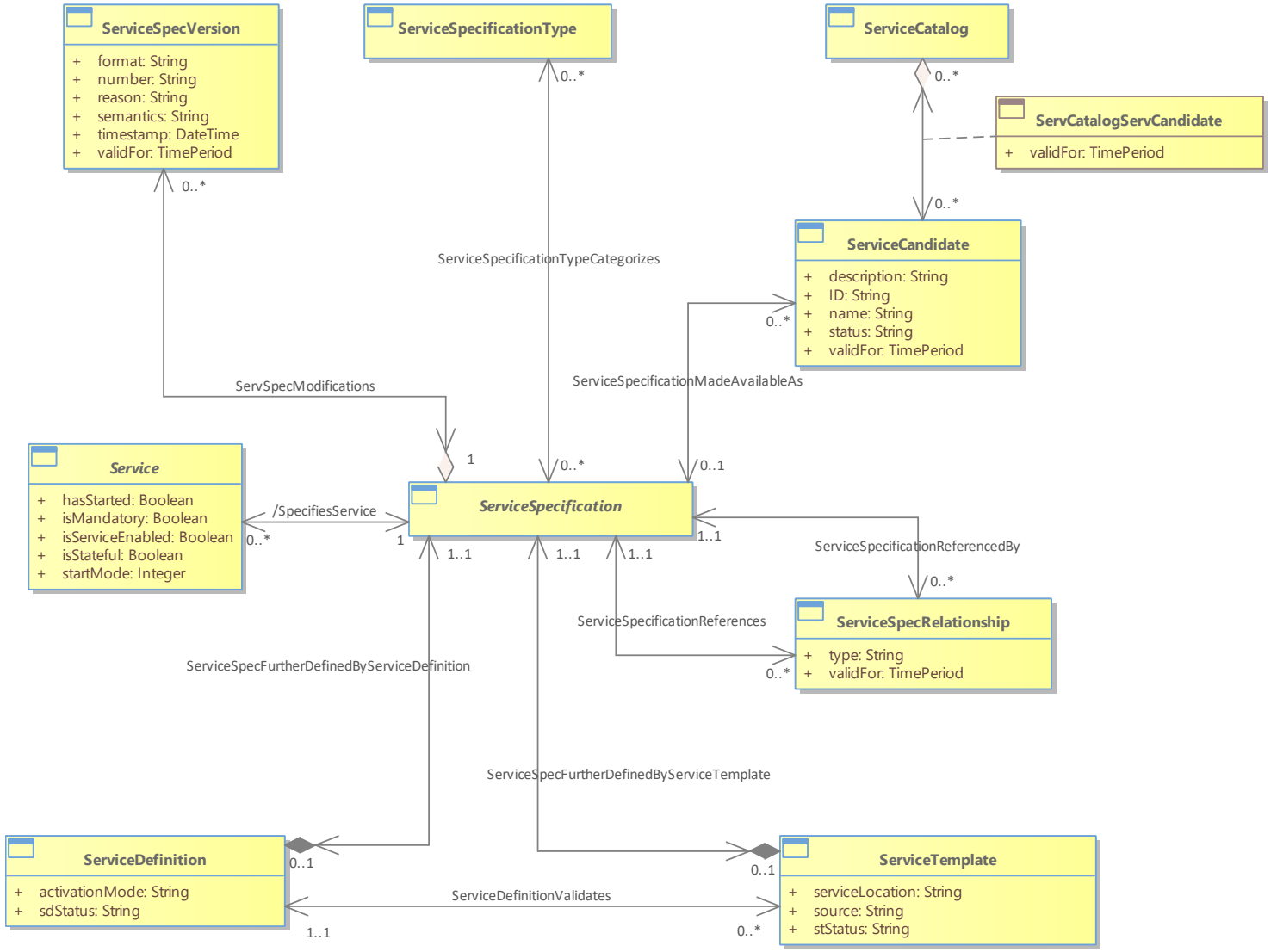

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Figure SO.31 TIP Service Management - Service Definition and Service Template

Figure SO.32 TIP Service Management - Service Characteristics

The ServiceSpecCharacteristic and the associated ServiceSpecCharacteristicValues pairs are used in both the Service Definition and the Service Template to describe, or characterize, a Service. The CharacteristicSpecification and the CharacteristicSpecValues pairs are used to characterize the Service Access Point through the association of the SapSpecification with the EntitySpecCharUse. These provide the actual characteristic / values pairs that would be found, for example, in a Bandwidth Profile Specification for an Ethernet Service User Network Interface (UNI) or Ethernet Virtual Connection (EVC) endpoint. The Bandwidth Profile Specification would contain the following characteristics: committed information rate (CIR), committed burst size (CBS), excess information rate (EIR), excess burst size (size), and possibly color marking (CM) and color forwarding (CF) indicators. By defining this specification, including the associated characteristic values, as an independent entity, the specification could be associated to a Sap or an EVC.

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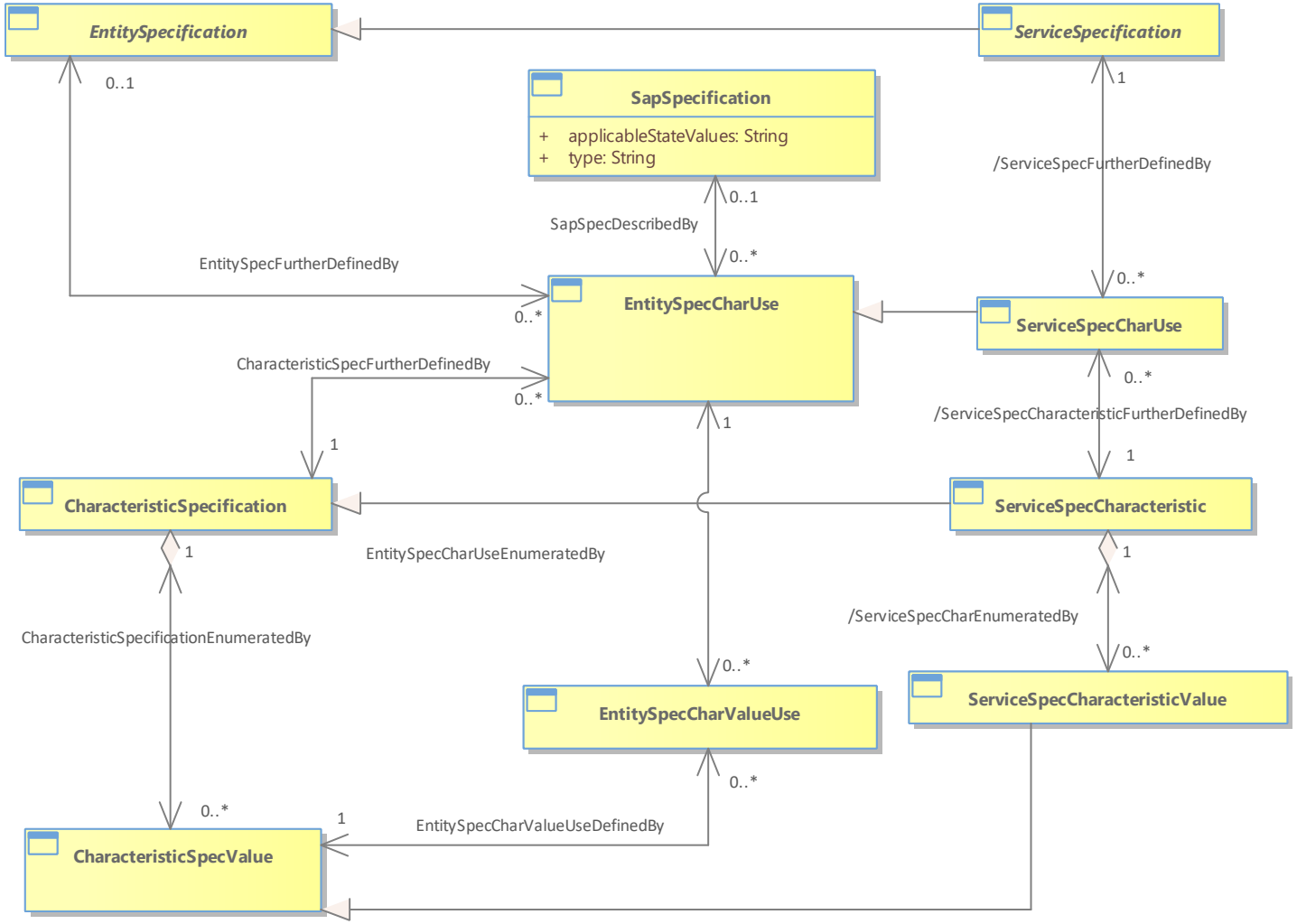

Figure SO.32 TIP Service Management - Service Characteristics

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Figure SO.33 TIP Service Management - Service Activation

The service configuration and activation process involves the instantiation of both Customer Facing Services, and associated Resource Facing Services. The CFS and RFS are constructed based upon their respective Service Definitions, ServiceTemplates, and real-time ServiceCharacteristicValues passed over the activation interface. The interface context will determine whether the Service Template being referenced is for a CFS (SAI) or an RFS (SCAI).

The CFS and RFS are accessed via one or more ServiceAccessPoints (SAPs) that are characterized by SapSpecifications. There is a relationship between the SAPs and the underlying Resources serving as access points. The following diagram illustrates the classes and relationships as required for Service Activation.

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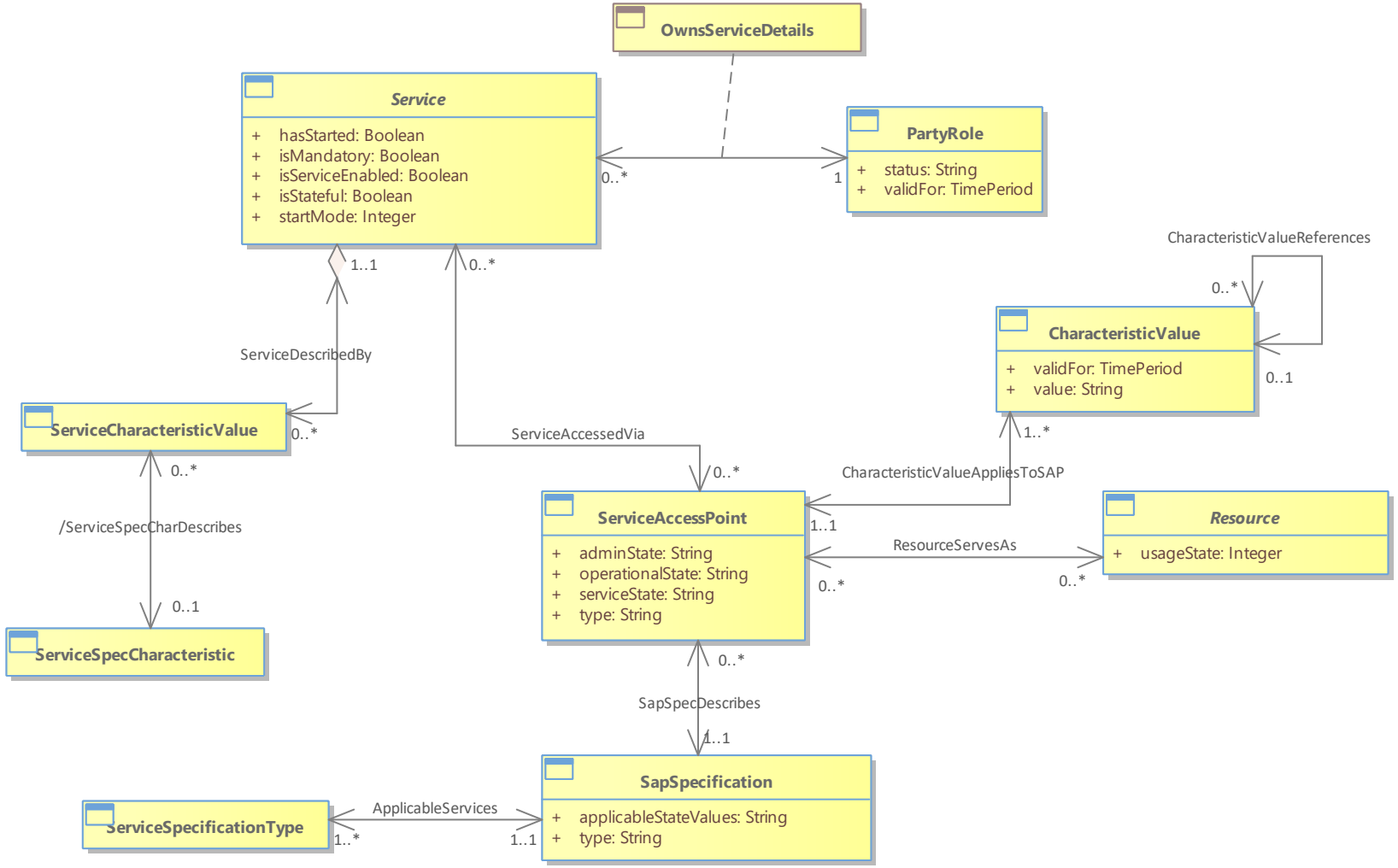

Figure SO.33 TIP Service Management - Service Activation

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Figure SO.33a - TIP Service Management - CommonServiceInfo

A number of the Service Management entities are extended via their association with the CommonServiceInfo entity and via inheritance with the CommonObjectInfo entity as shown in Figure SO.33a.

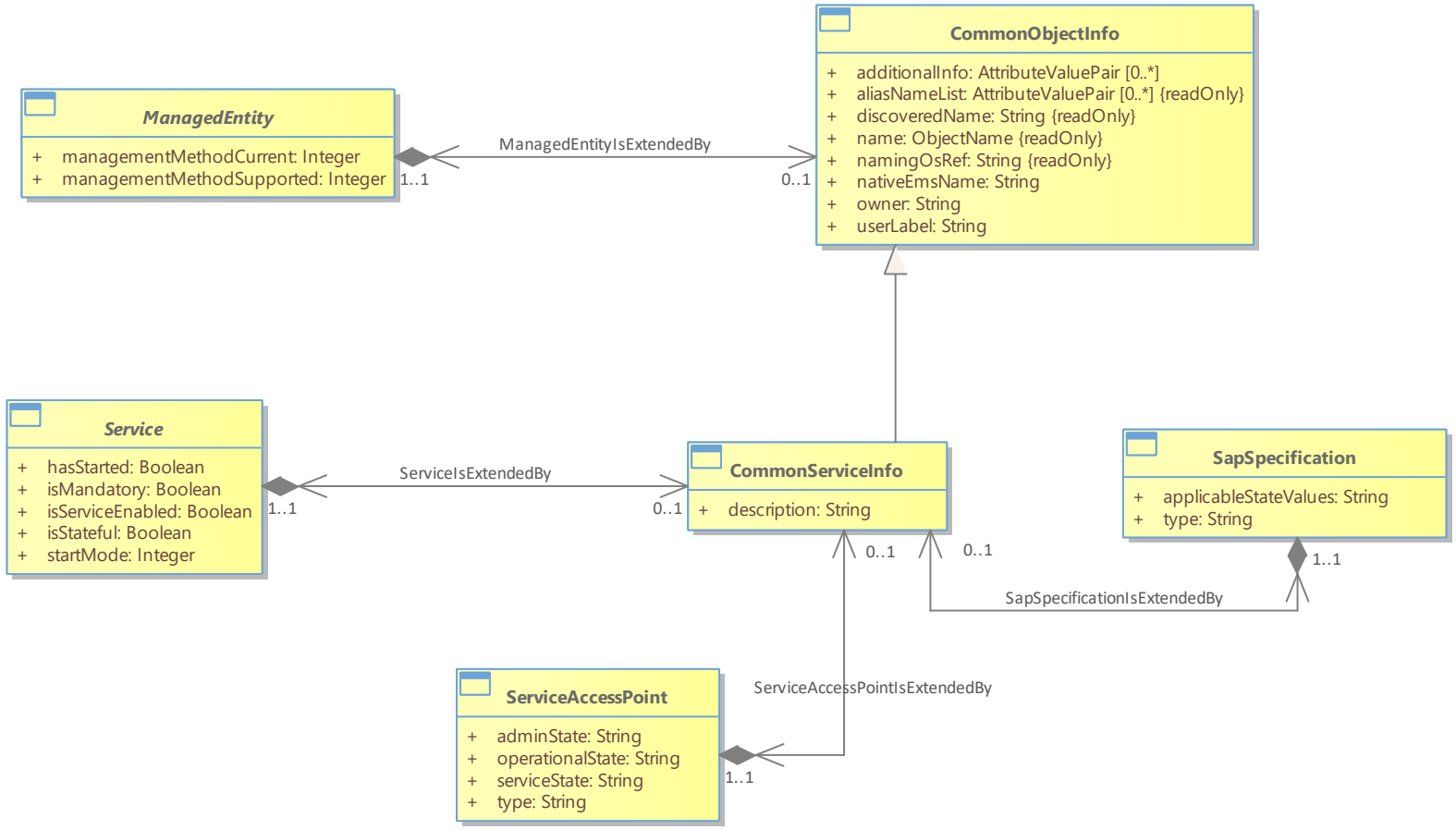

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Figure SO.33a - TIP Service Management - CommonServiceInfo

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Figure SO.34 - Example of Ethernet Service CFS/RFS

Ethernet Services Example

The following example is based upon multiple service providers involved in the delivery of wholesale and retail services. The Network Service Provider (NSP) offers triple-play IP services to his residential customers, represented by an IP CFS. This CFS is instantiated over the Service Activation Interface (SAI). The NSP Service Management OSS decomposes the CFS into different resource facing service components that require management, one of them being the Ethernet Connectivity RFS (which may also be split into Ethernet access, and Ethernet Aggregation and Backhaul).

The NSP purchases both Ethernet Access and Aggregation & Backhaul Products from the Broadband Ethernet Access (BEA) and Broadband Aggregation & Backhaul (BAB) providers respectively; these are represented in the BEA/BAB domains as “products”. They are instantiated as Ethernet Access Line CFS and Ethernet Backhaul CFS over the BEA/BAB Service Activation Interfaces in their respective OSS/BSS domains. The BEA and BAB Service Management OSS decompose these CFS’s into resource facing services that get instantiated over Service Component Activation interfaces. As a result, the underlying resources to support the component RFS are provisioned. The BEA and the BAB notify the NSP that their respective services are available and operational.

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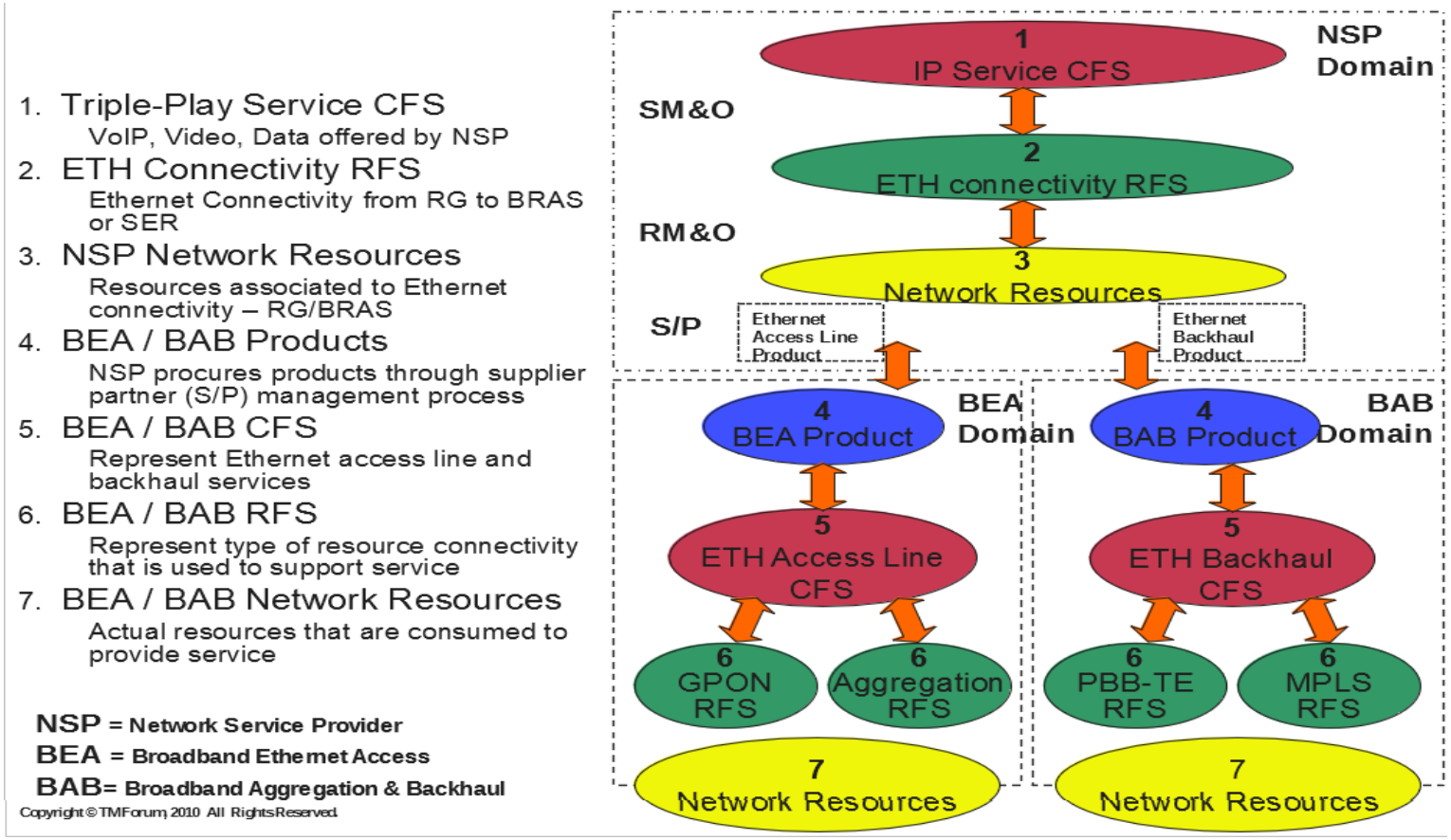

Figure SO.34 - Example of Ethernet Service CFS/RFS

5 Administrative Appendix

This Appendix provides additional background material about the TM Forum and this document. In general, sections may be included or omitted as desired; however, a Document History must always be included.

About this document

This is a TM Forum Guidebook. The guidebook format is used when:

The document lays out a 'core' part of TM Forum's approach to automating business processes. Such guidebooks would include the Telecom Operations Map and the Technology Integration Map, but not the detailed specifications that are developed in support of the approach.

Information about TM Forum policy, or goals or programs is provided, such as the Strategic Plan or Operating Plan.

Information about the marketplace is provided, as in the report on the size of the OSS market.

5.1 Document History

Version History

Version Number	Date	Modified by	Purpose
0.1	September 2002	J. Strassner	First rough draft
0.2	October 2002	J. Strassner	Final population using DEN-ng high-level service model, which contains mined information from DEN, OSS/J and MetaSolv

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Version Number	Date	Modified by	Purpose
0.3	October 2002	J. Strassner	Final draft ready for modeling team to review
0.4	November 2002	J. Strassner	Updated in response to detailed comments from John Reilly and Chris Hartley
0.5	November 2002	J. Strassner	Updated to incorporate additional comments from John Reilly and Cliff Faurer.
1.0	November 2002	TMF	Formatted for TMF Publishing
1.1	April 2003	J. Strassner	Added ServicePackage and ServiceBundle, which link to the QoS model. Minor corrections to get Service model better in sync with other models
1.2	May 2003	J. Strassner	Incorporated comments from the SID team, including detailed comments from Cliff and John Reilly. Excised ProductItem, and enhanced the link to the QoS Addendum.
1.3	May 2003	J. Strassner	Incorporated final comments from Cliff, updated for consistency with other Addenda
1.4	June 2003	J. Strassner	Updated PartyRole interaction section
2.0	July 2003	TMF	Formatted for Member Review with NGOSS R3.5
2.1	Aug 2004	TMF	To reflect TMF Approved Status
3.0	Mar 2008	J Reilly	Updates to Characteristics and miscellaneous updates to synchronize addendum with UML model.

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Version Number	Date	Modified by	Purpose
7.5	Jan 2008	John Reilly	Updated for ITU-T submission
7.6	May 2008	Tina O'Sullivan	Converted to correct template and other corrections prior to posting.
7.7	July 2008	John Reilly	Updates to Characteristics and miscellaneous updates to synchronize addendum with UML model.
7.8	July 2008	John Reilly	Updates after new template applied
7.9	July 2008	Tina O'Sullivan	Final changes prior to posting.
8.0	May 2009	Alicja Kawecki	Minor updates to reflect TM Forum Approved status
9.0	January 2010	John Reilly	Updates to reflect SID Phase IX Change Requests
9.1	April 2010	Alicja Kawecki	Minor cosmetic corrections for web posting and ME
9.2	June 2010	Alicja Kawecki	Updated Notice
9.3	Oct 2010	Alicja Kawecki	Updated to reflect TM Forum Approved status
9.4	January 2011	Josh Salomon	Updates to reflect SID release 9.5 change requests
9.5	March 2011	John Reilly	Correct figure references and corrupt table.

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Version Number	Date	Modified by	Purpose
9.6	March 2011	Alicja Kawecki	Updated Notice, minor formatting corrections prior to web posting and ME
9.7	September 2011	Alicja Kawecki	Updated to reflect TM Forum Approved status
9.8	June 2013	Cécile Ludwichowski	Incorporation of CR: Artf2381 (Figure SO.45) Artf1431 (Figure SO.31, SO-33) Artf1788
9.9	July 2013	Alicja Kawecki	Updated cover & notice, corrected footer prior to posting
9.10	Sep 2013	Alicja Kawecki	Added IPR Mode to cover, updated notice to reflect TM Forum Approved status
9.10.1	Nov 2013	Alicja Kawecki	Updated cover, header, footer & Notice to reflect TM Forum Approved status
14.5.0	October 2014	John Reilly	Updated Figure SO.7 to align with model
14.5.1	March 2015	Alicja Kawecki	Updated cover, header, footer and Notice to reflect TM Forum Approved status
16.0	May 2016	John Reilly	Aligned model and guidebook, removed data dictionary, and aligned with changes in other domains.

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Version Number	Date	Modified by	Purpose
16.0.1	26 May 2016	Alicja Kawecki	Minor cosmetic edits prior to publication for Fx16
17.0.0	March 2017	Cécile Ludwichowski	Remove SID framework figure from the guidebook
17.0.1	27 June 2017	Alicja Kawecki	Applied rebranding and minor cosmetic edits prior to publication for Fx17
17.0.2	16 November 2018	Adrienne Walcott	Updated cover, notice and minor cosmetic changes prior to publishing
17.5.0	30 June 2017	Cécile Ludwichowski	Correct typo for AdministerResourceDetails (FP-462) Correct all figures numbering Change CFSSpec, RFSSpec, CFS, RFS - Composite / Atomic to concrete Class fixed typo
17.5.1	19 Dec 2017	Adrienne Walcott	Formatting/style edits prior to publishing
18.0.0	05 Jun 2018	Cécile Ludwichowski	Replaces the relation between ProductSpecification and ProductOffering by a derived relationship that can be seen as an overview Characteristic model clean-up for Product and ProductSpecification

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Version Number	Date	Modified by	Purpose
18.0.1	12 Jul 2018	Adrienne Walcott	Formatting/style edits prior to R18 publishing
18.0.2	10-Oct-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status
18.5.0	03-Dec-2018	Cécile Ludwichowski	Updated according to FP-745 SID-Characteristics clean-up for Service, Resource and Usage.
18.5.1	05-Mar-2019	Cécile Ludwichowski	Updated to reflect TM Forum Approved Status
20.0.0	03-June-2020	Cécile Ludwichowski	Guidebook generated from RSA with BIRT report.
21.0.0	04-June-2021	Cécile Ludwichowski	This release includes changes coming from the following Change Request - Split Capacity ABE in each horizontal Domain concerned
23.0.0	16-June-2023	Cécile Ludwichowski	This release includes diagram review and cleanup given the move from RSA to Sparx EA modeling tool per ISA-772
24.5.0	10-Jan-2025	Kevin Scaggs	This release includes Usage ABE grouping updates per ISA-969, Level 1 ABE description updates per ISA-1113, and diagrams for ABEs with no diagrams per ISA-1202
25.0.0	18-July-2025	Kevin Scaggs	This release of the Service Guidebook includes updates as follows: • Adding concept of UsageVolumeBalance per ISA-1129, and • Updating TIP Service Management ABE to "likely to be depreciated" per ISA-1243.

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Release History

Release Number	Date Modified	Modified by:	Description of changes
8.0	17/July/2008	John Reilly	Updates to Characteristics and miscellaneous updates to synchronize addendum with UML model.
9.0	January 2010	John Reilly	Updates to reflect SID Phase IX Change Requests
9.5	January 2011	Josh Salomon	Updates to reflect SID release 9.5 change requests
13.0	June 2013	Cécile Ludwichowski	Incorporation of CR: Artf2381 (Figure SO.45) Artf1431 (Figure SO.31, SO-33 Artf1788
14.5.0	October 2014	John Reilly	Updated Figure SO.7 to align with model and added ServiceLevelSpecExpression entity and figure.
16.0	May 2016	John Reilly	Aligned model and guidebook, removed data dictionary, and aligned with changes in other domains.
17.0	March 2017	Cécile Ludwichowski	Remove SID framework figure from the guidebook
17.1	November 2017	Adrienne Walcott	Updated to reflect TM Forum Approved Status

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Release Number	Date Modified	Modified by:	Description of changes
17.5	December 2017	Cécile Ludwichowski	Correct typo for AdministerResourceDetails (FP-462) Correct all figures numbering Change CFSSpec, RFSSpec, CFS, RFS - Composite / Atomic to concrete Class (FP-627) fixed typo
18.0.0	05 June 2018	Cécile Ludwichowski	Replaces the relation between ProductSpecification and ProductOffering by a derived relationship that can be seen as an overview Characteristic model clean-up for Product and ProductSpecification Remove all Entities identified as “Likely to be removed” in 17.0
18.0.1	10-Oct-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status
18.5.0	03-Dec-2018	Cécile Ludwichowski	Updated according to FP-745 SID-Characteristics clean-up for Service, Resource and Usage.
18.5.1	05-Mar-2019	Adrienne Walcott	Updated to reflect TM Forum Approved Status
20.0.0	03-June-2020	Cécile Ludwichowski	Guidebook generated from RSA with BIRT report.
21.0.0	04-June-2021	Cécile Ludwichowski	This release includes changes coming from the following Change Request -Split Capacity ABE in each horizontal Domain concerned

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Release Number	Date Modified	Modified by:	Description of changes
23.0.0	16-June-2023	Cécile Ludwichowski & Kevin Scaggs	This release includes diagram review and cleanup given the move from RSA to Sparx EA modeling tool per ISA-772.
24.5.0	10-Jan-2025	Kevin Scaggs	This release includes Usage ABE grouping updates per ISA-969, Level 1 ABE description updates per ISA-1113, and diagrams for ABEs with no diagrams per ISA-1202.
25.0.0	18-July-2025	Kevin Scaggs	This release of the Service Guidebook includes updates as follows: • Adding concept of UsageVolumeBalance per ISA-1129, and • Updating TIP Service Management ABE to "likely to be depreciated" per ISA-1243.

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5.3 Acknowledgments

This document was prepared by the members of the TM Forum Information Framework (SID) team.

The Shared Information/Data Model is a genuinely collaborative effort. The TM Forum would like to thank the following people for contributing their time and expertise to the production of this document. It is just not possible to recognize all the organizations and individuals that have contributed or influenced the introduction. We apologize to any person or organization we inadvertently missed in these acknowledgments.

Key individuals that reviewed, provided input, managed, and determined how to utilize inputs coming from all over the world, and really made this document happen were:

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Jean-Marie Magueur	Orange
Emmanuel Otchere	Huawei
Kevin Scaggs	TM Forum

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